

INDUCTANCE

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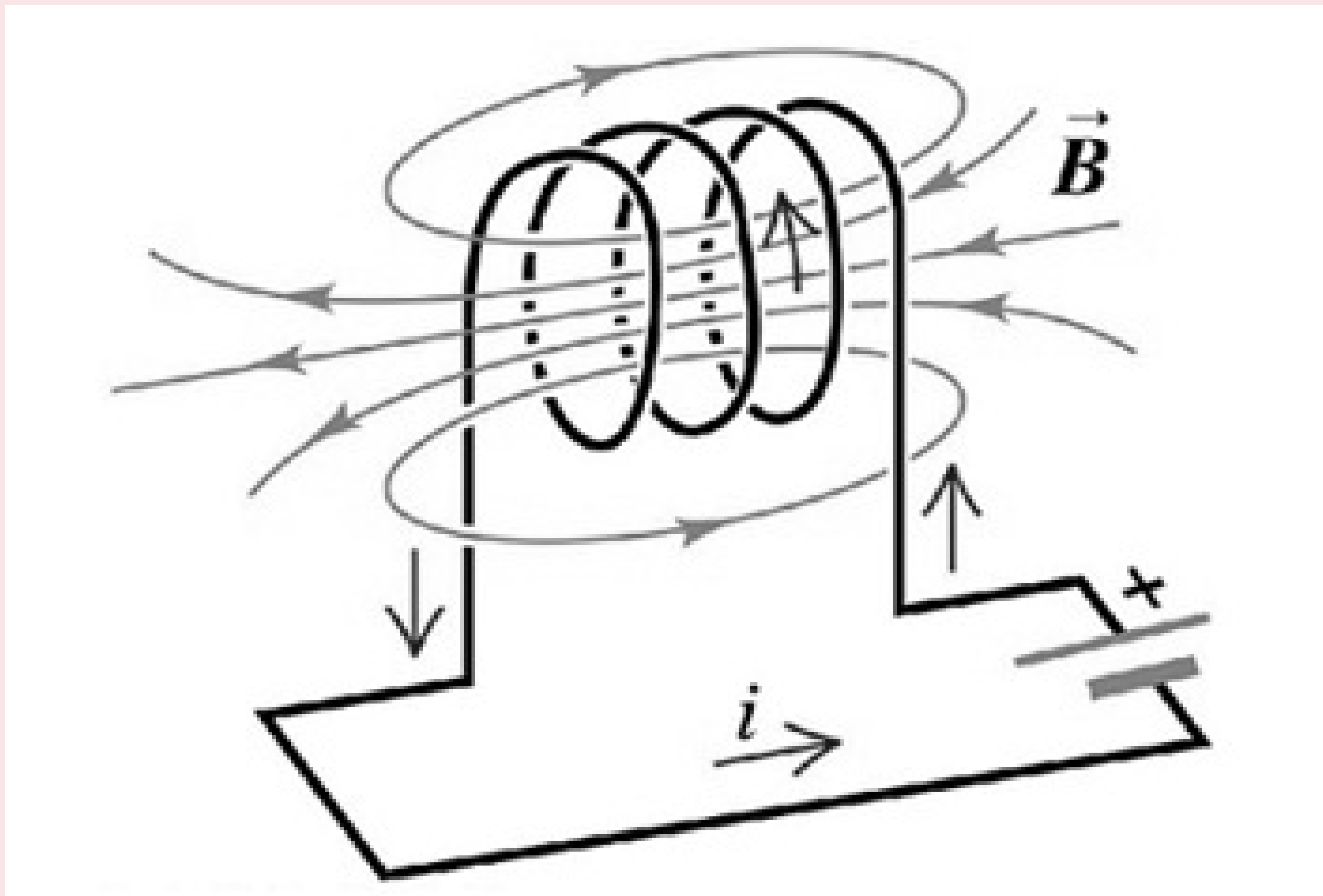
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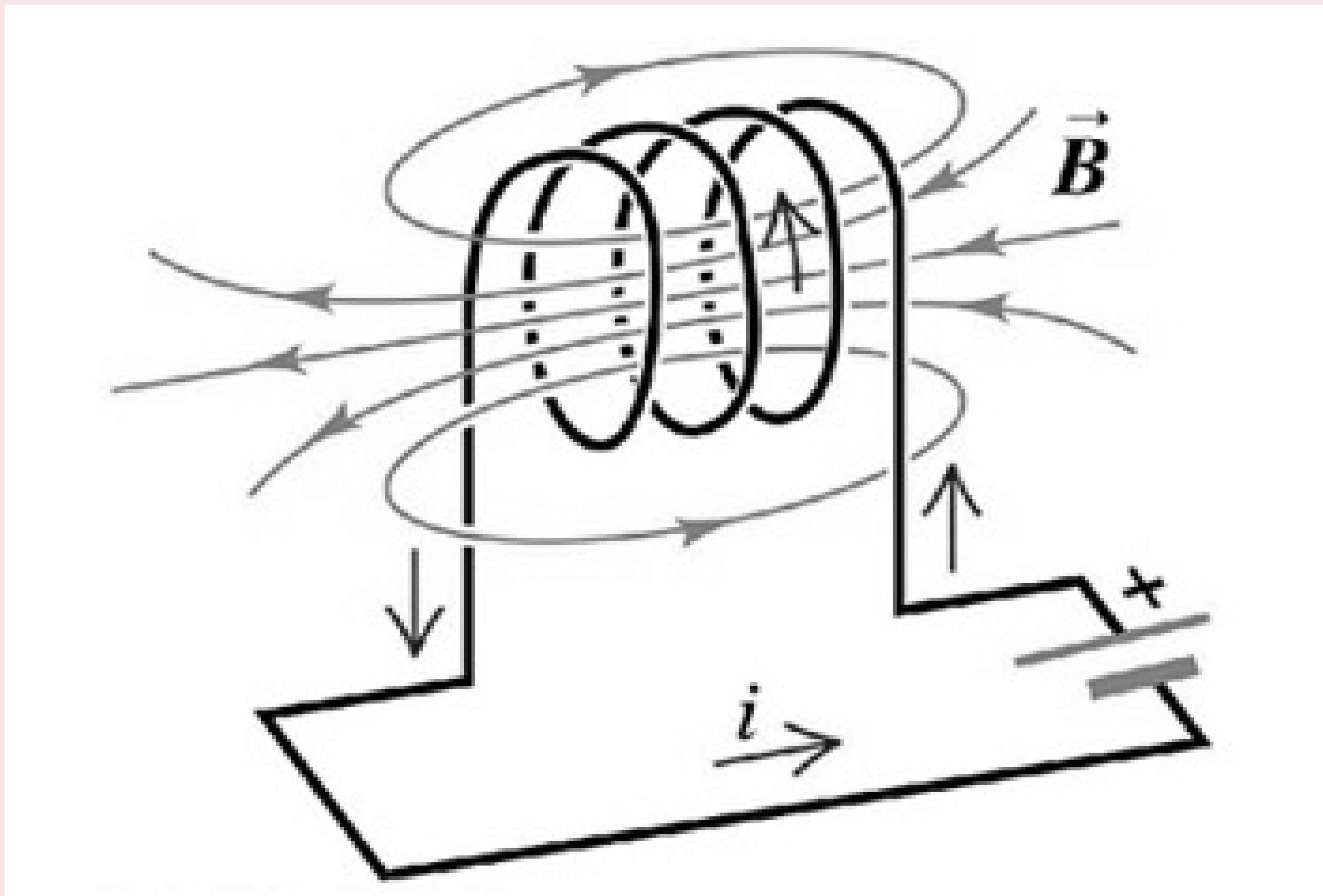
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- **Inductance** is the **tendency of an electrical conductor to oppose a change in the electric current** flowing through it.
- **Inductance** is the **property of a device that tells us how effectively it induces an emf in another device**.

Self-Inductance

- **An induced EMF appears in a single coil if the current in that coil is changing.** This phenomenon is known as **self-inductance**.

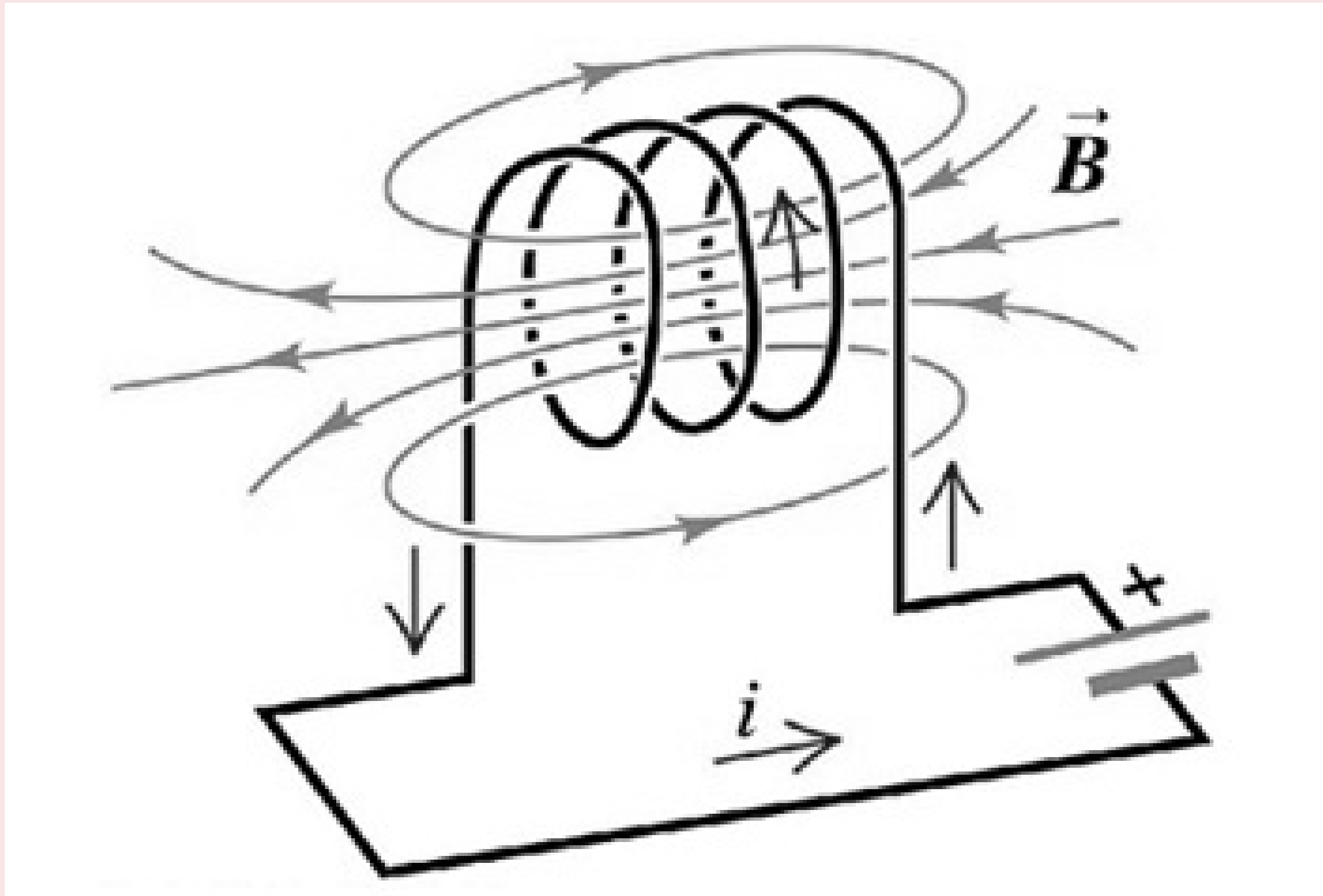


Self-Inductance



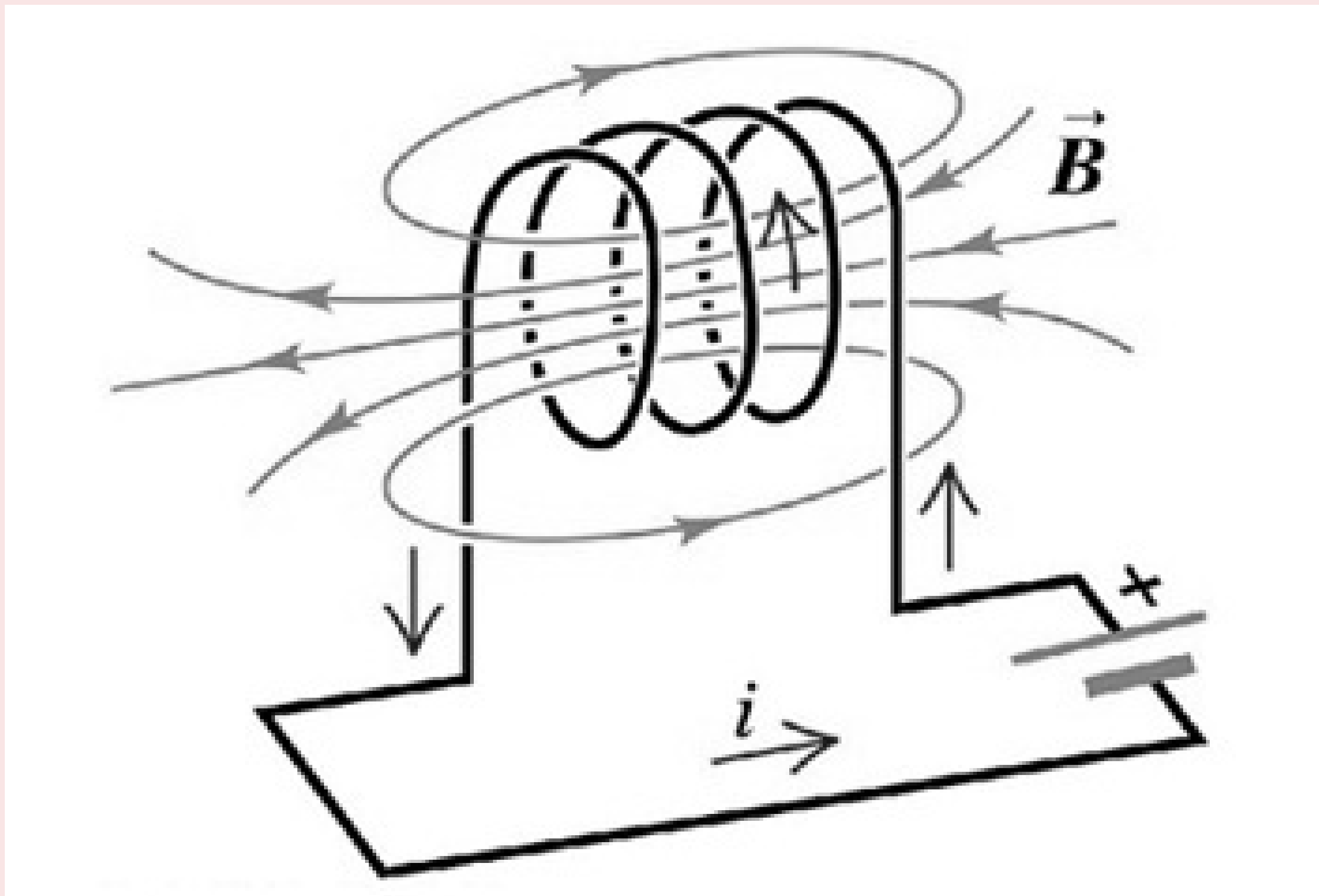
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- This EMF is called the self induced EMF and it obeys Faraday's law of induction.

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- **self-induced emf** always **opposes any change** (increase or decrease) of **current** in the coil.

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- The **self-induced emf** is **also called** the **back emf** as it opposes any change in the current in a circuit.
- **self-inductance** plays the role of **inertia**. It is the **electromagnetic analogue of mass** in mechanics.

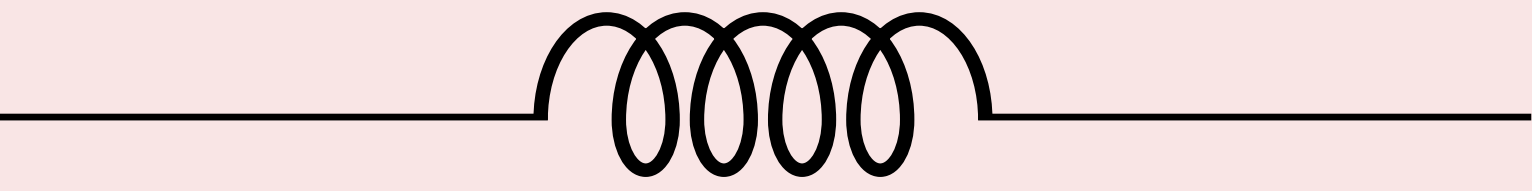
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Self-Inductance of a long Solenoid

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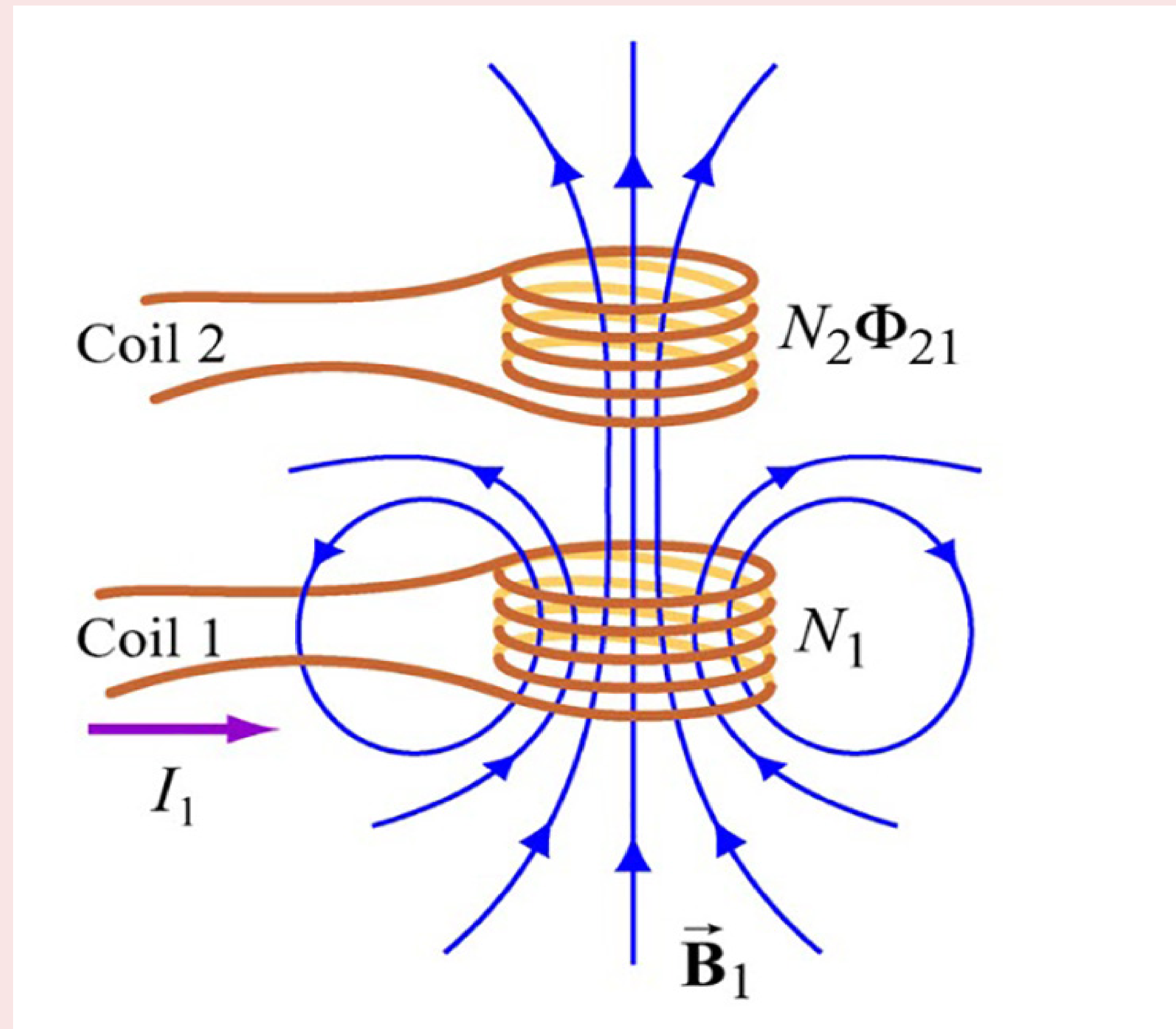
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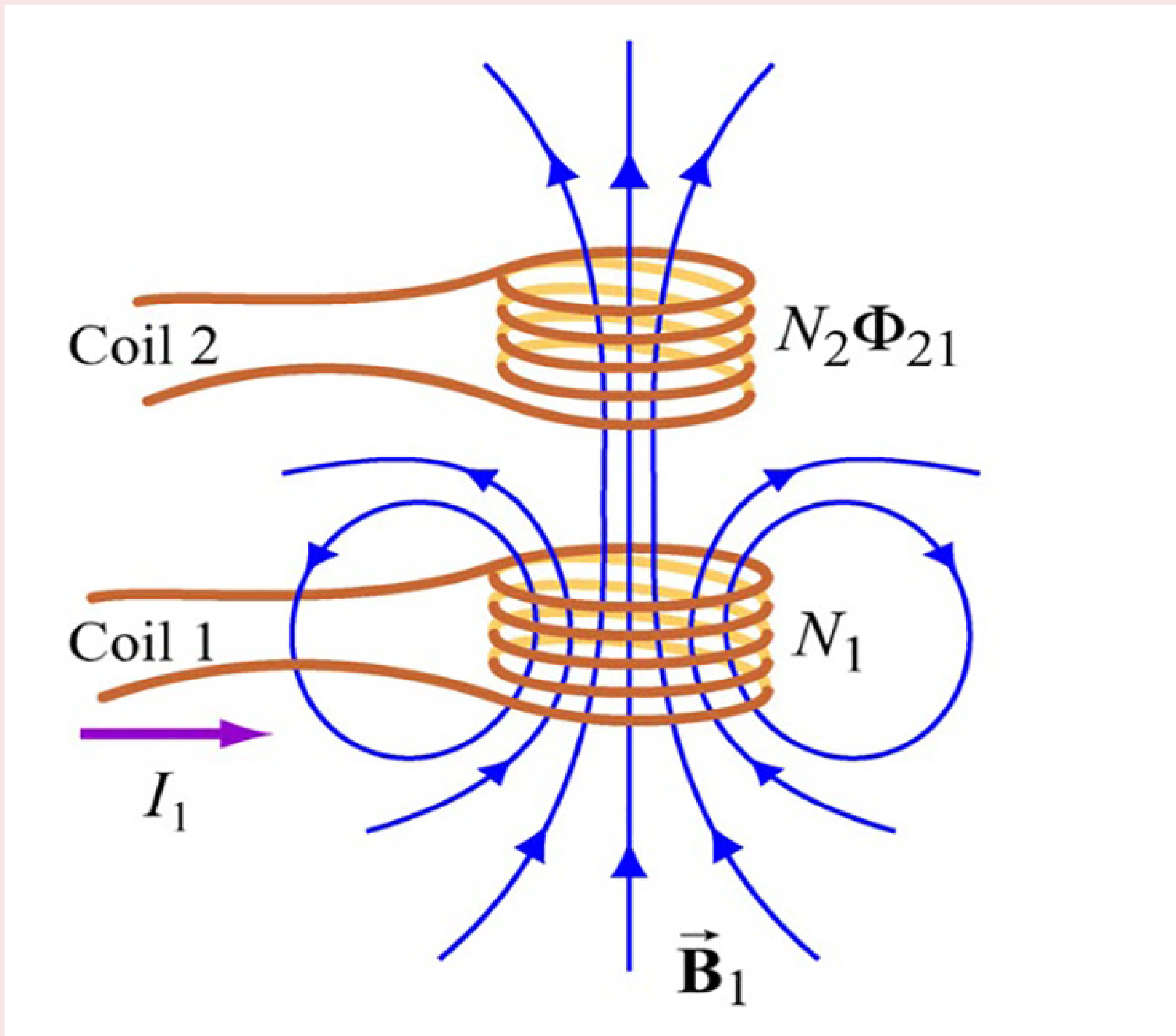
- **Self inductance of solenoid** : $L = \mu_0 n^2 l A$ or $L = \frac{\mu_0 N^2 A}{l}$

Mutual-Inductance

- **Mutual inductance** is the phenomenon in which a **change in current in one coil induces an EMF in another nearby coil** due to the shared magnetic field.

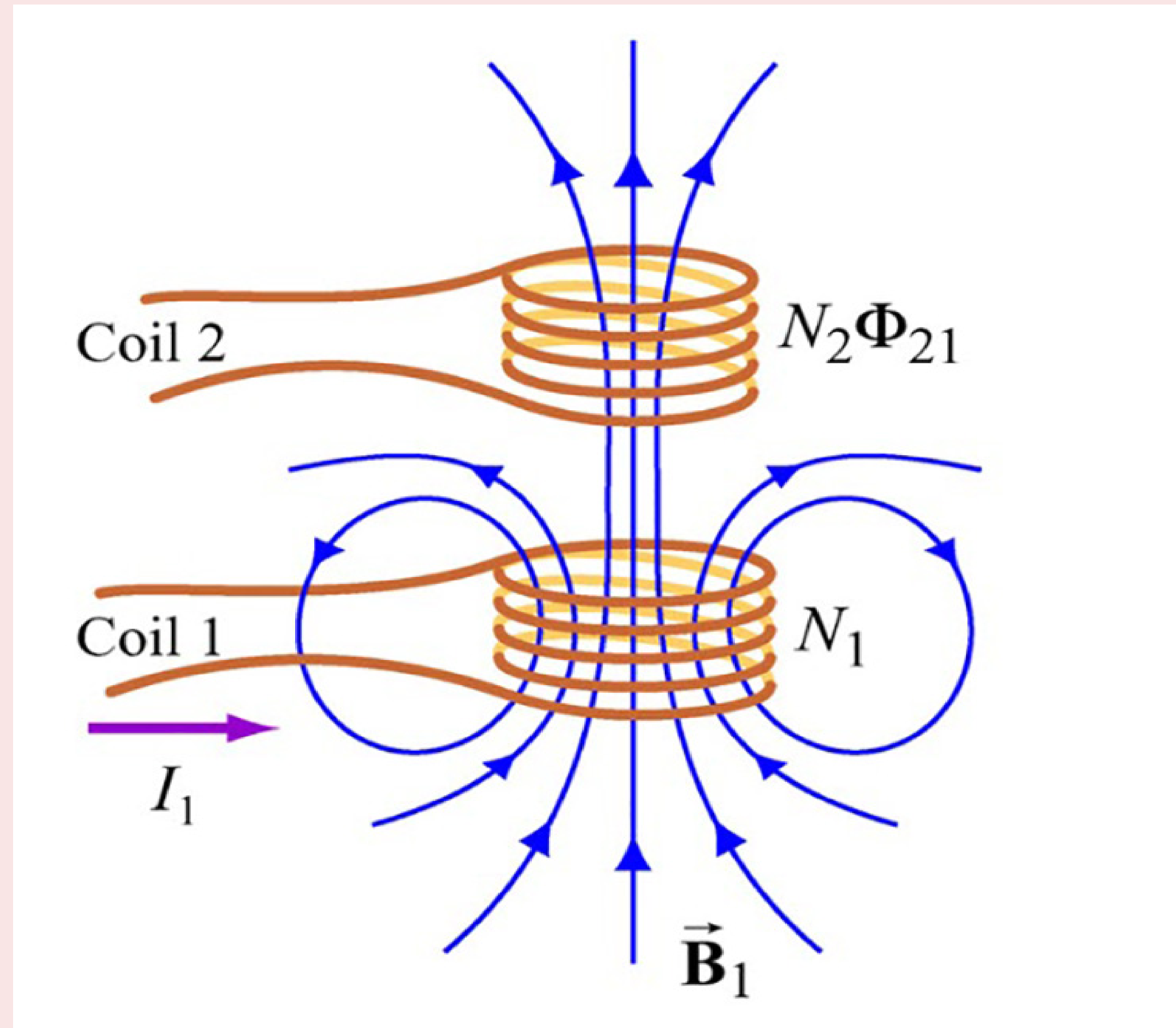


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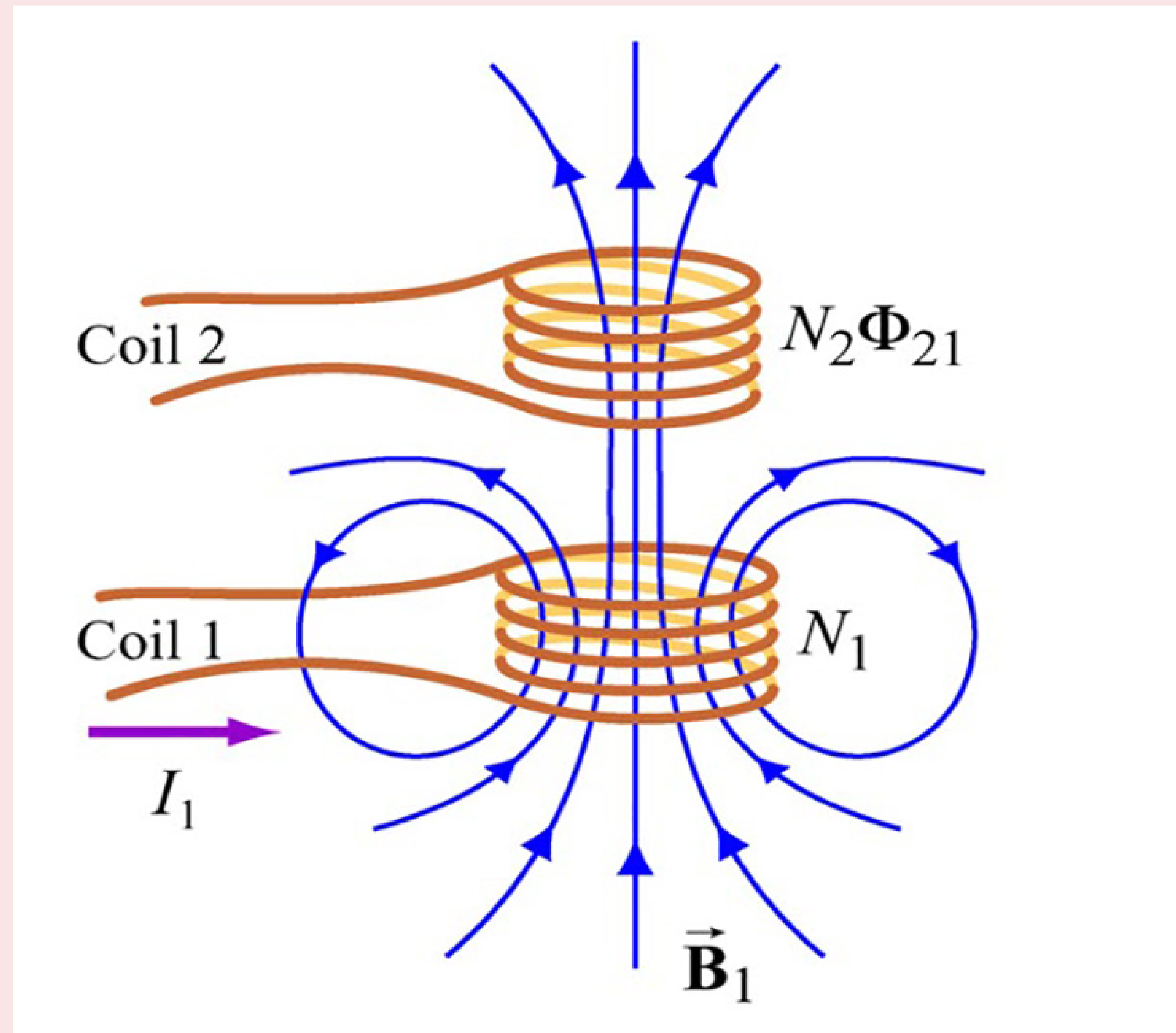
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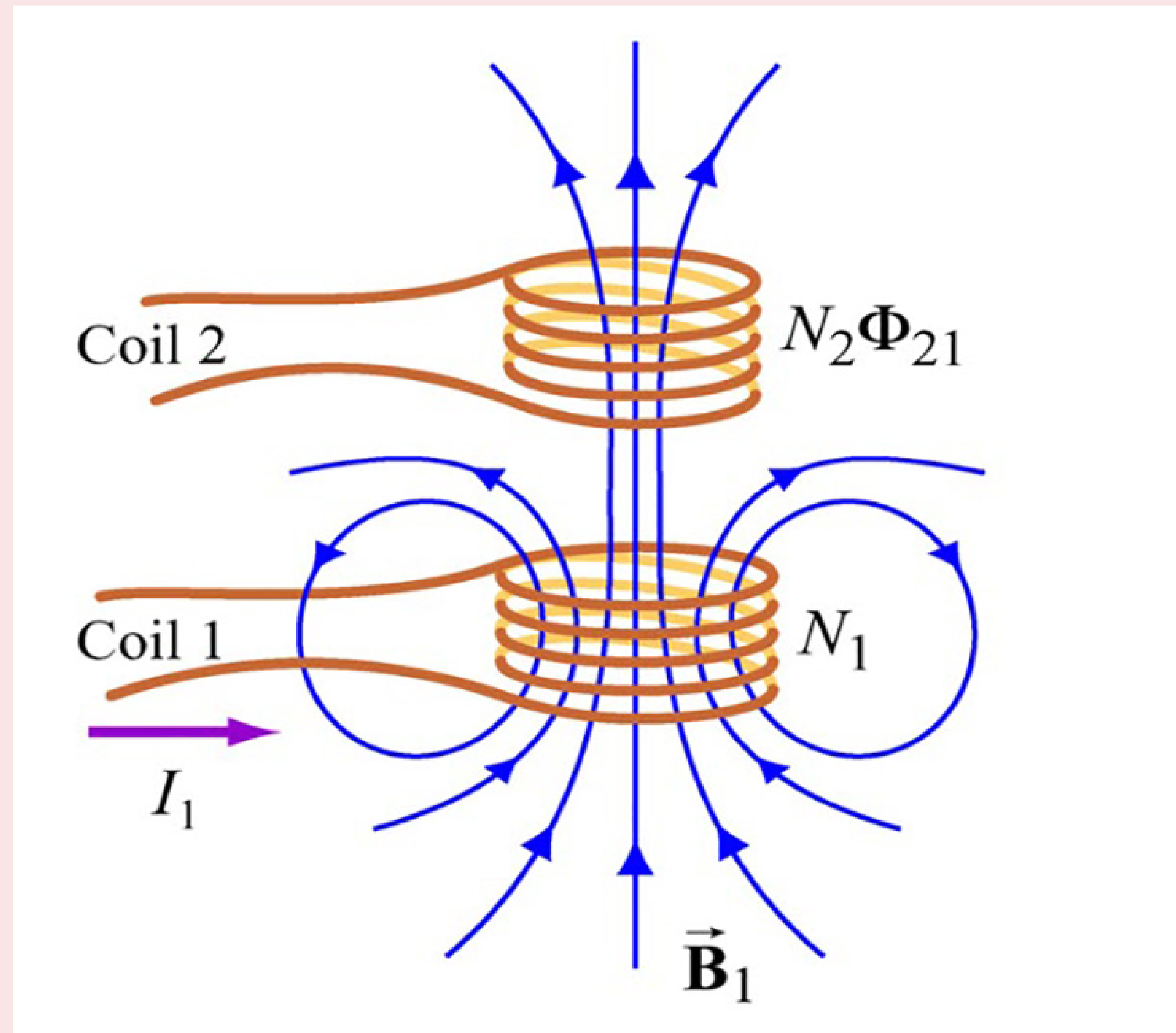
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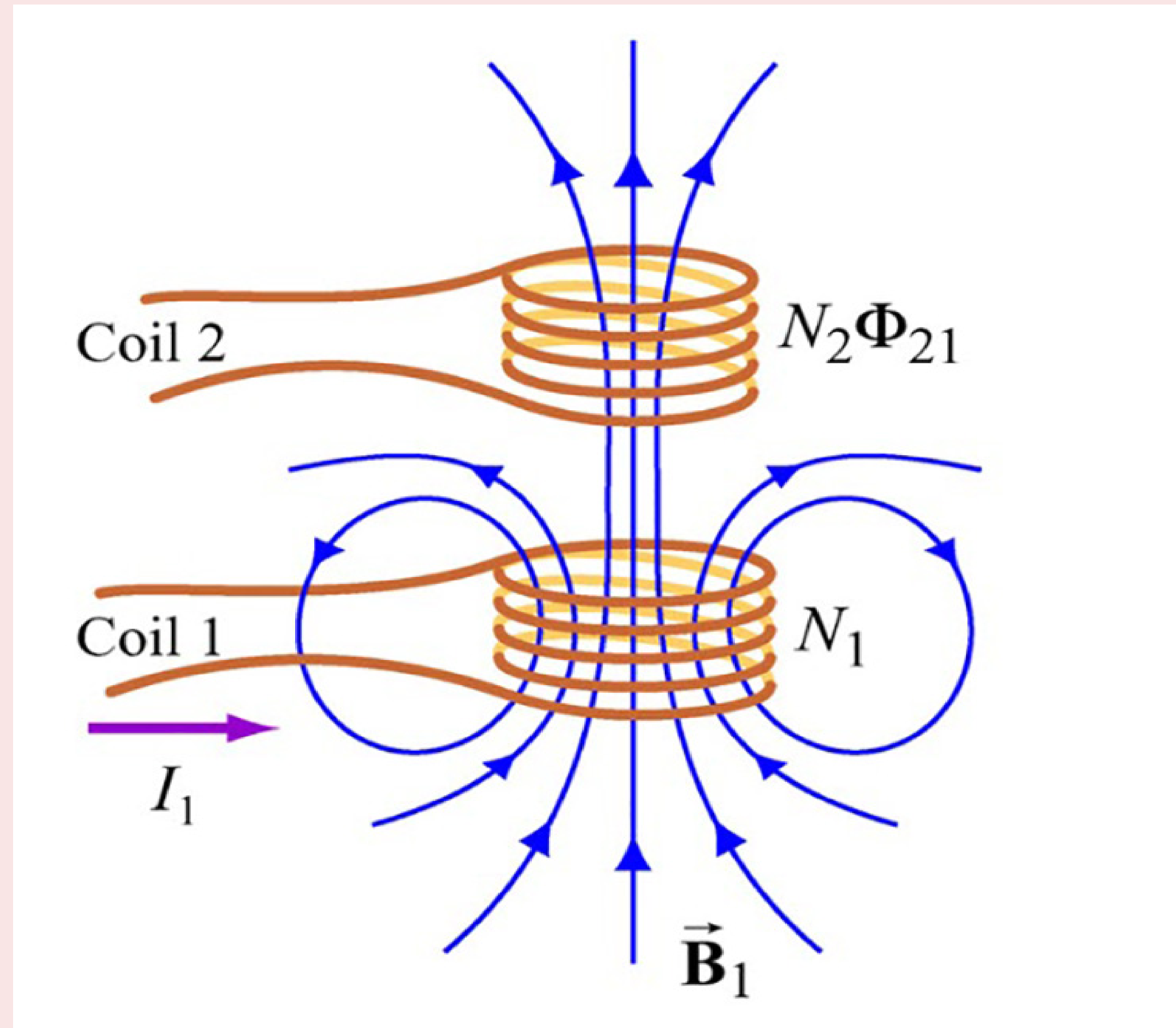
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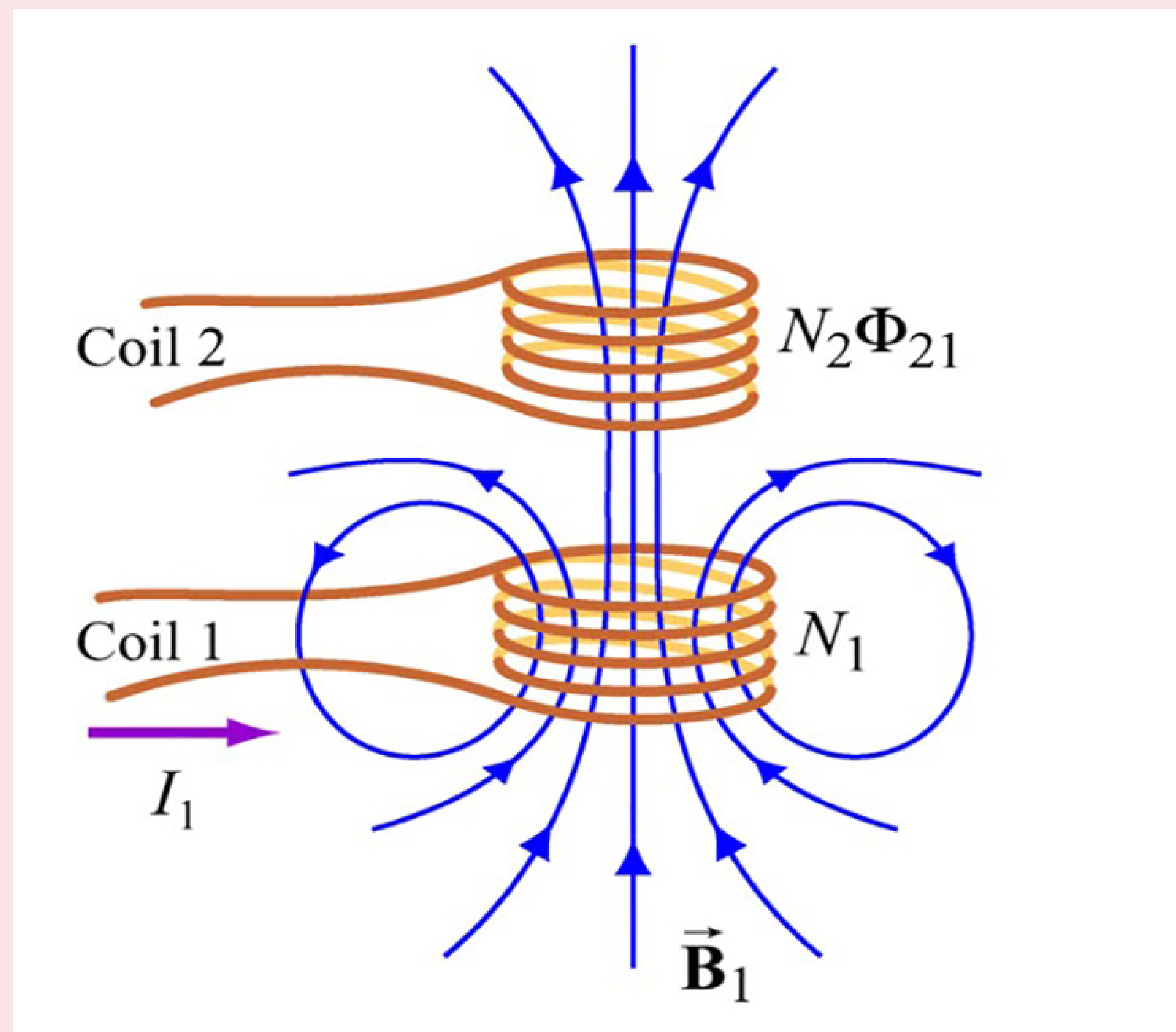
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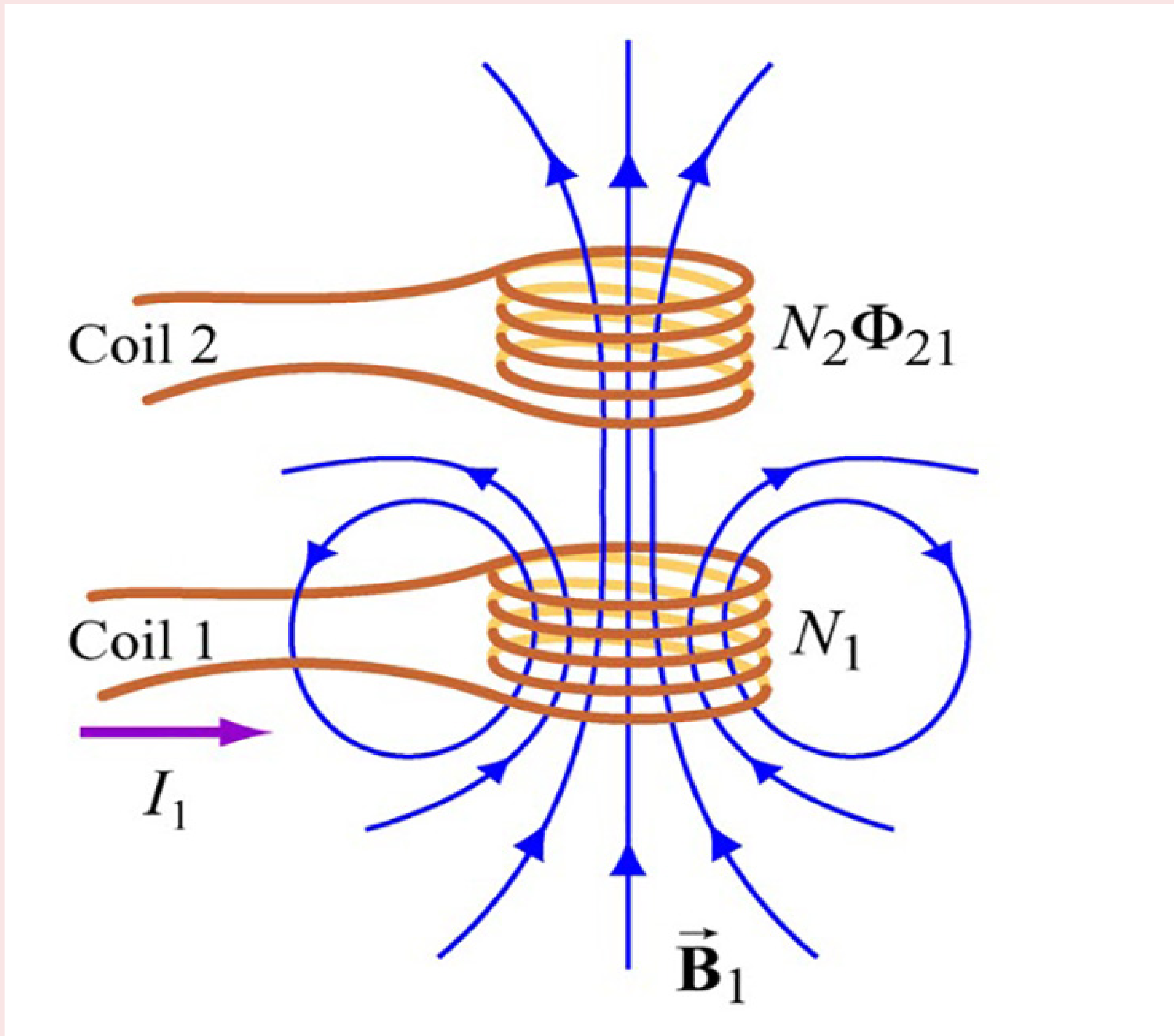
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- Here, M_{21} is called **mutual inductance of second coil 2 w.r.t coil 1**.

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- The emf developed in either coil is found by combining Faraday's law and the definition of mutual inductance. Since $N_2\phi_{21}$ is the total flux through coil 2 due to I_1 , we obtain

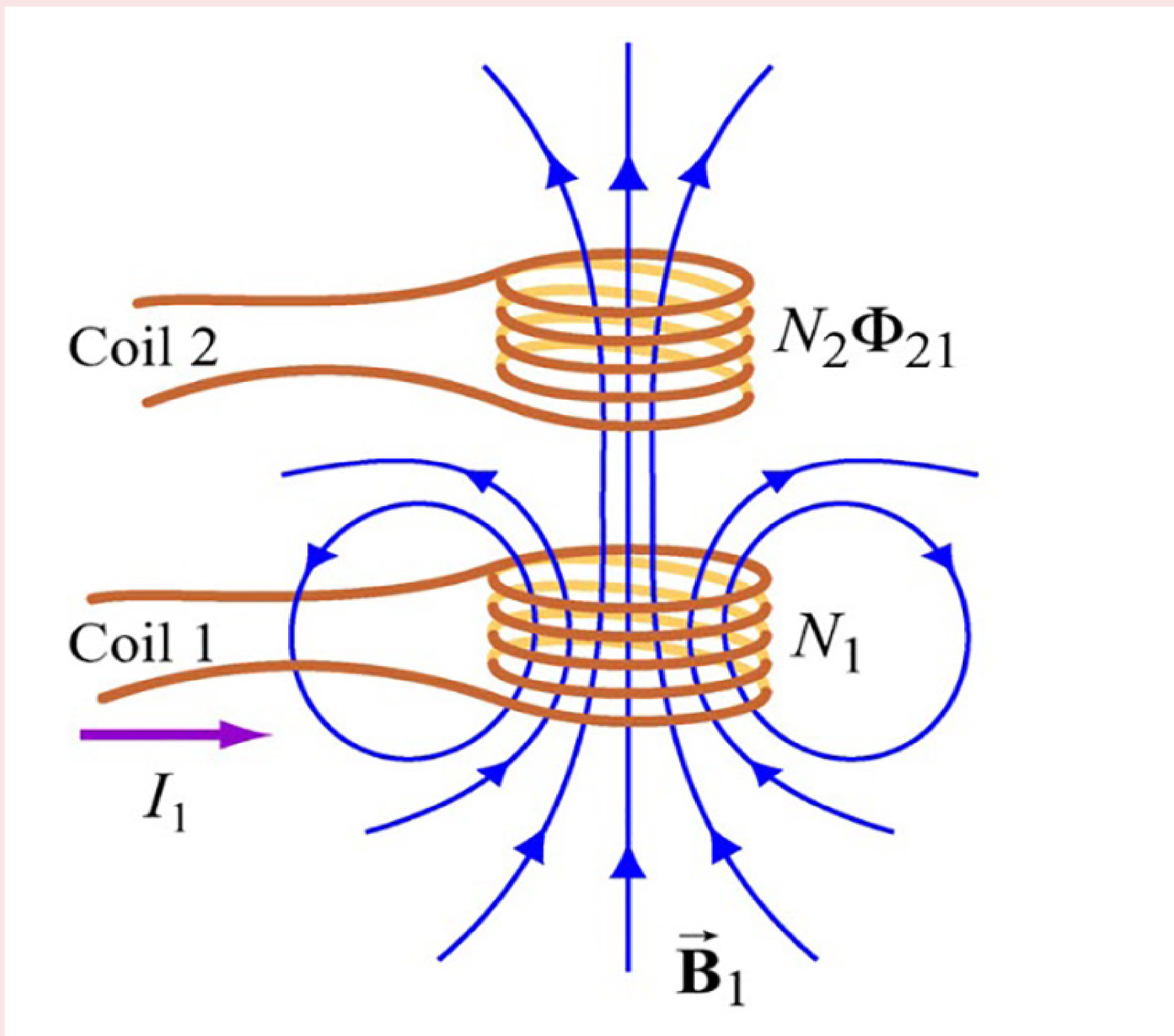
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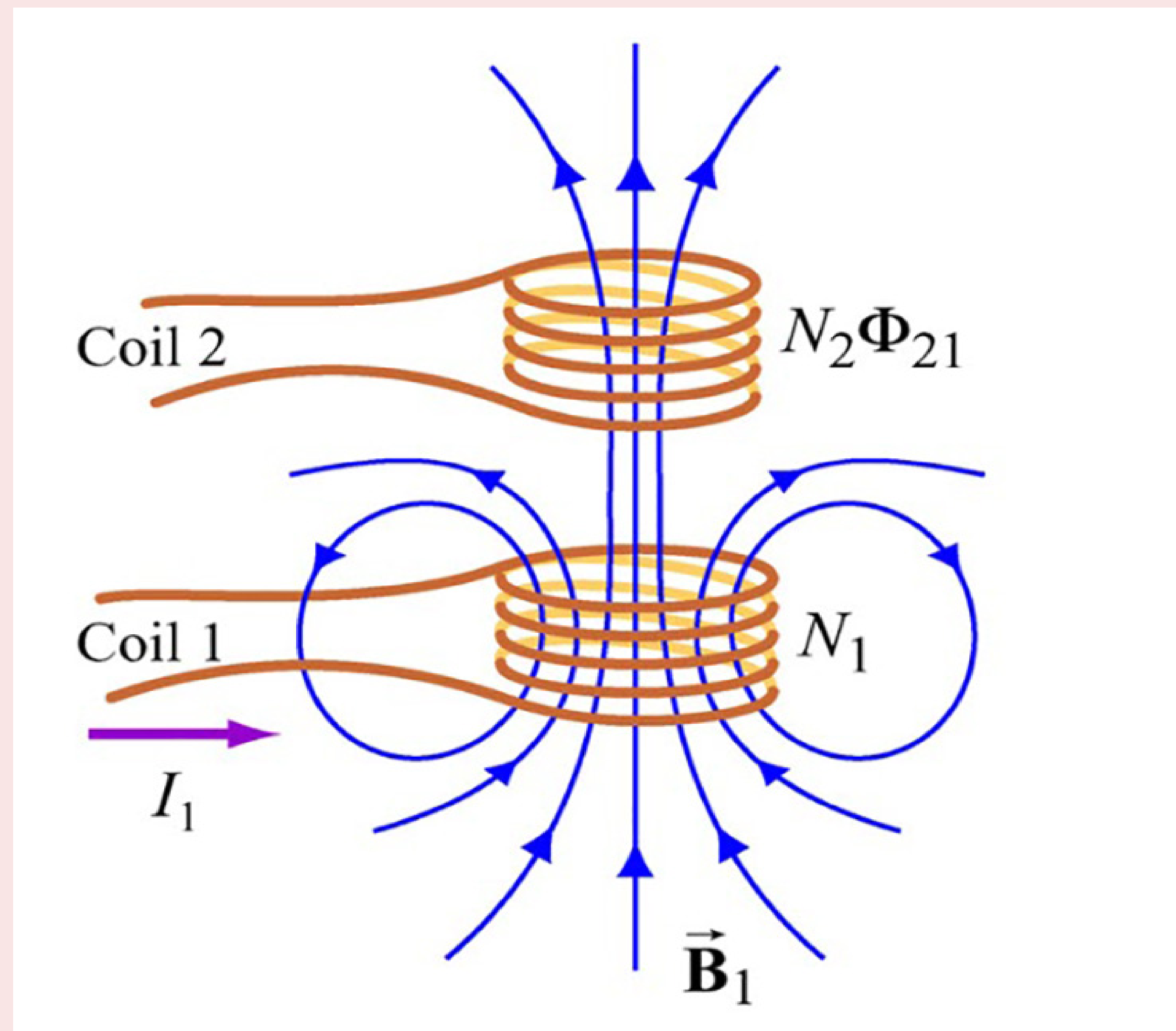
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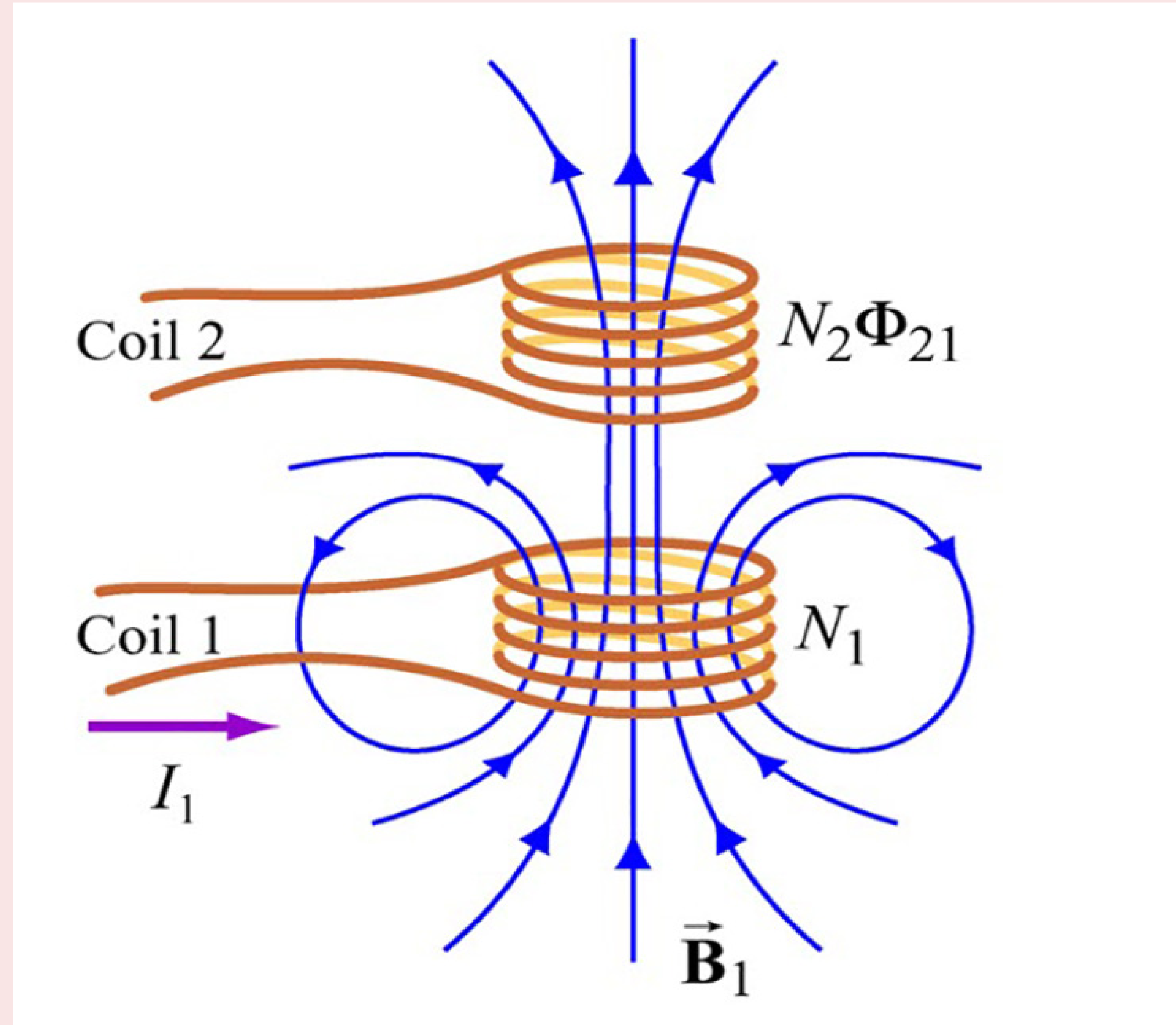
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- **Definition :** Mutual inductance is the **magnetic flux associated with the secondary coil** due to the **flow of unit current in the primary coil.**



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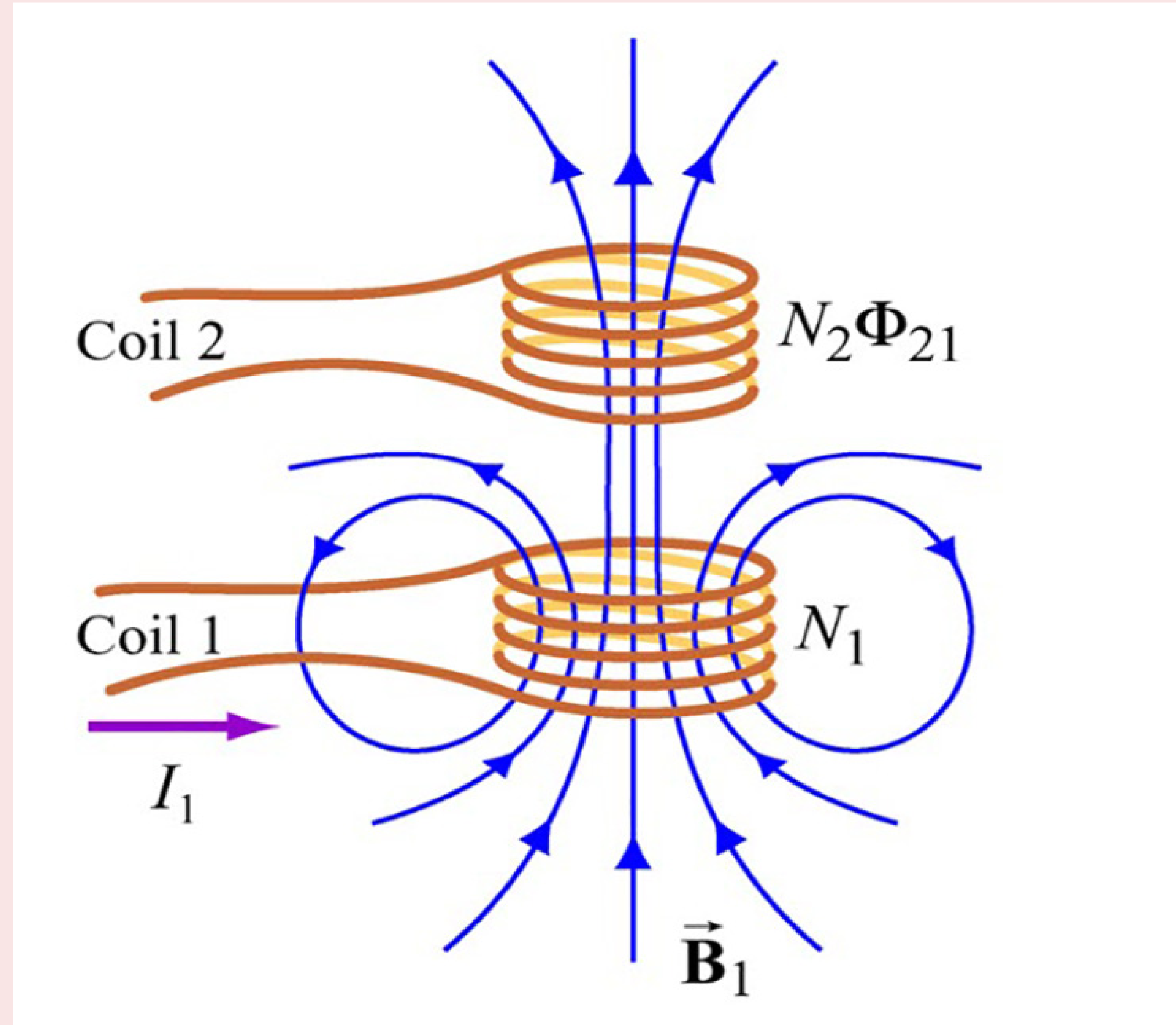


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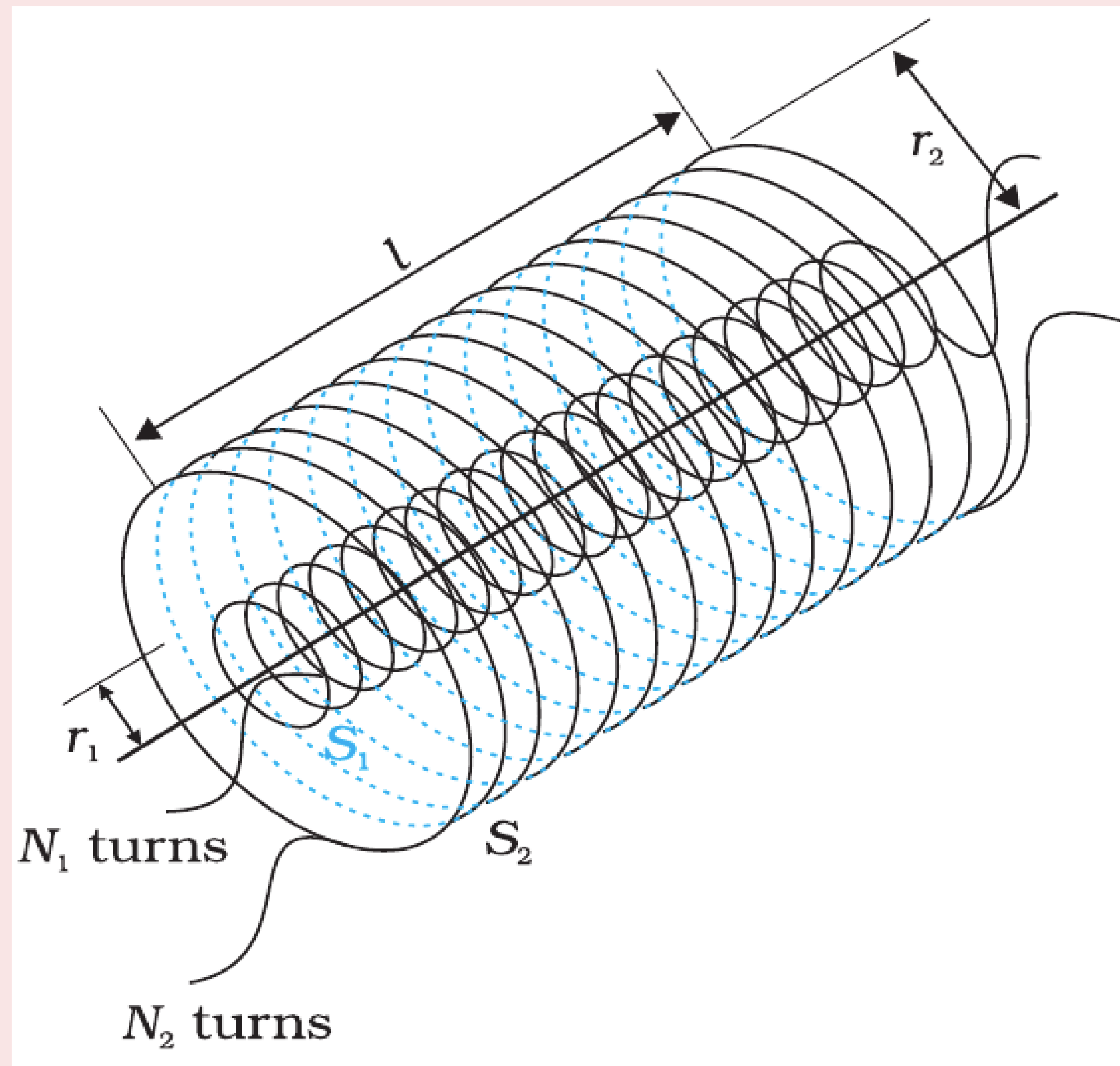
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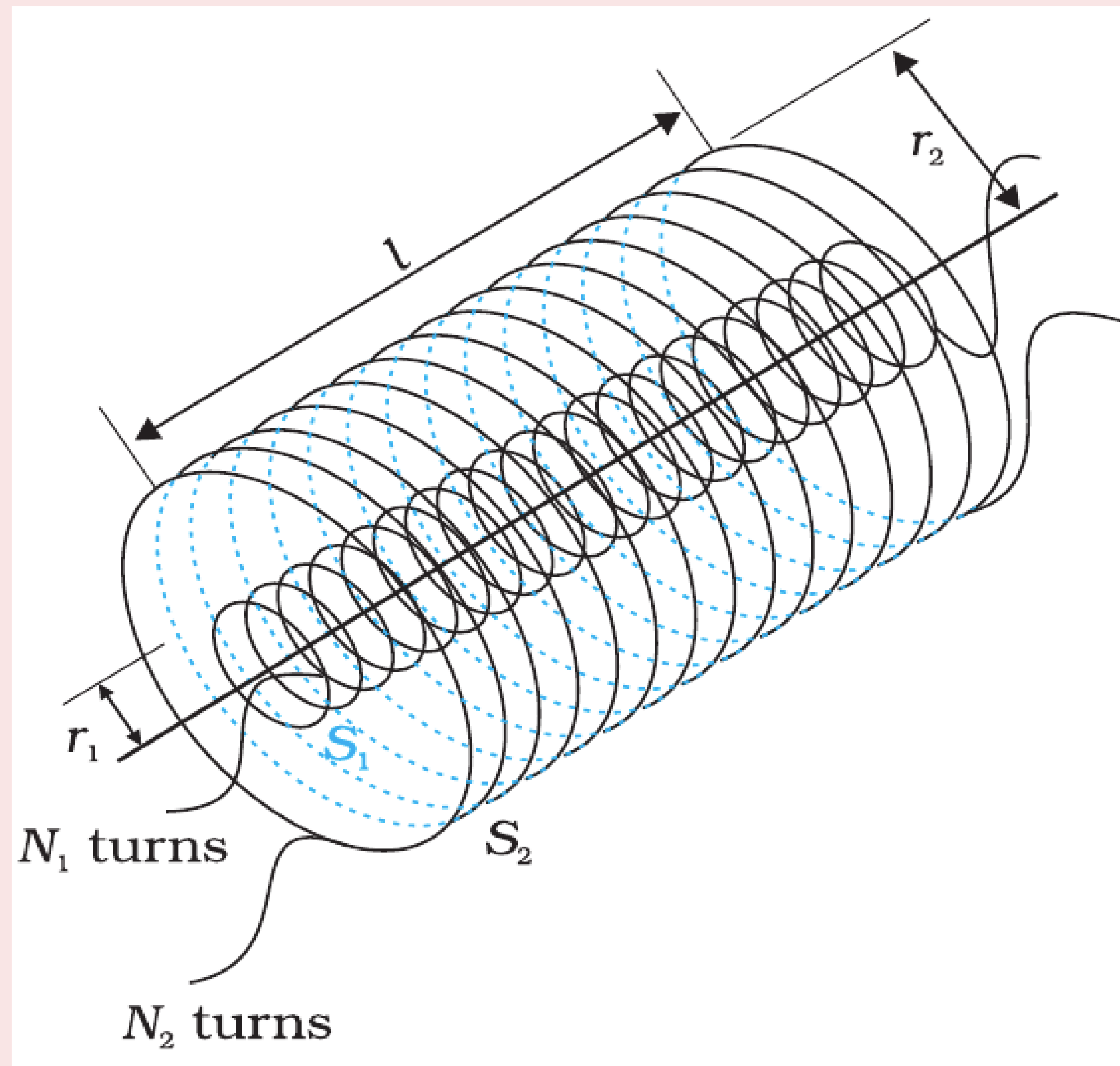
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 - The **mutual-inductance** of pairs coils or solenoids etc., **depends on their geometry, separation** as well as their relative **orientation**, and on the **permeability** of the medium, but is **independent of the currents** in the coils.

Mutual-Inductance Two Long Co-axial Solenoids



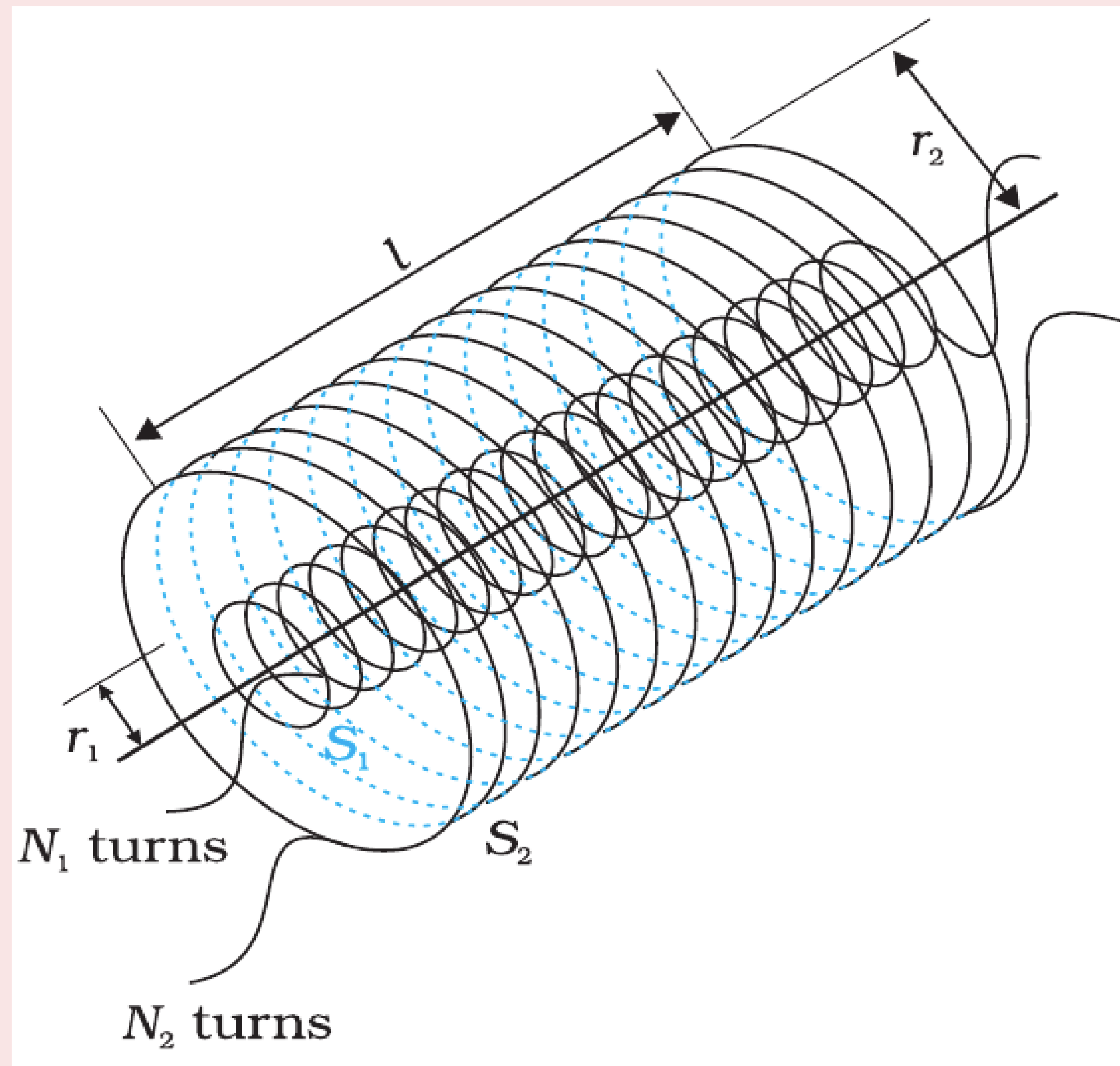
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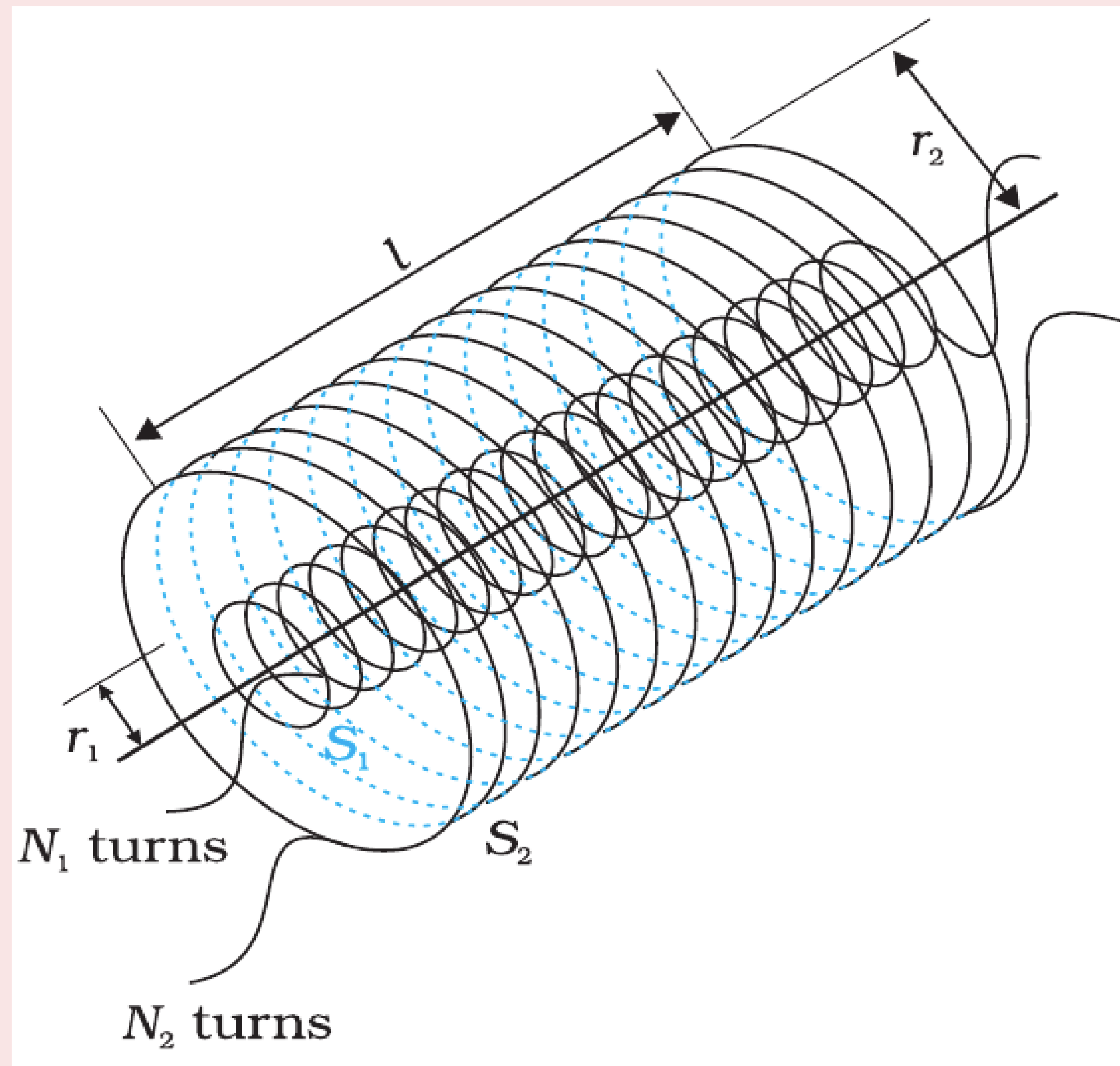
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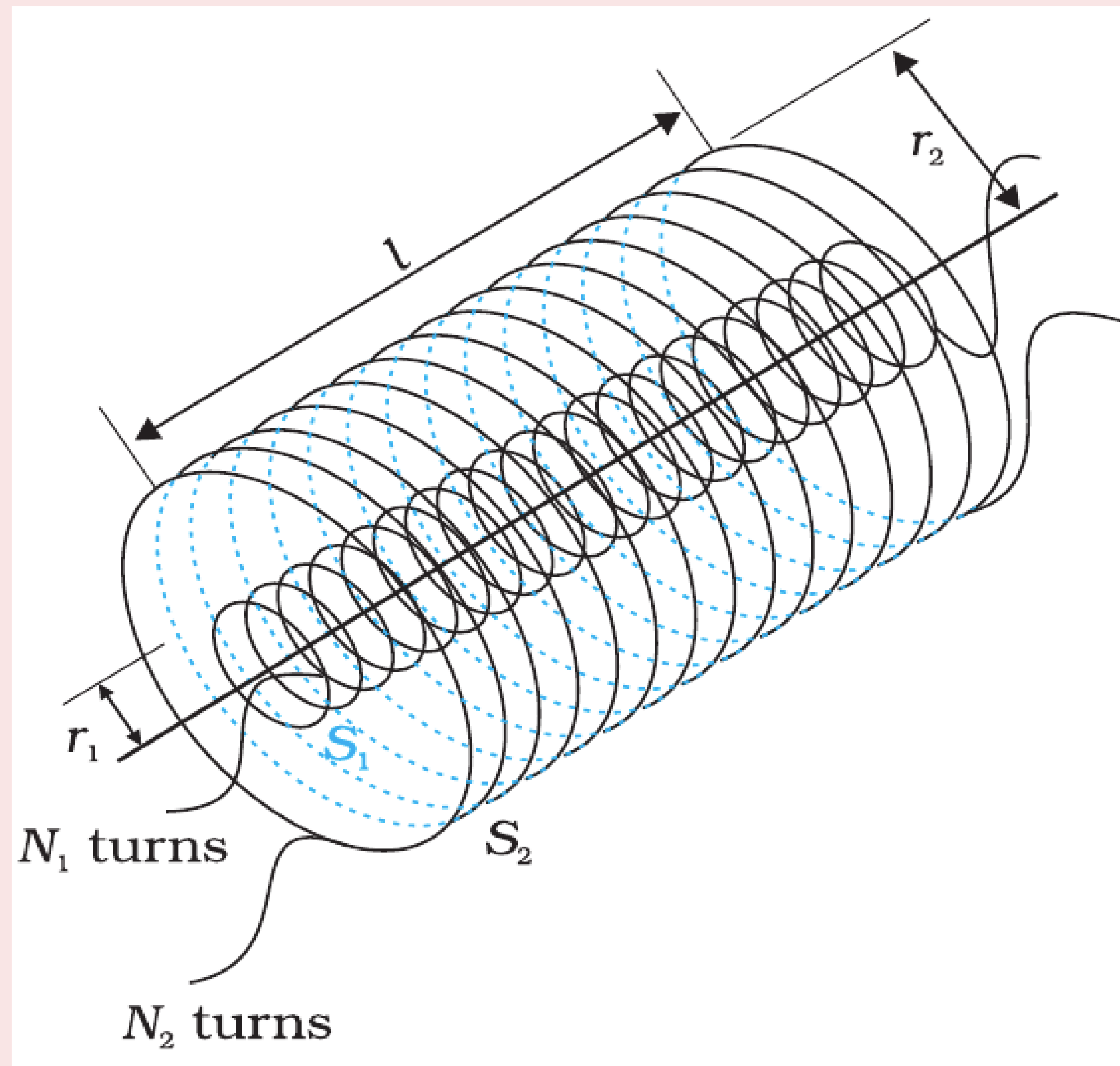
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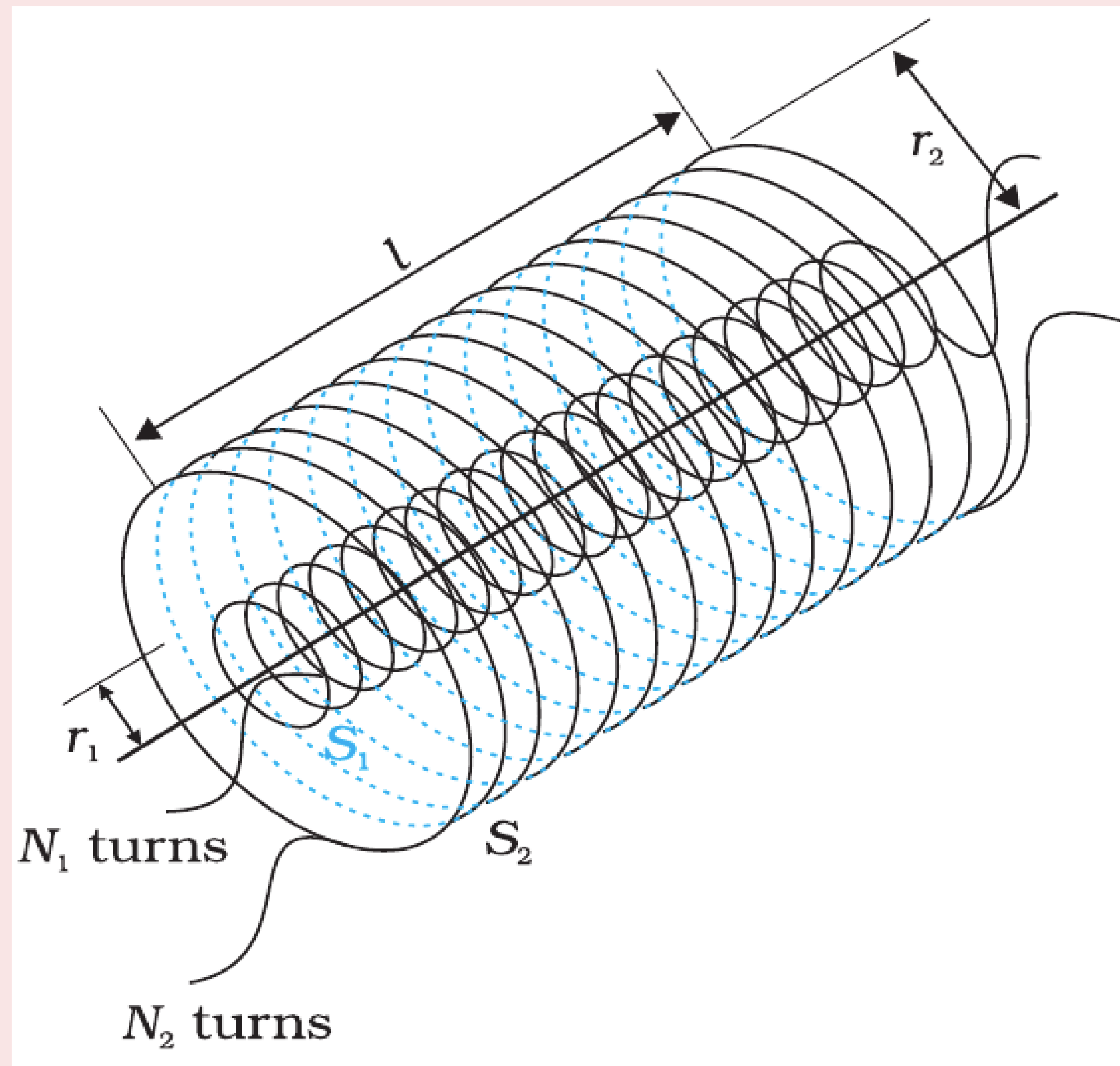
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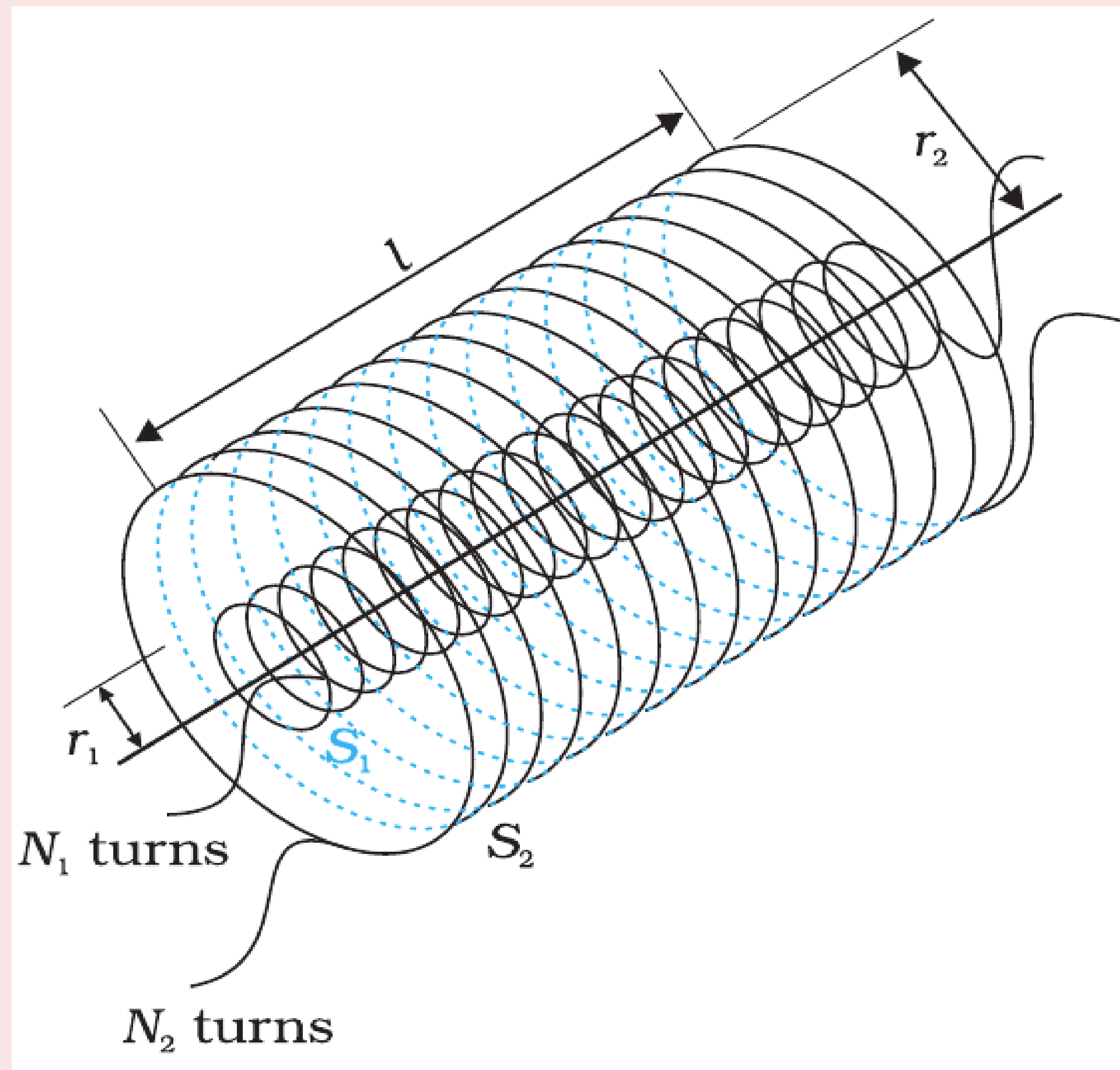
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- Total flux linkage with solenoid S_1 is $N_1 \phi_{12} = N_1 B_2 A_1$

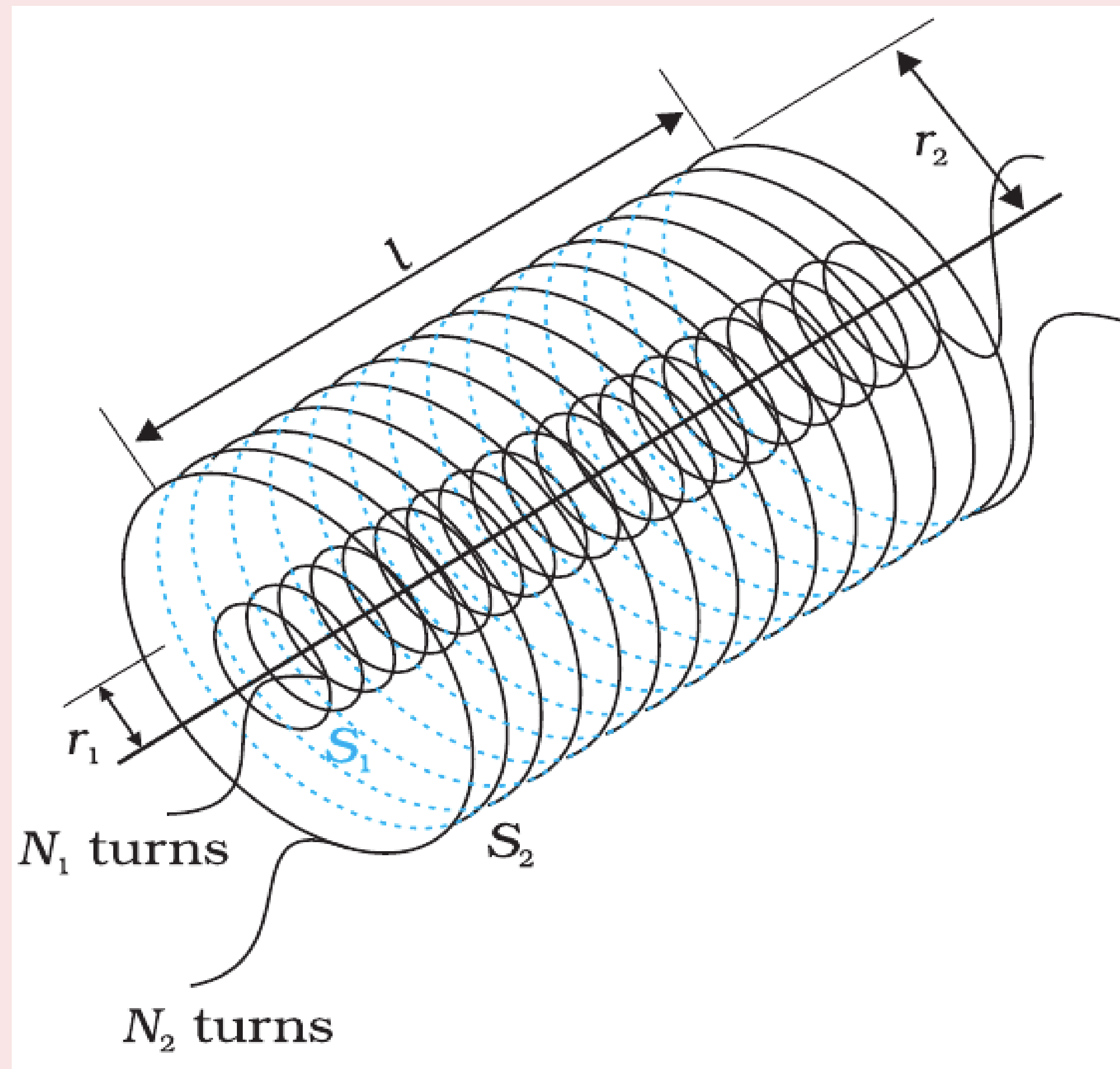
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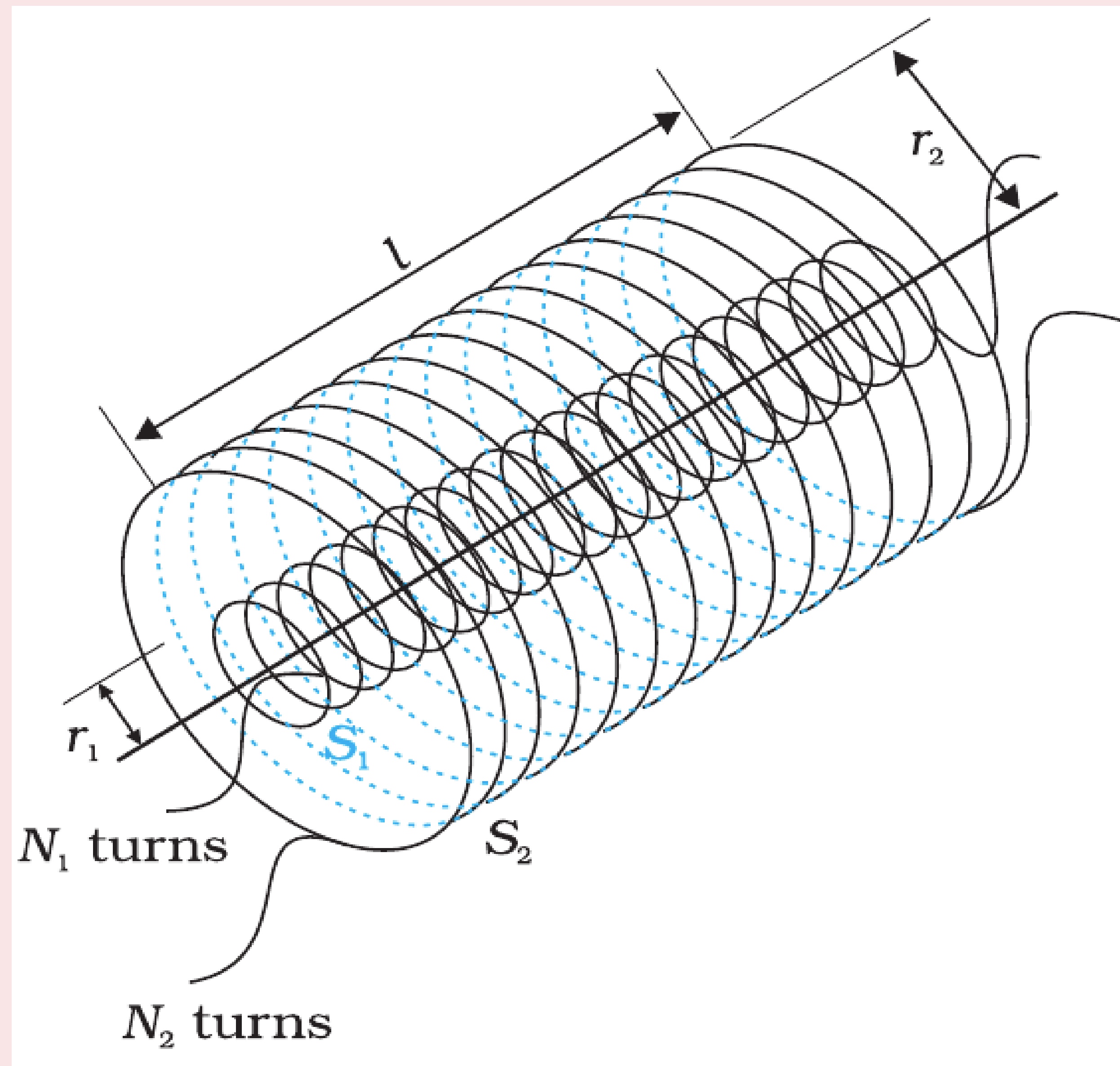
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$$N_1 \phi_{12} = N_1 B_2 A_1 = (n_1 l) (\mu_0 n_2 I_2) (\pi r_1^2)$$

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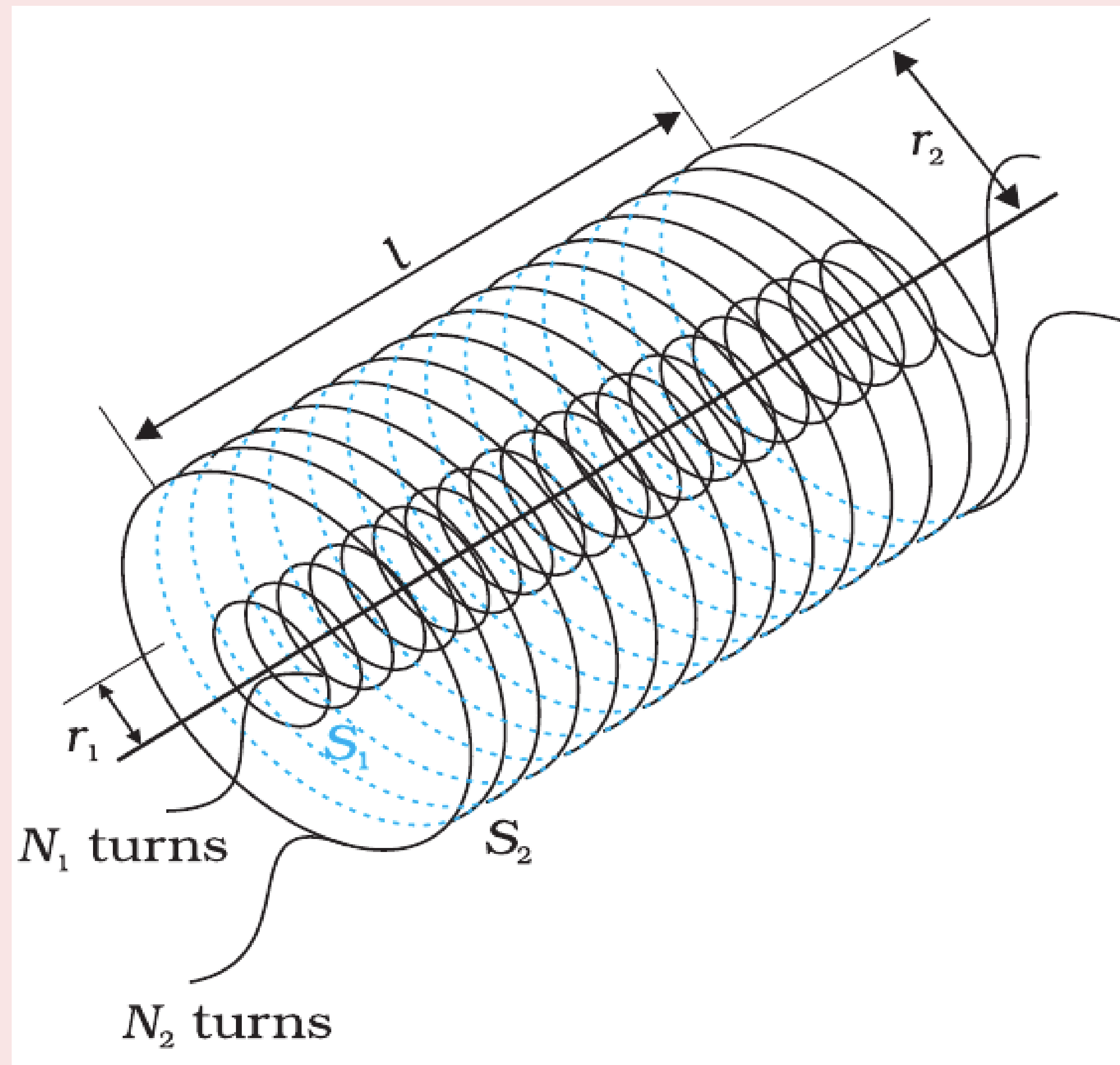


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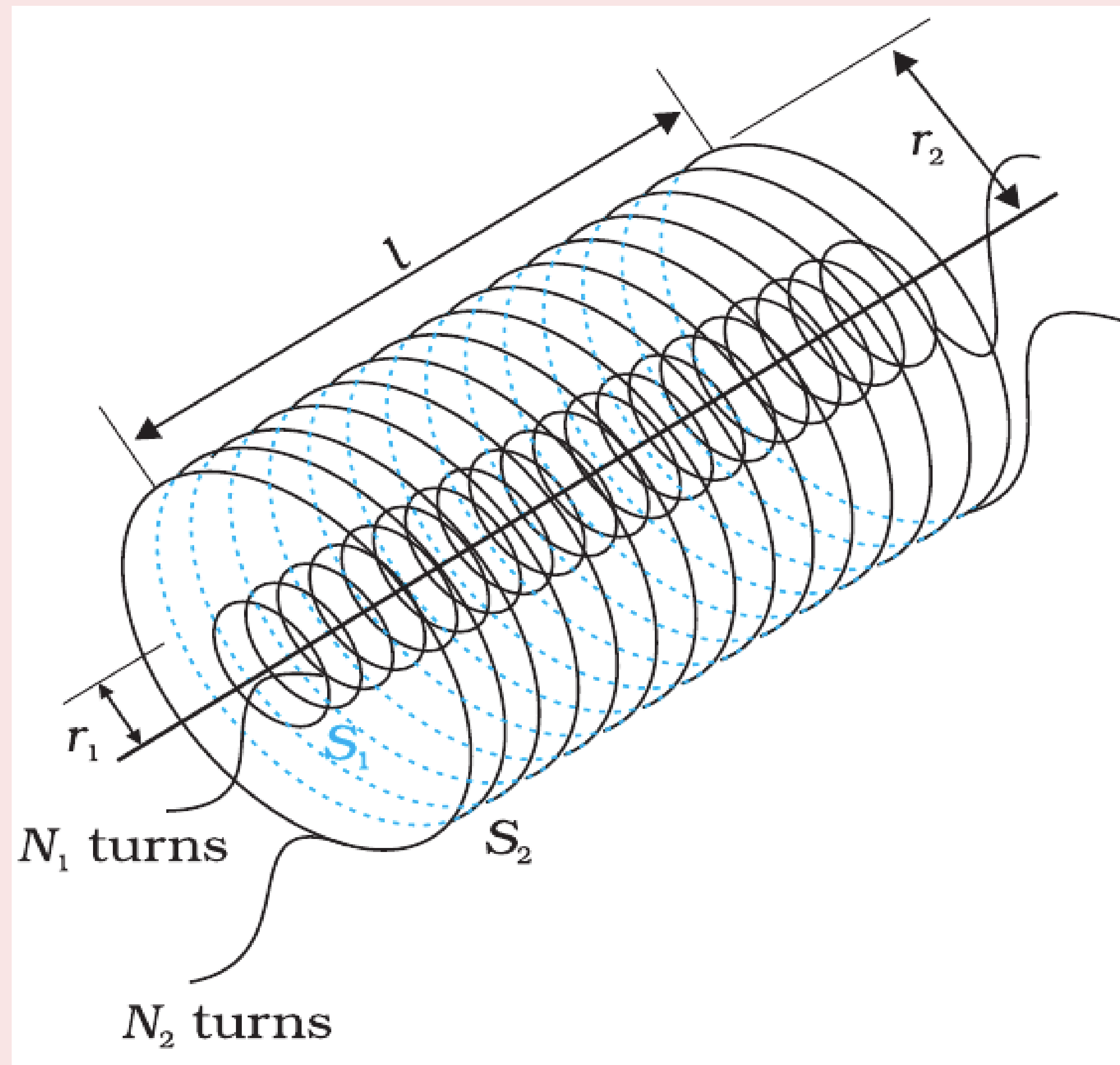
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- Comparing with $N_1 \phi_{12} = M_{12} I_2$ we get
- Mutual inductance of solenoid S_1 with respect to S_2

$$M_{12} = \mu_0 n_1 n_2 l \pi r_1^2 \quad \dots\dots(1)$$

Mutual-Inductance Two Long Co-axial Solenoids

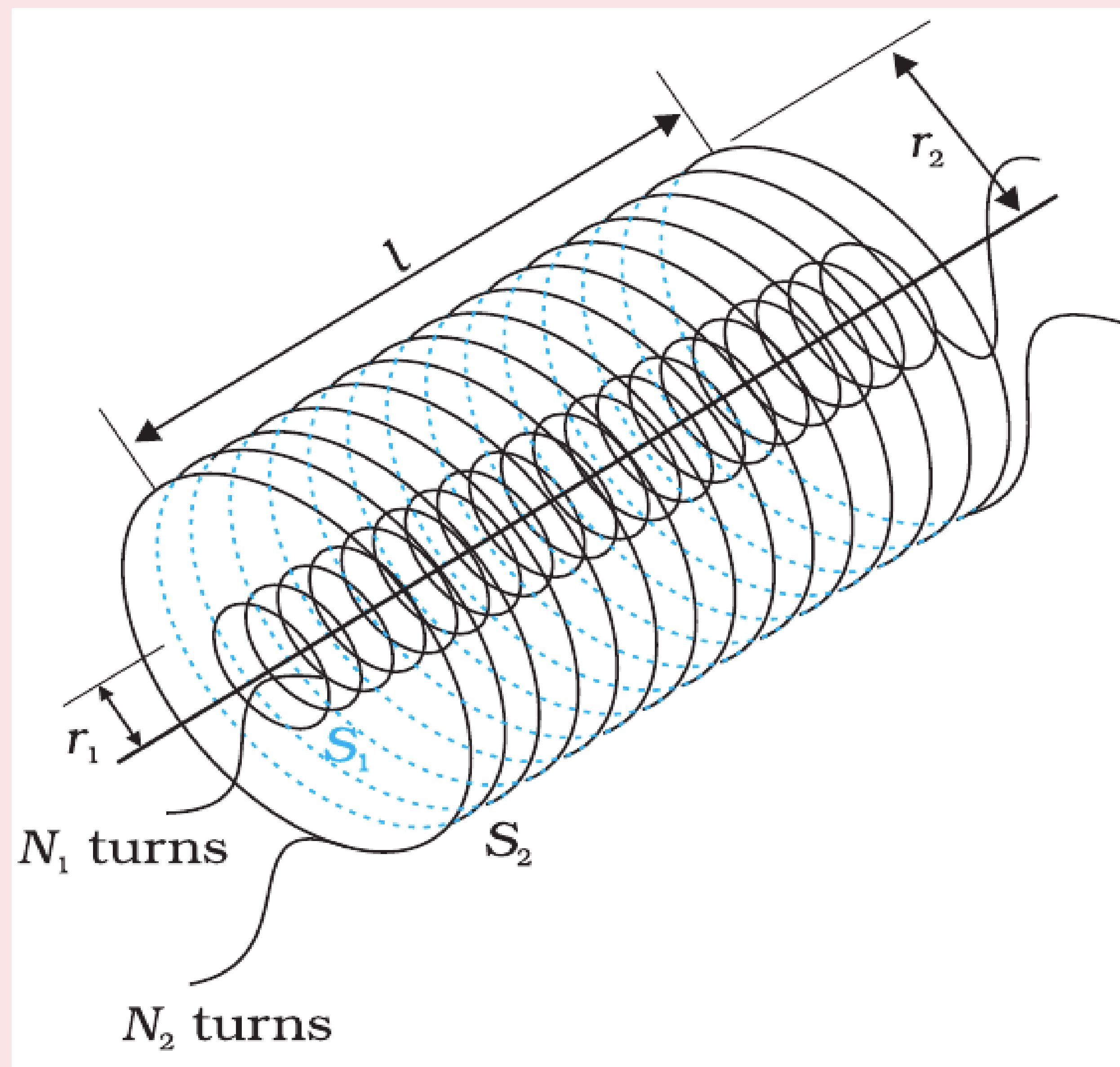


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- Mutual inductance of solenoid S_1 with respect to S_2

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Mutual-Inductance Two Long Co-axial Solenoids

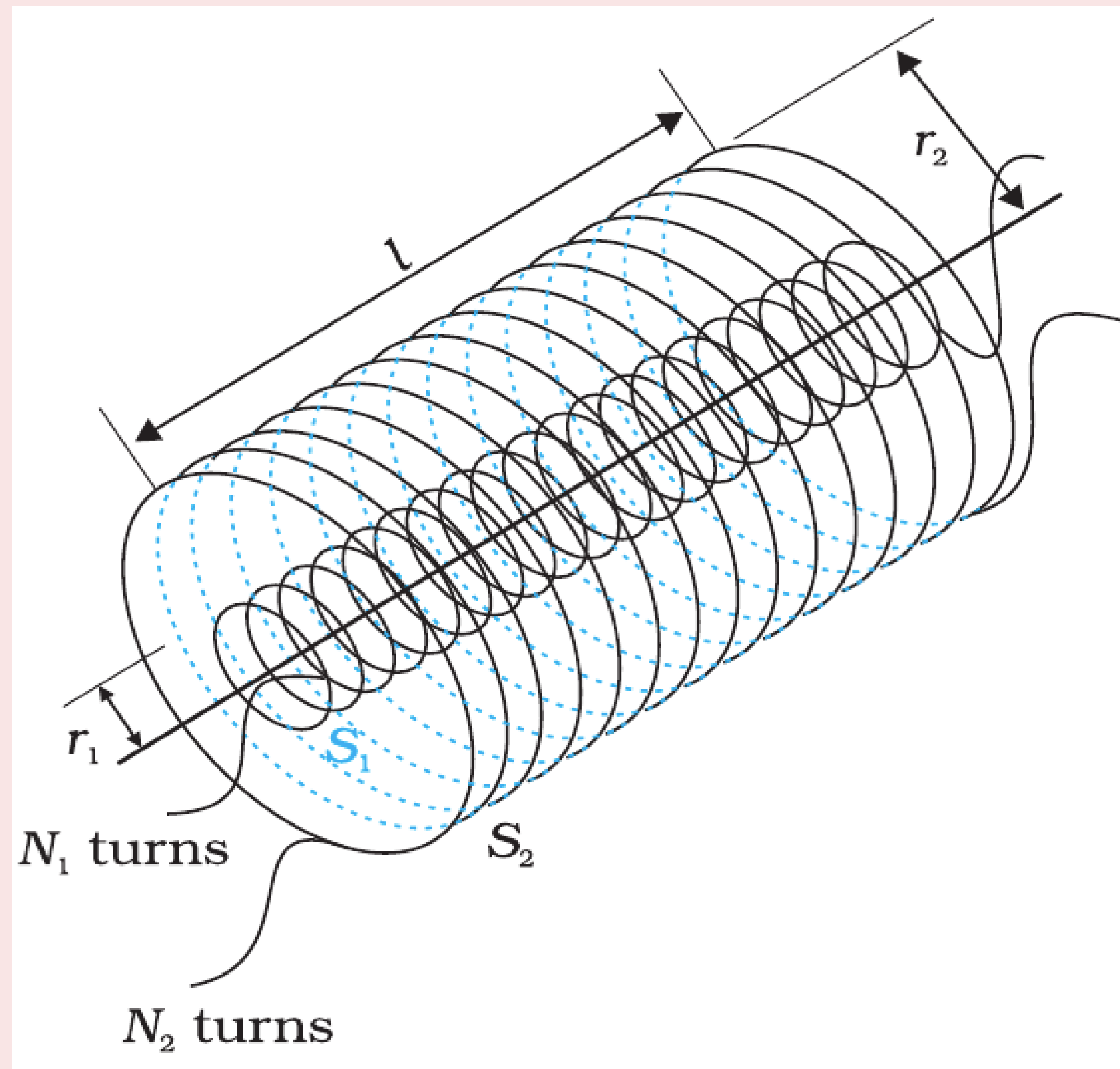


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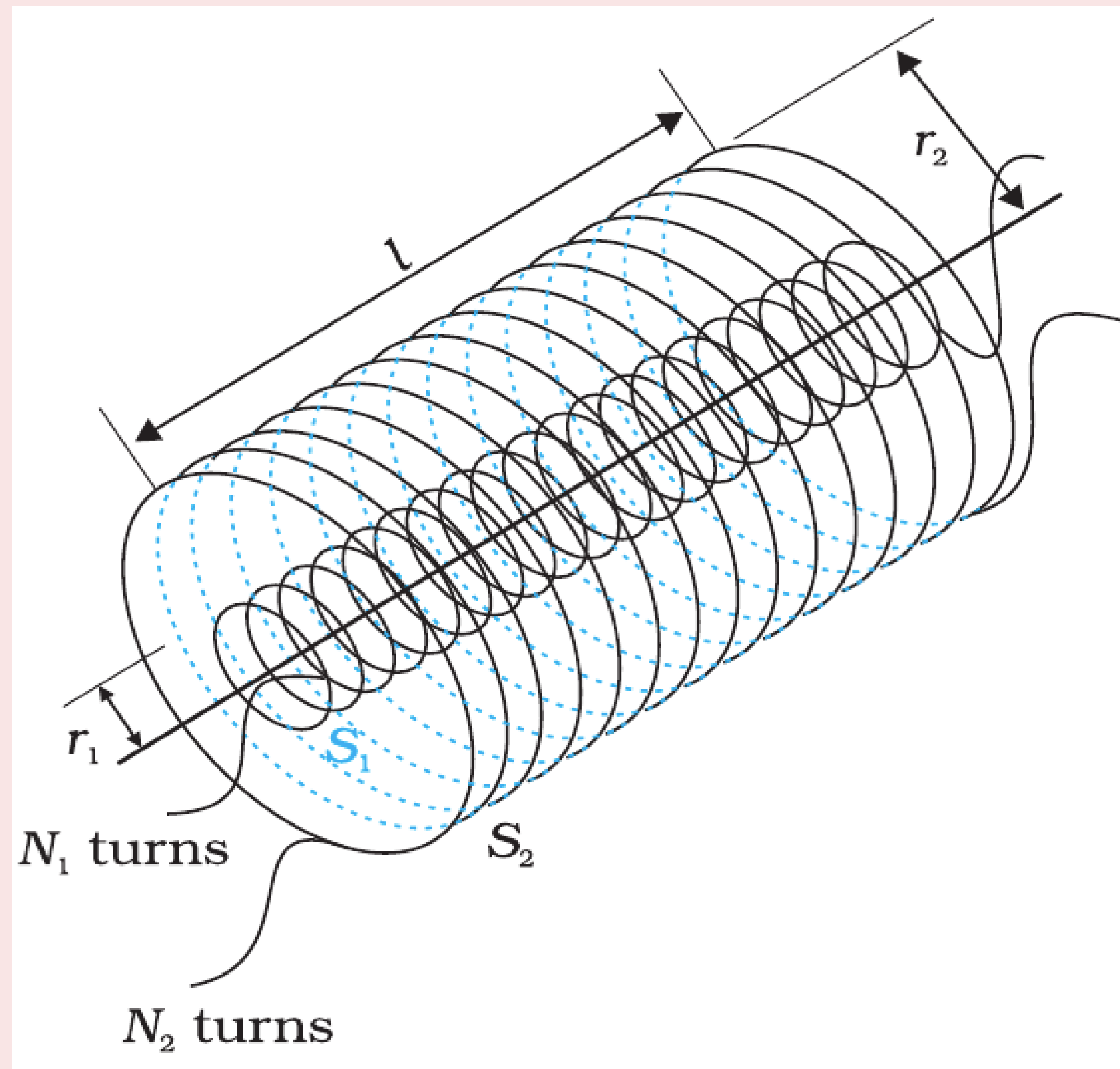


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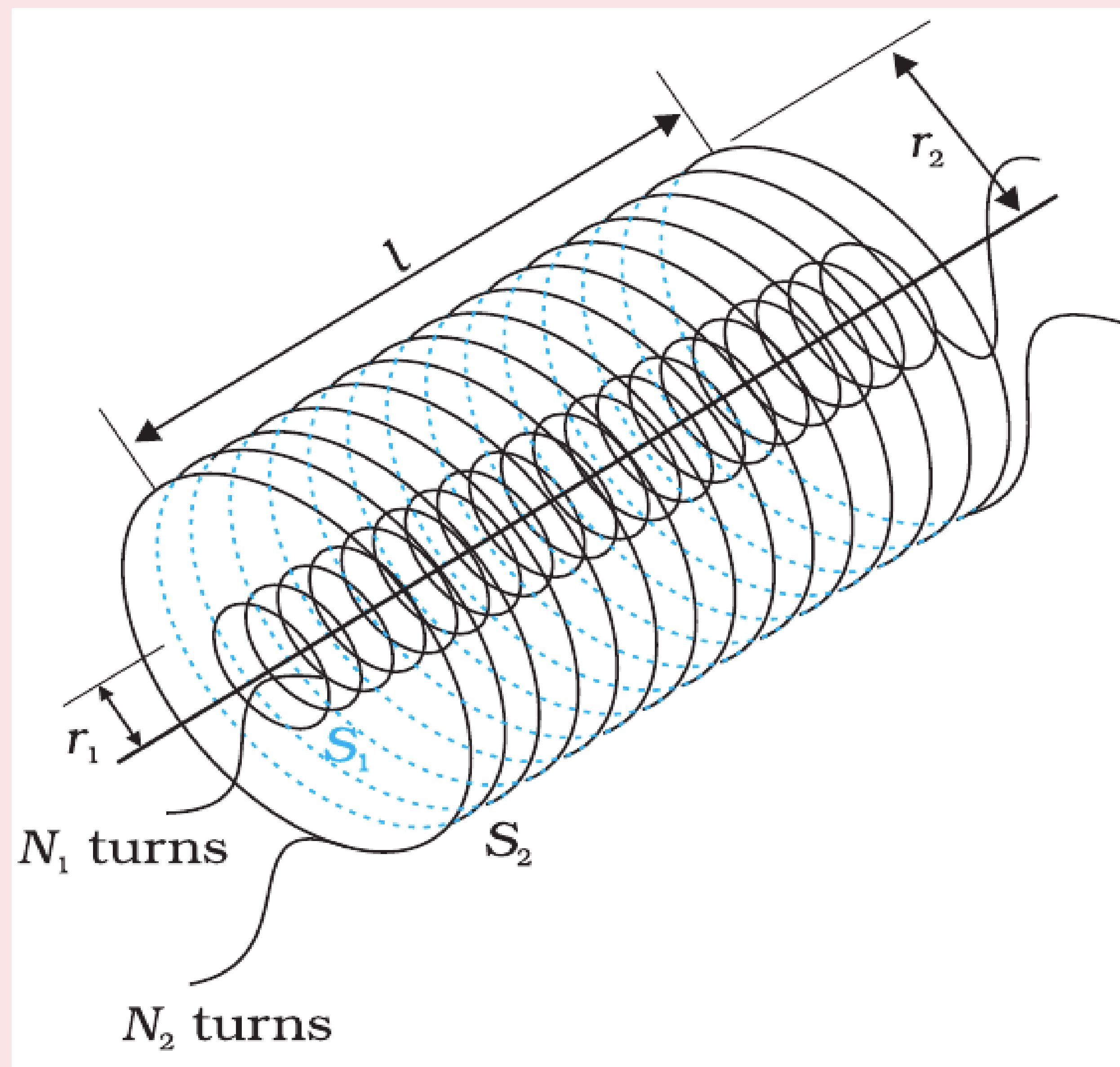


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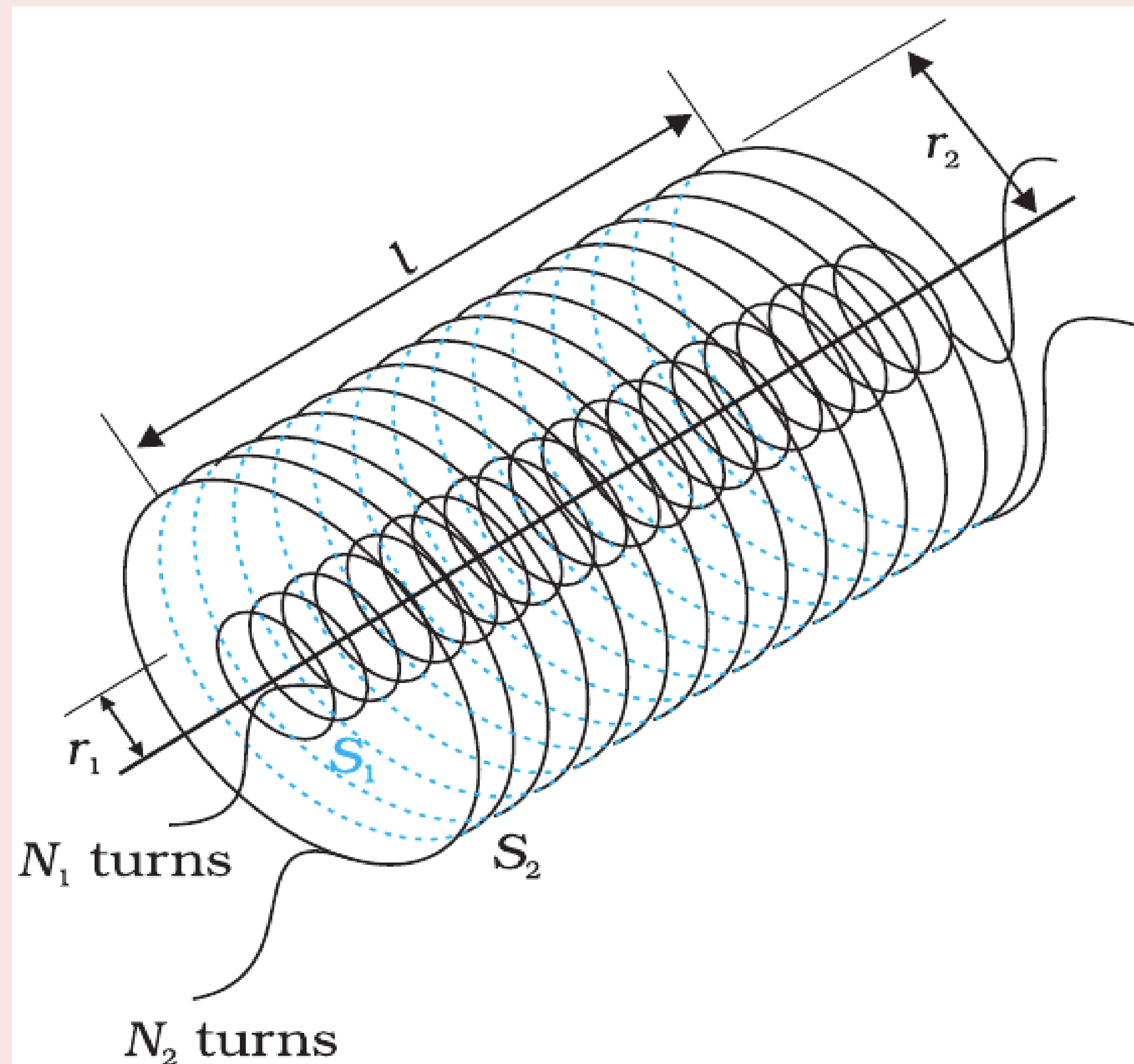
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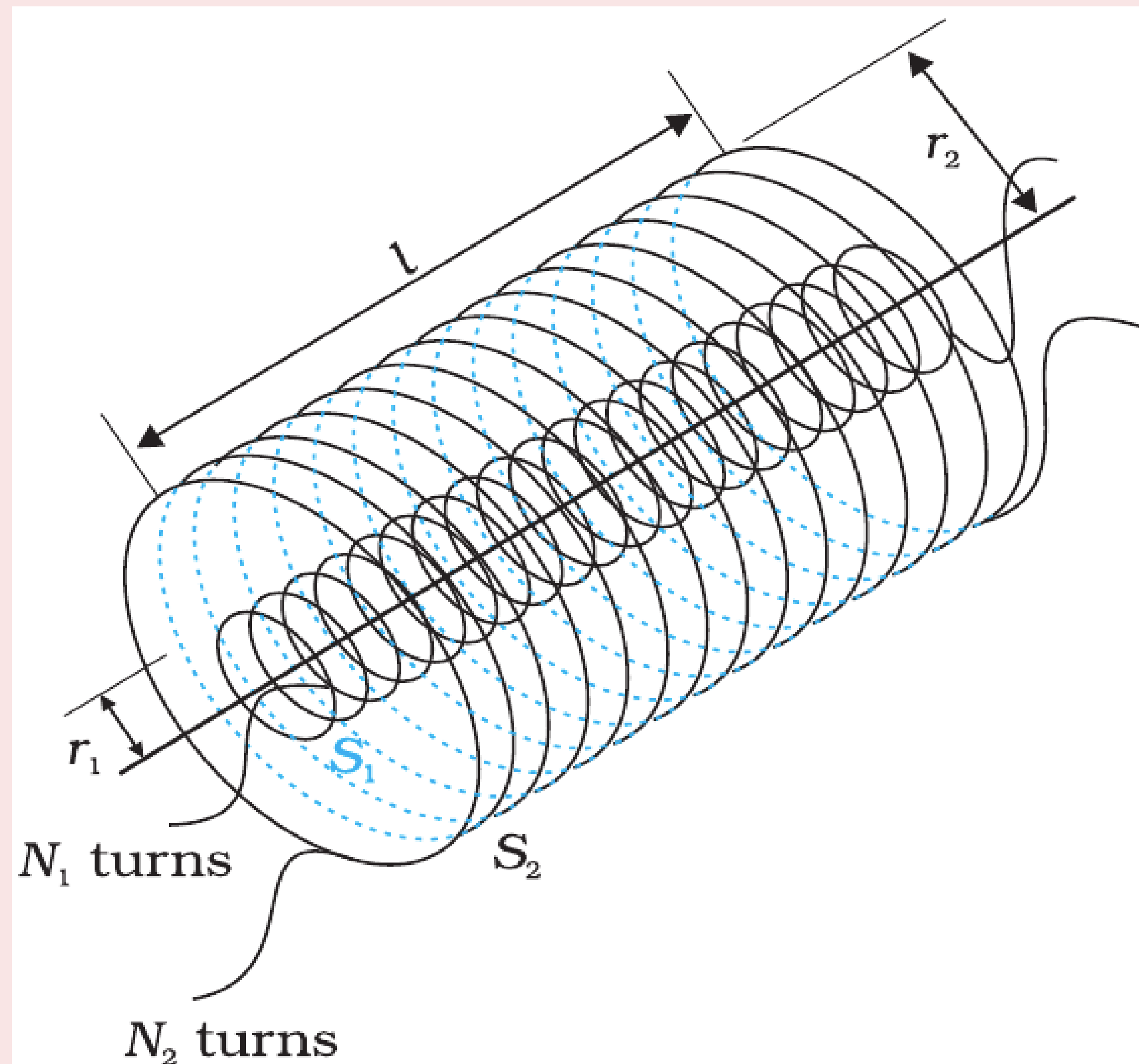
Mutual-Inductance Two Long Co-axial Solenoids

- Mutual inductance of solenoid S_2 with respect to S_1

$$M_{21} = \mu_0 n_1 n_2 l \pi r_1^2 \dots\dots(2)$$



Mutual-Inductance Two Long Co-axial Solenoids



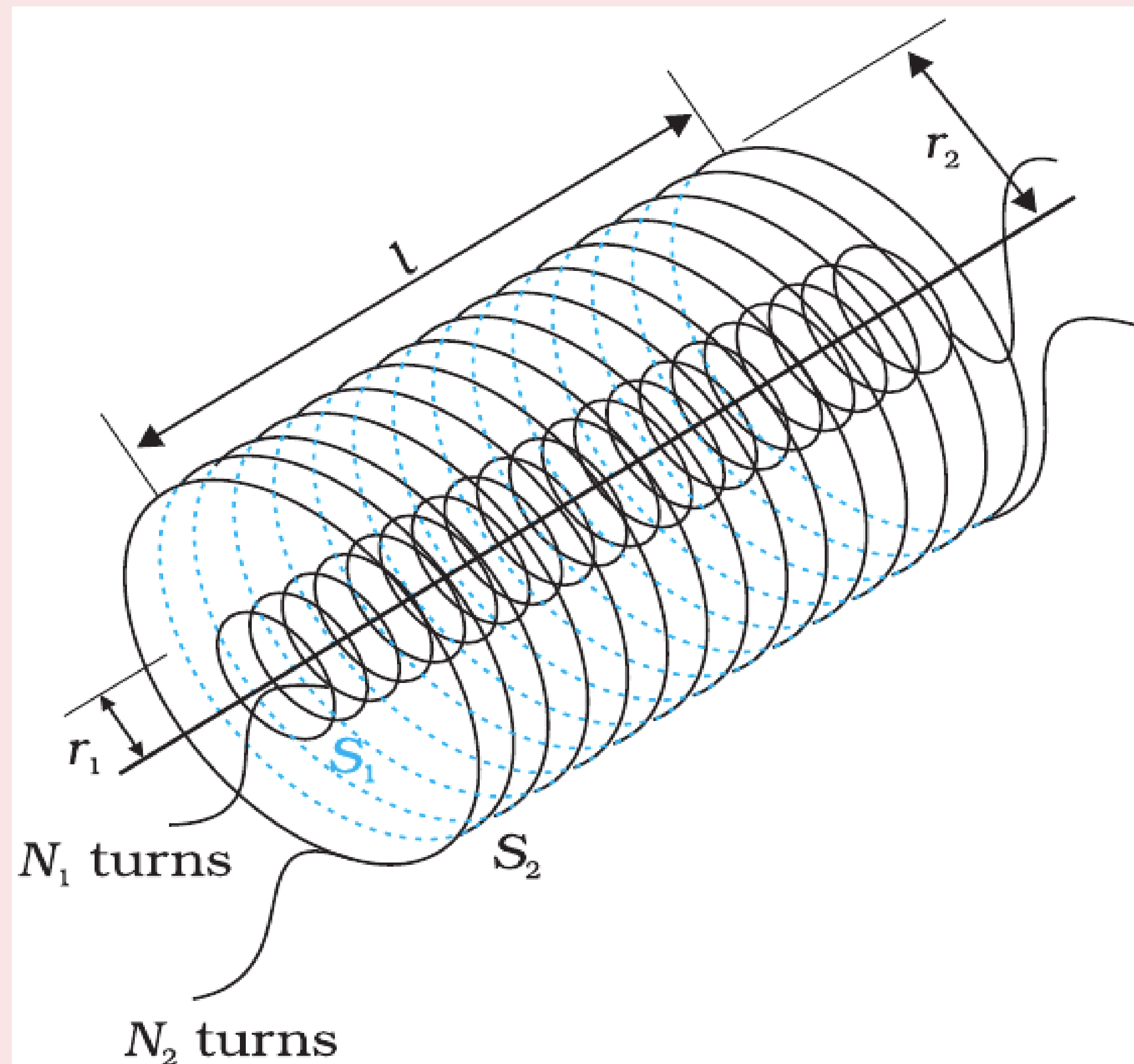
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- From equation (1) and (2) We get,

$$M_{12} = M_{21} = M$$

Mutual-Inductance Two Long Co-axial Solenoids



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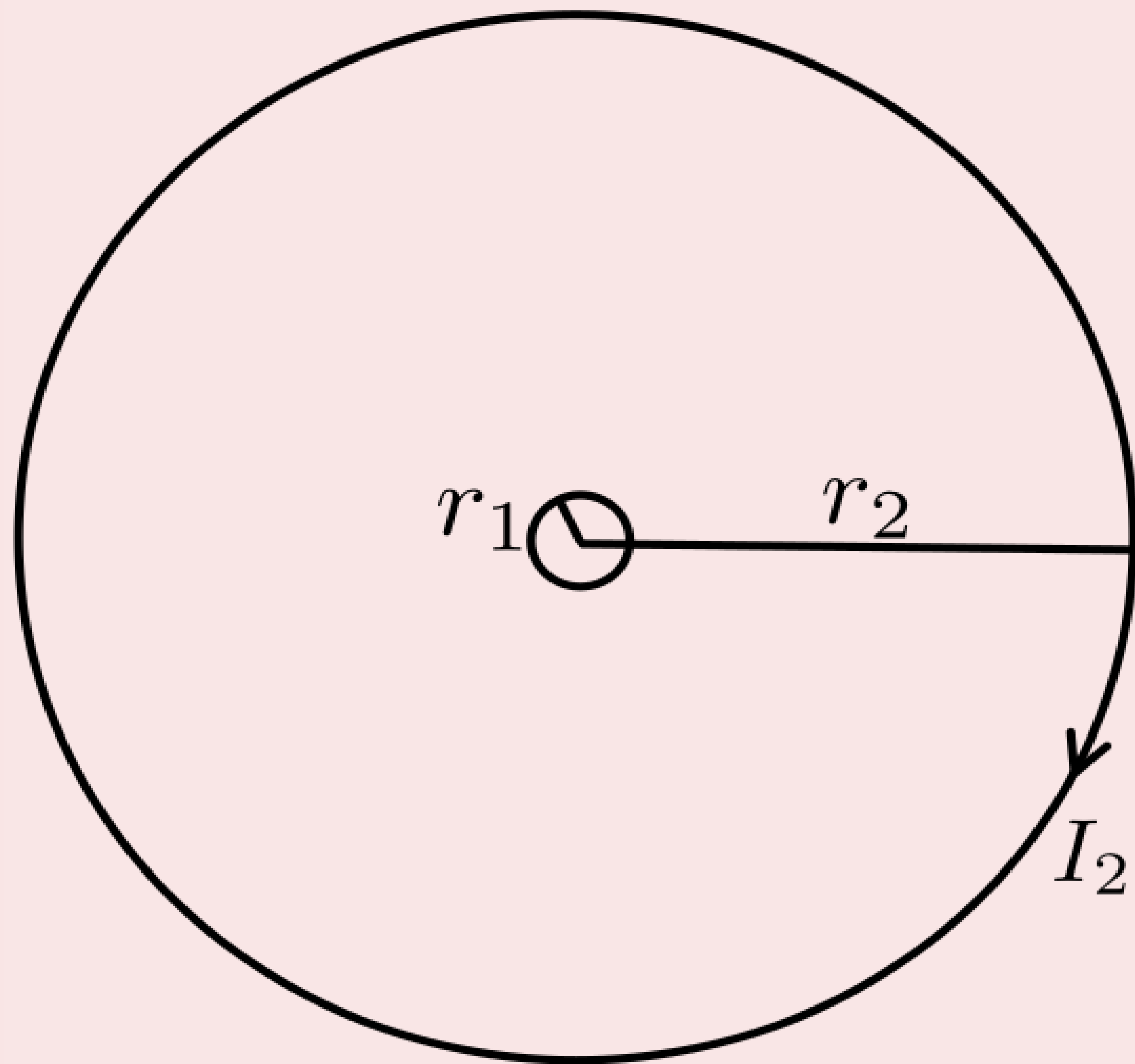
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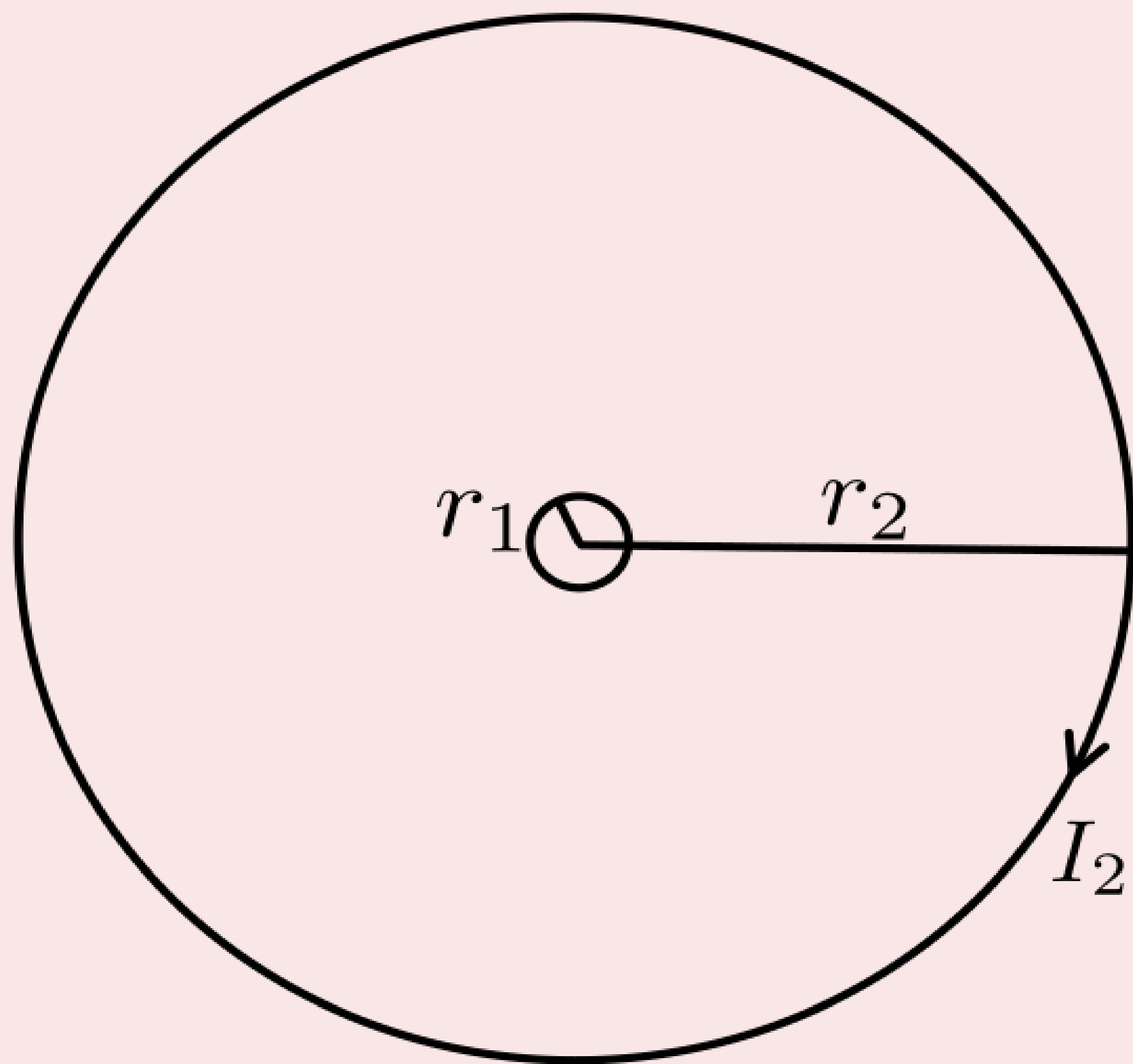
- If a medium of relative permeability μ_r had been present, the mutual inductance would be

Mutual-Inductance Two concentric circular coils ($r_1 \ll r_2$)

- Let a **current I_2** flow through the **outer circular coil** .

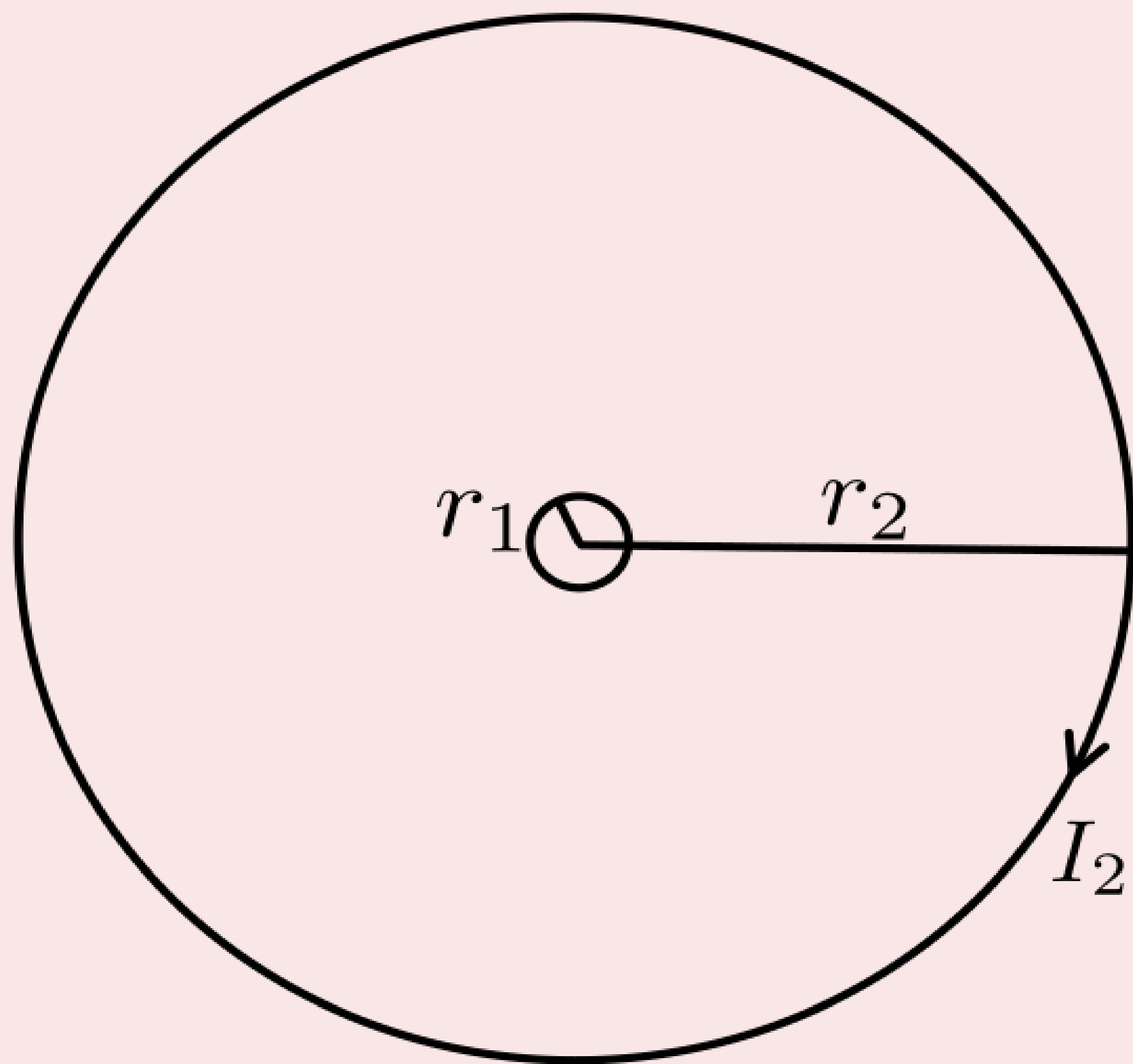


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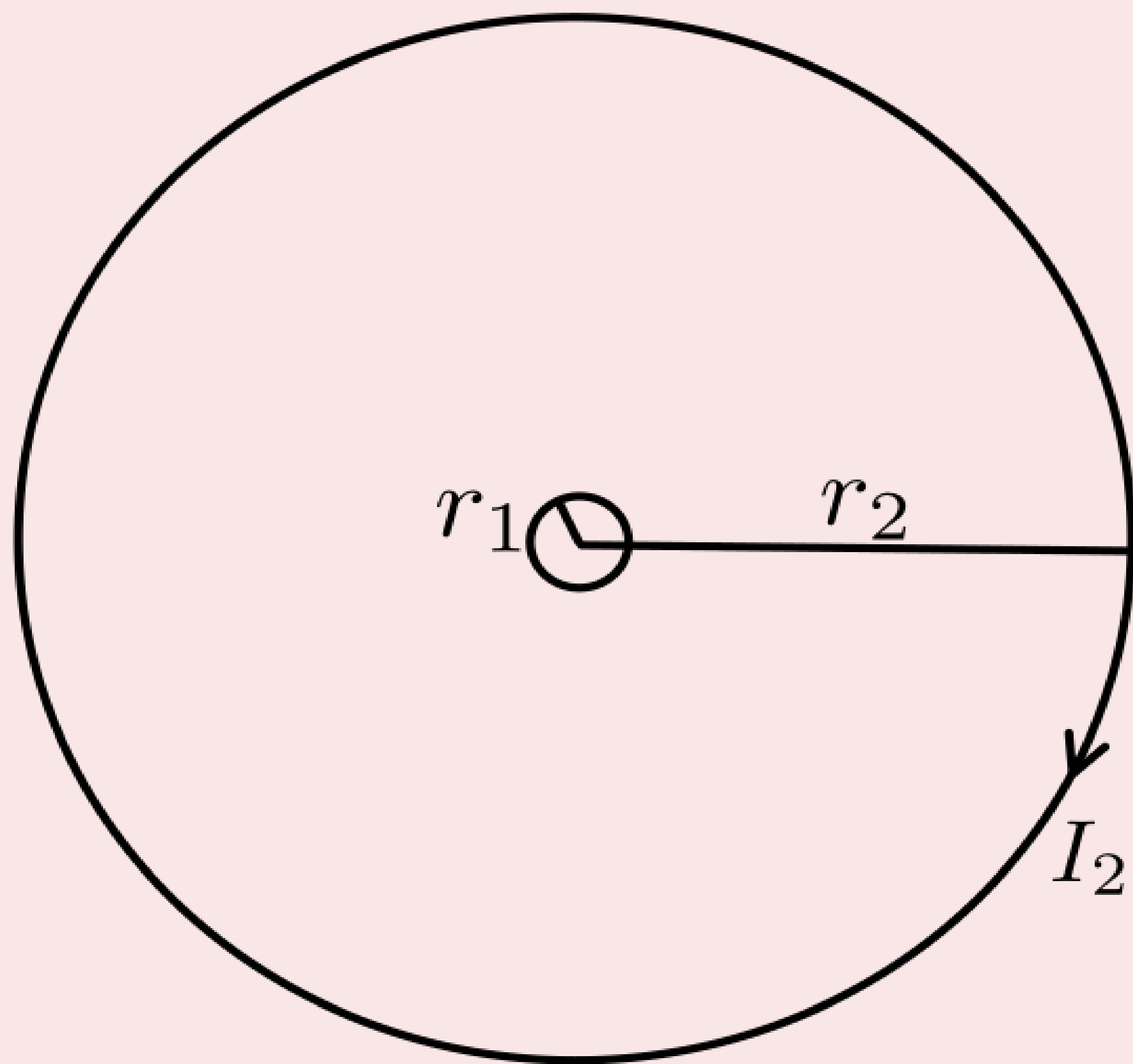
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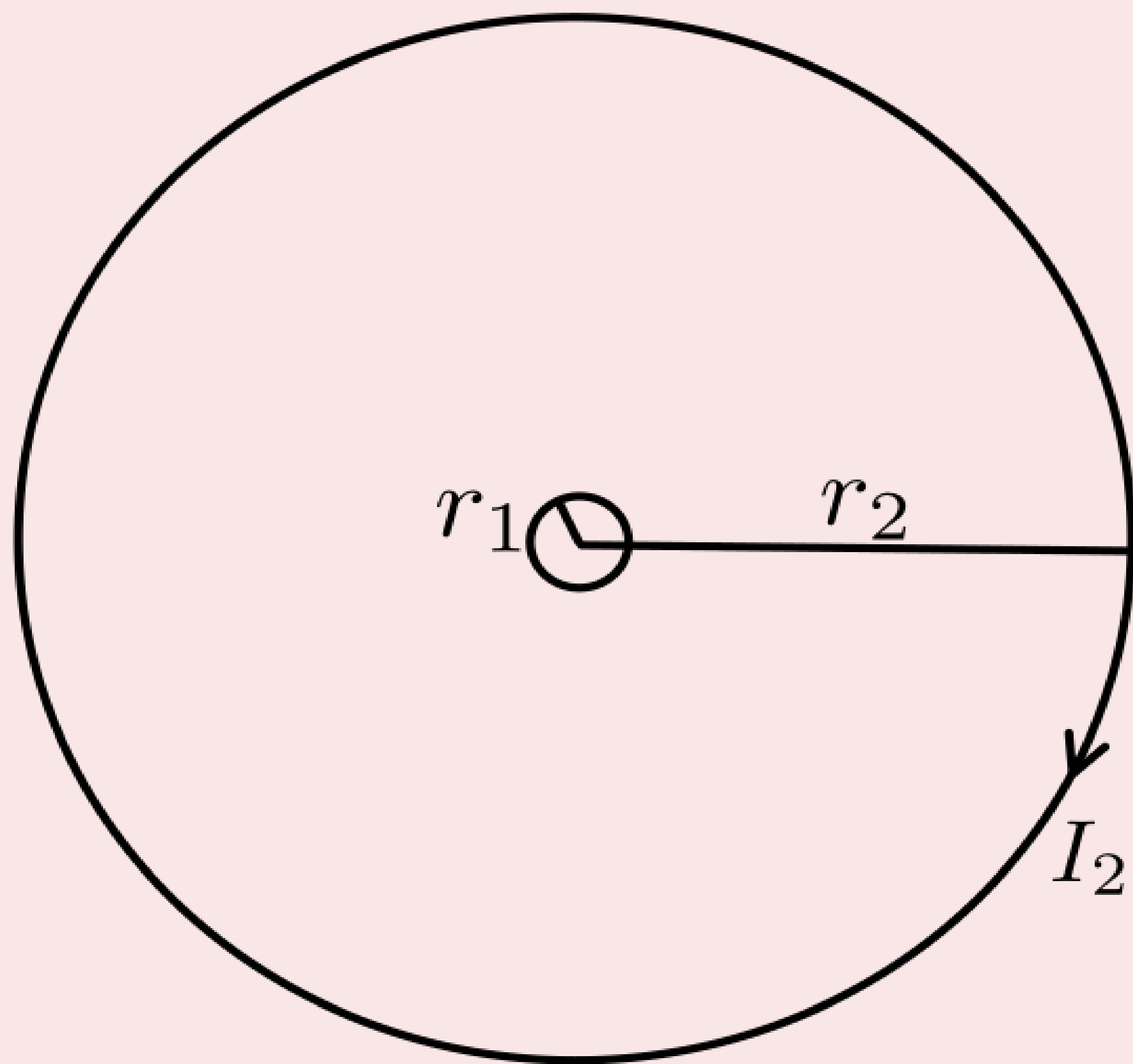


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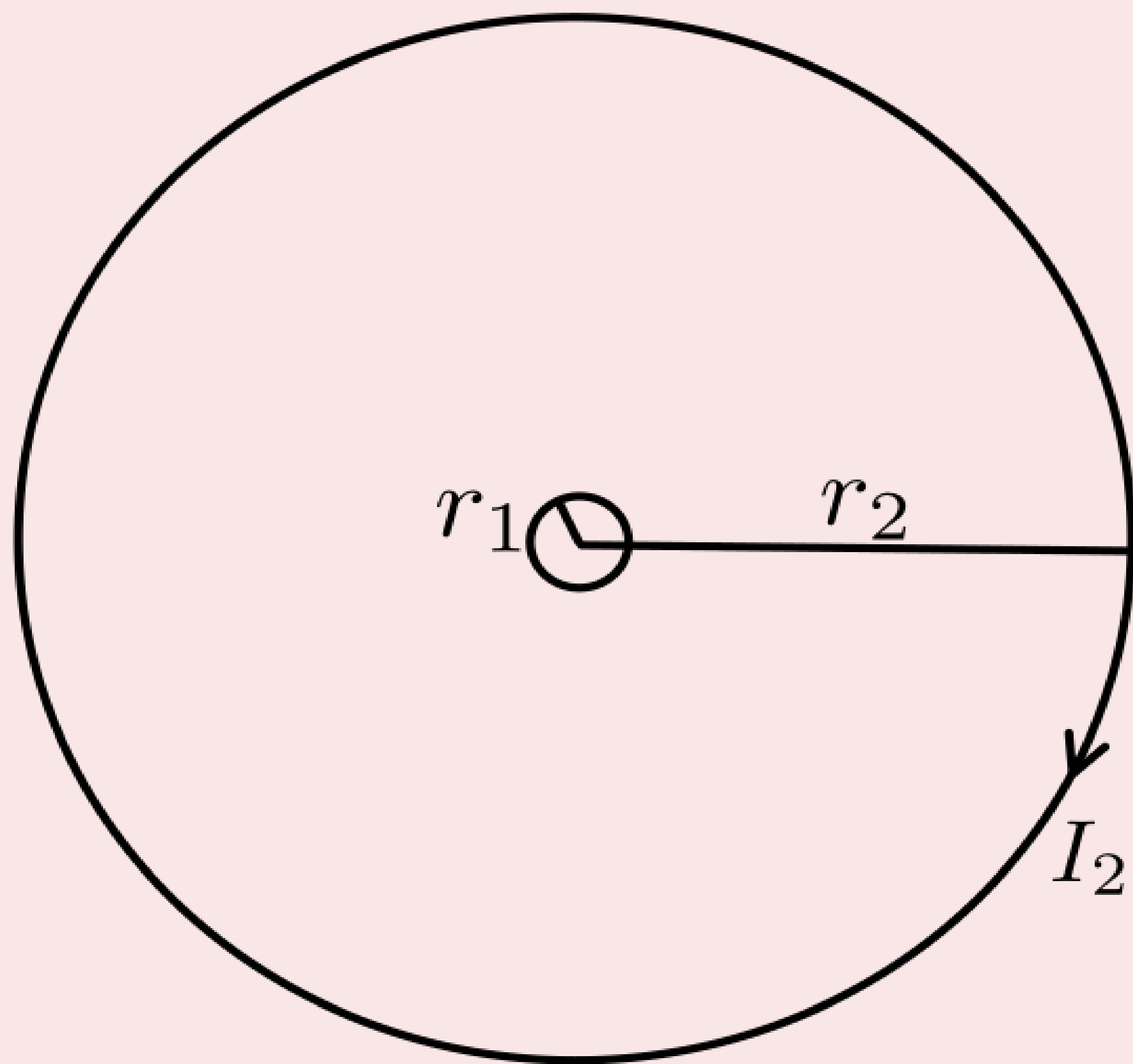
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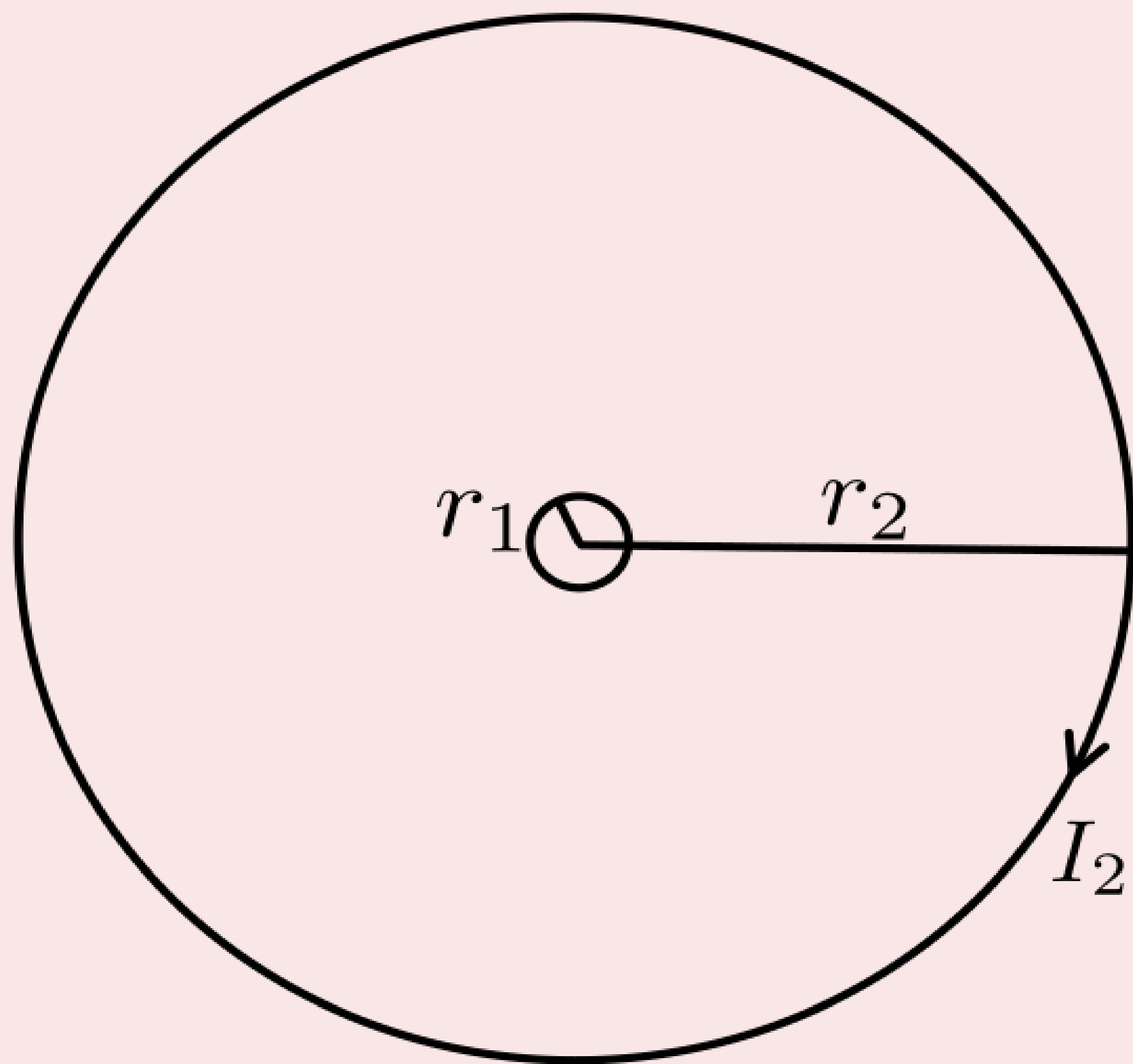
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Energy Stored in an Inductor

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Example 6.10 (a) Obtain the expression for the magnetic energy stored in a solenoid in terms of magnetic field B , area A and length l of the solenoid. (b) How does this magnetic energy compare with the electrostatic energy stored in a capacitor?

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Solution

(a) From Eq. (6.19), the magnetic energy is

$$\begin{aligned} U_B &= \frac{1}{2} LI^2 \\ &= \frac{1}{2} L \left(\frac{B}{\mu_0 n} \right)^2 && \text{(since } B = \mu_0 nI, \text{ for a solenoid)} \\ &= \frac{1}{2} (\mu_0 n^2 Al) \left(\frac{B}{\mu_0 n} \right)^2 && \text{[from Eq. (6.17)]} \\ &= \frac{1}{2\mu_0} B^2 Al \end{aligned}$$

(b) The magnetic energy per unit volume is,

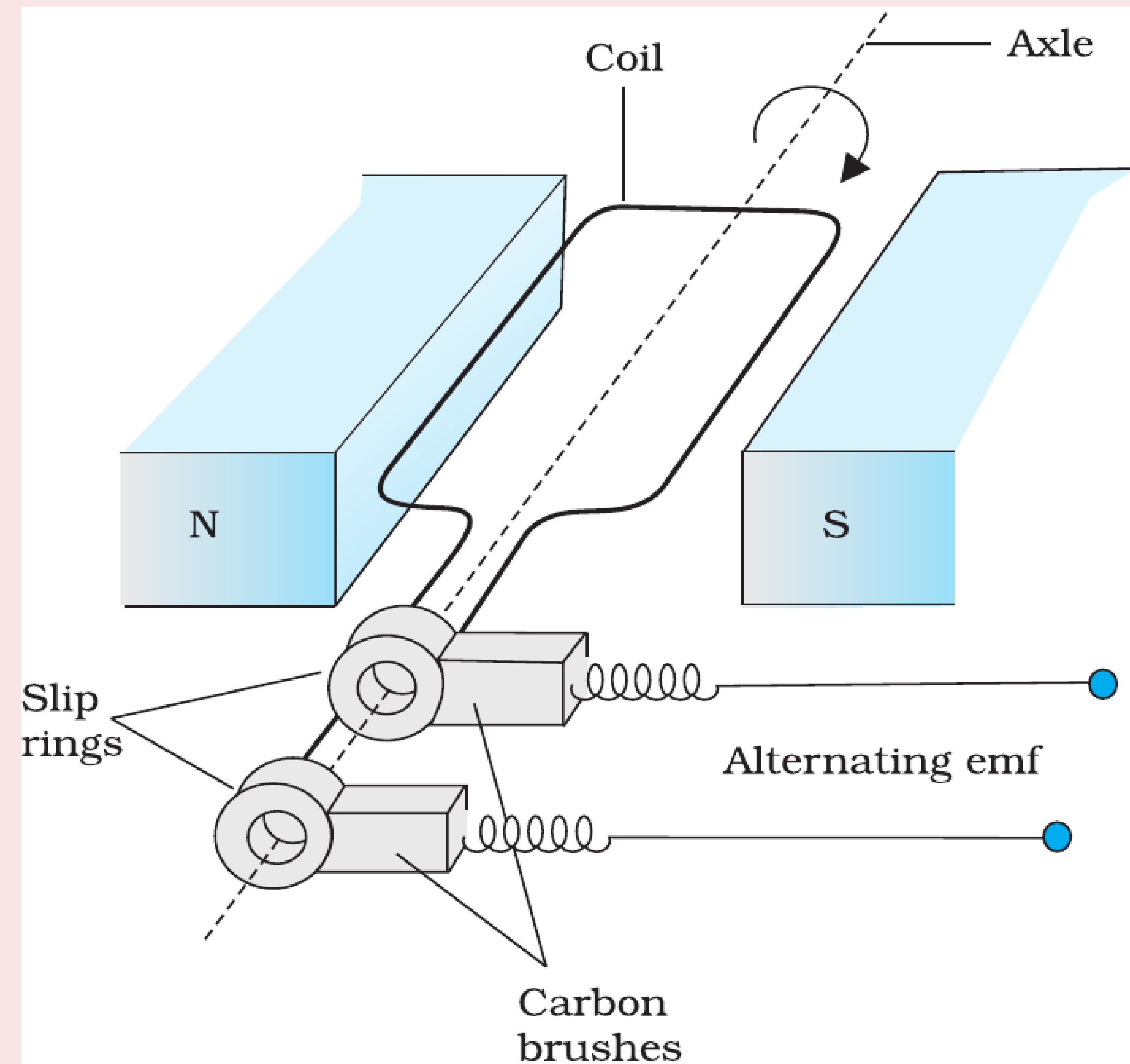
$$\begin{aligned} u_B &= \frac{U_B}{V} && \text{(where } V \text{ is volume that contains flux)} \\ &= \frac{U_B}{Al} \\ &= \frac{B^2}{2\mu_0} \end{aligned} \tag{6.20}$$

We have already obtained the relation for the electrostatic energy stored per unit volume in a parallel plate capacitor (refer to Chapter 2, Eq. 2.77),

$$u_E = \frac{1}{2} \epsilon_0 E^2 \tag{2.77}$$

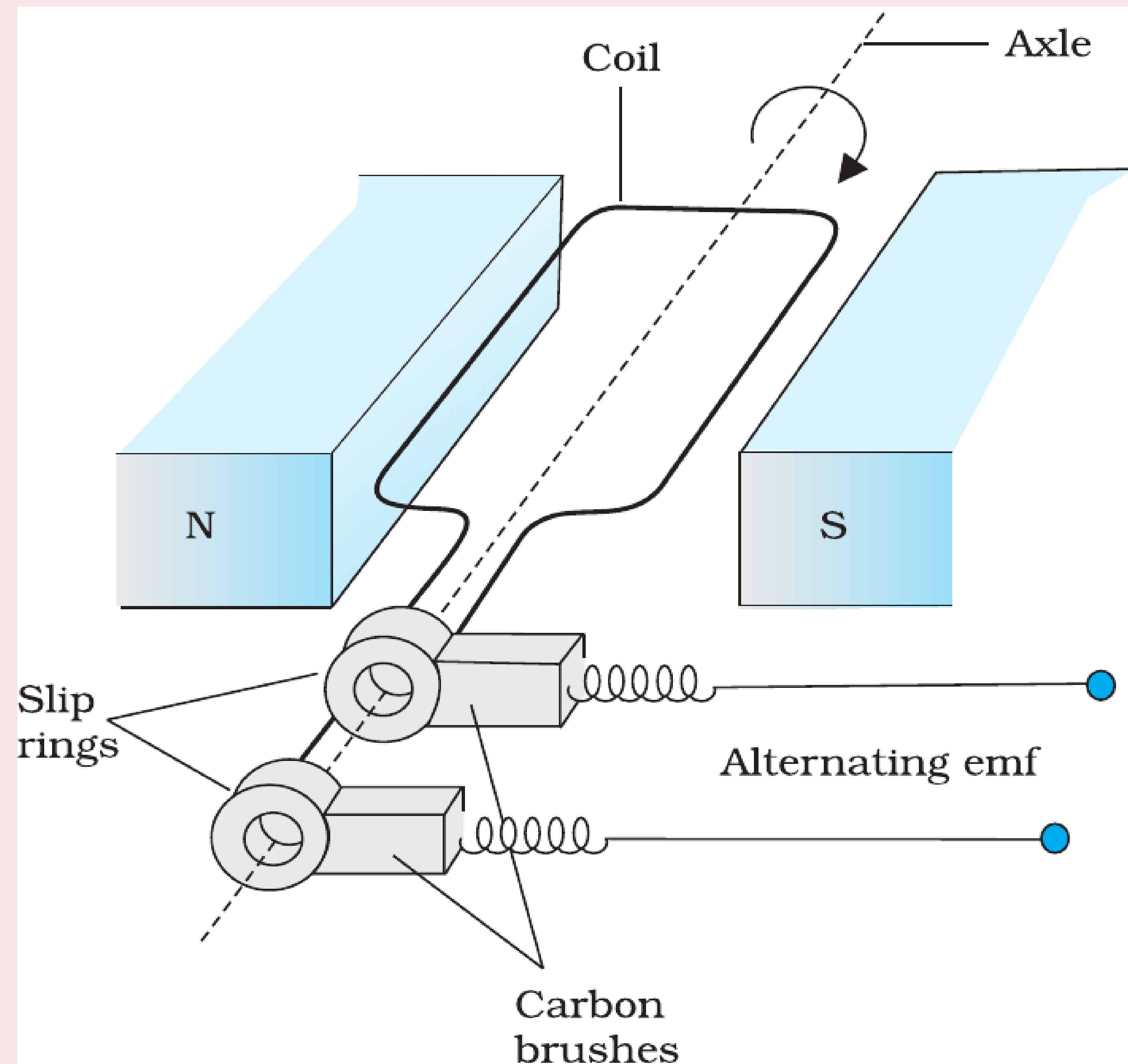
In both the cases energy is proportional to the square of the field strength. Equations (6.20) and (2.77) have been derived for special cases: a solenoid and a parallel plate capacitor, respectively. But they are general and valid for any region of space in which a magnetic field or/and an electric field exist.

AC GENERATOR



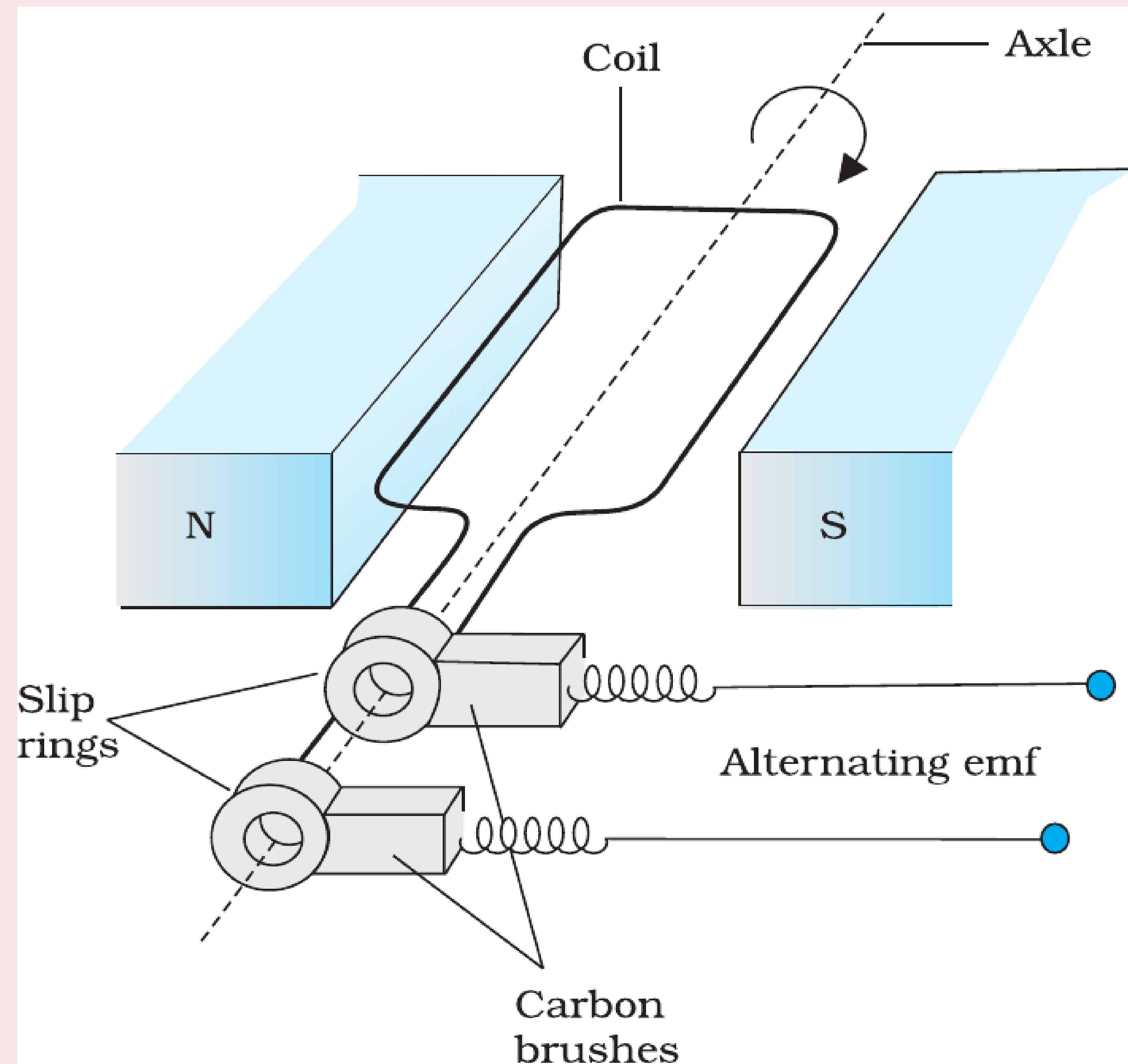
- An ac generator **converts mechanical energy into electrical energy.**

AC GENERATOR



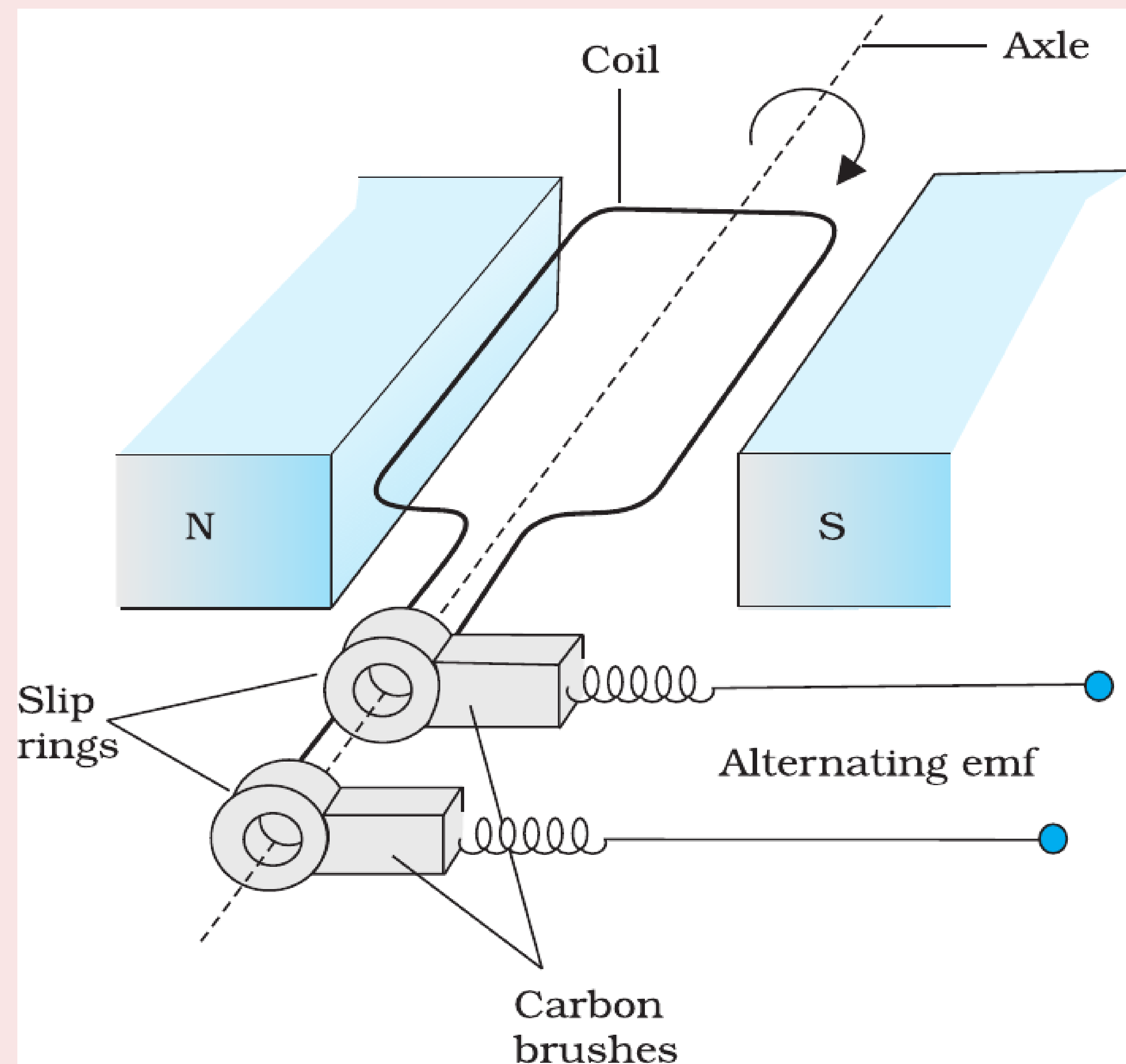
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AC GENERATOR



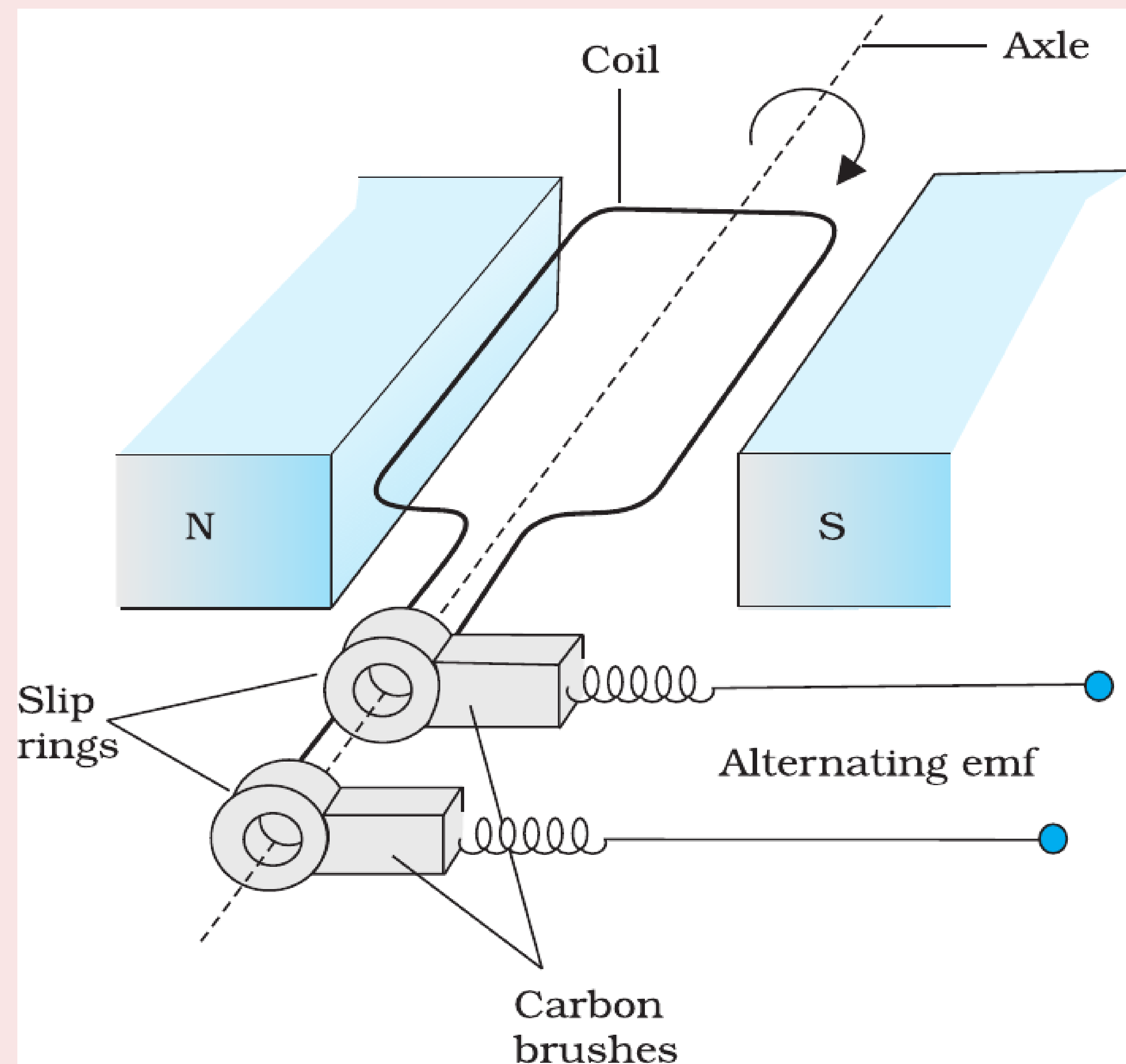
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AC GENERATOR



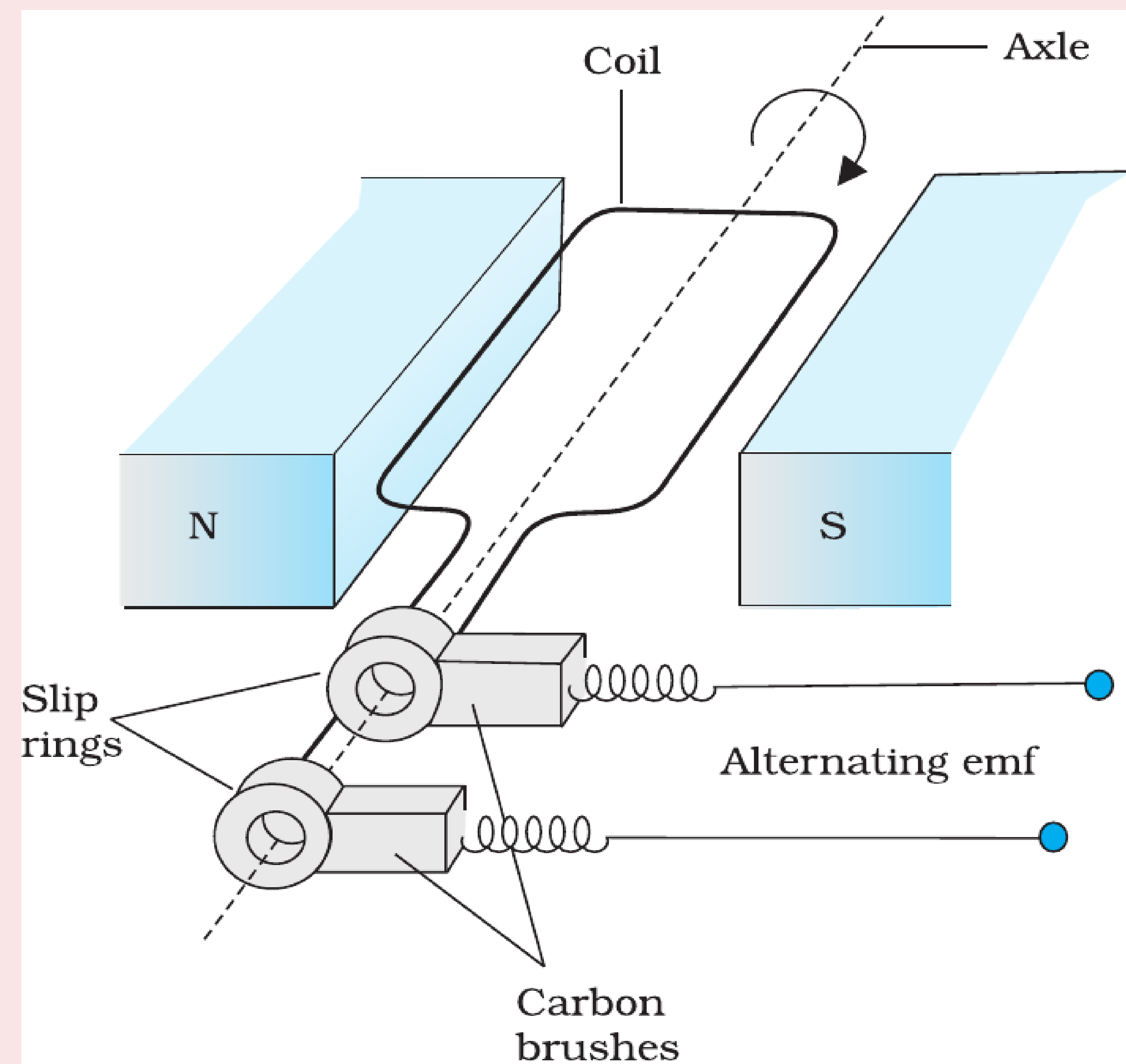
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AC GENERATOR



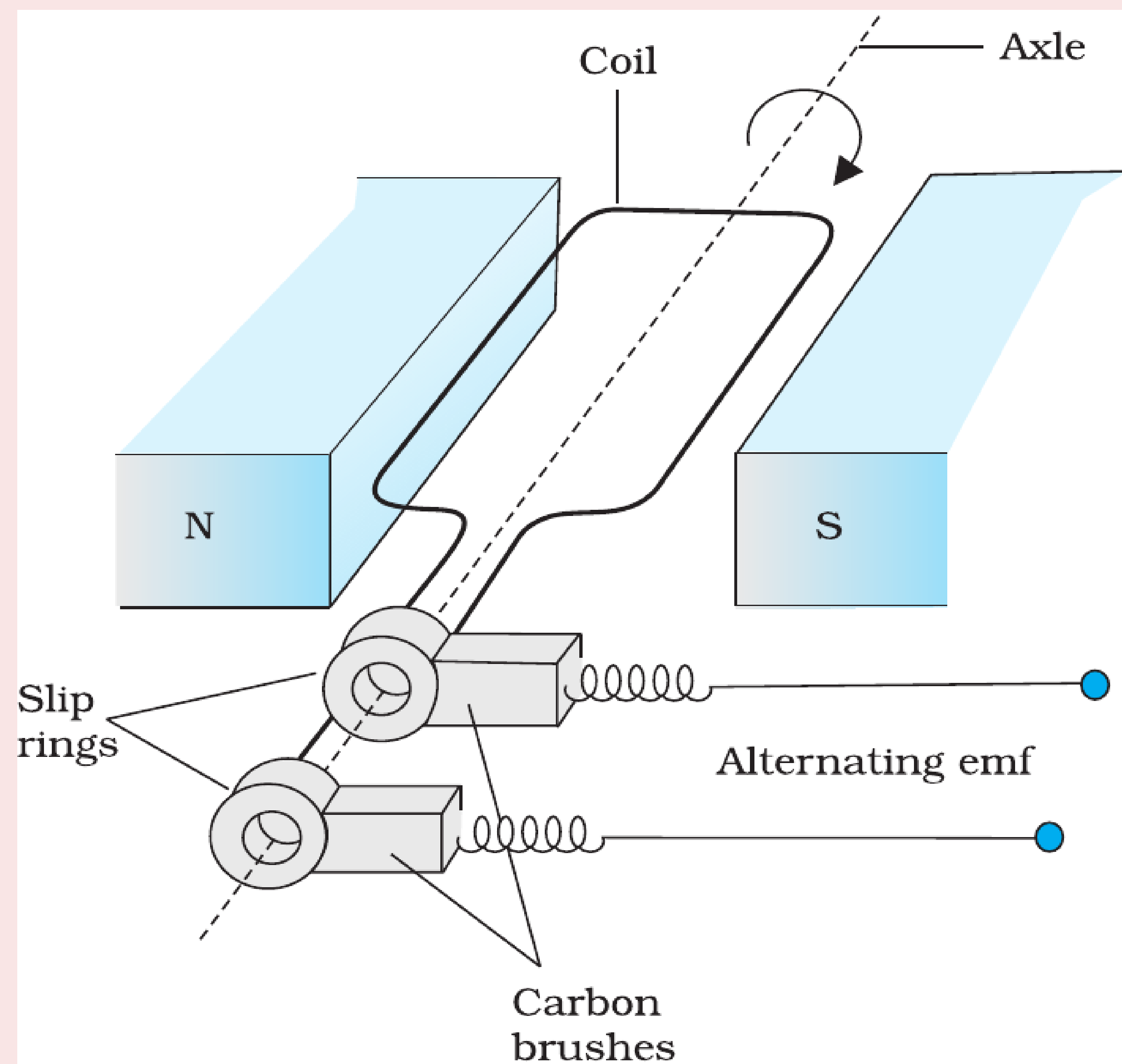
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 - 1 **Magnet :** It may be a **permanent magnet** or an **electromagnet** and provide a strong uniform magnetic field.
 - 2 **Armature :** It is a **coil** generally **wound over a soft-iron core**. The **core increases the magnetic field** due to its magnetization.

AC GENERATOR



- **3 Slip Rings :** The **two ends of the coil are connected to two slip rings** . The coil together with the rings can rotate in the magnetic field.

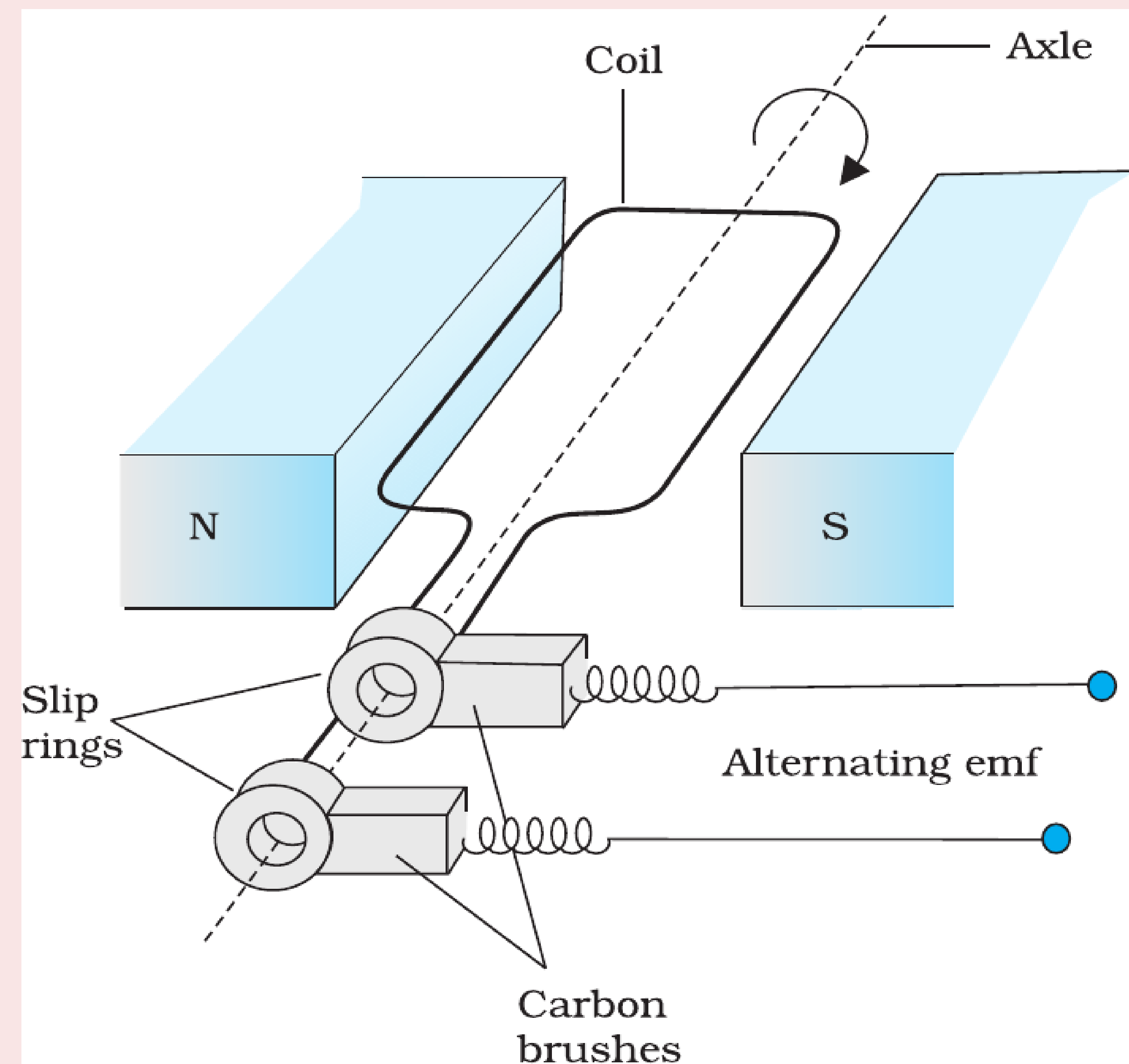
AC GENERATOR



- **3 Slip Rings :** The **two ends of the coil are connected to two slip rings** . The coil together with the rings can rotate in the magnetic field.
- **4 Carbon Brushes :** **Two graphite brushes permanently touch the slip rings**. As the armature rotates, the slip rings slip against the brushes so that the **contact is maintained all the time**. These brushes are connected to the external circuit.

AC GENERATOR

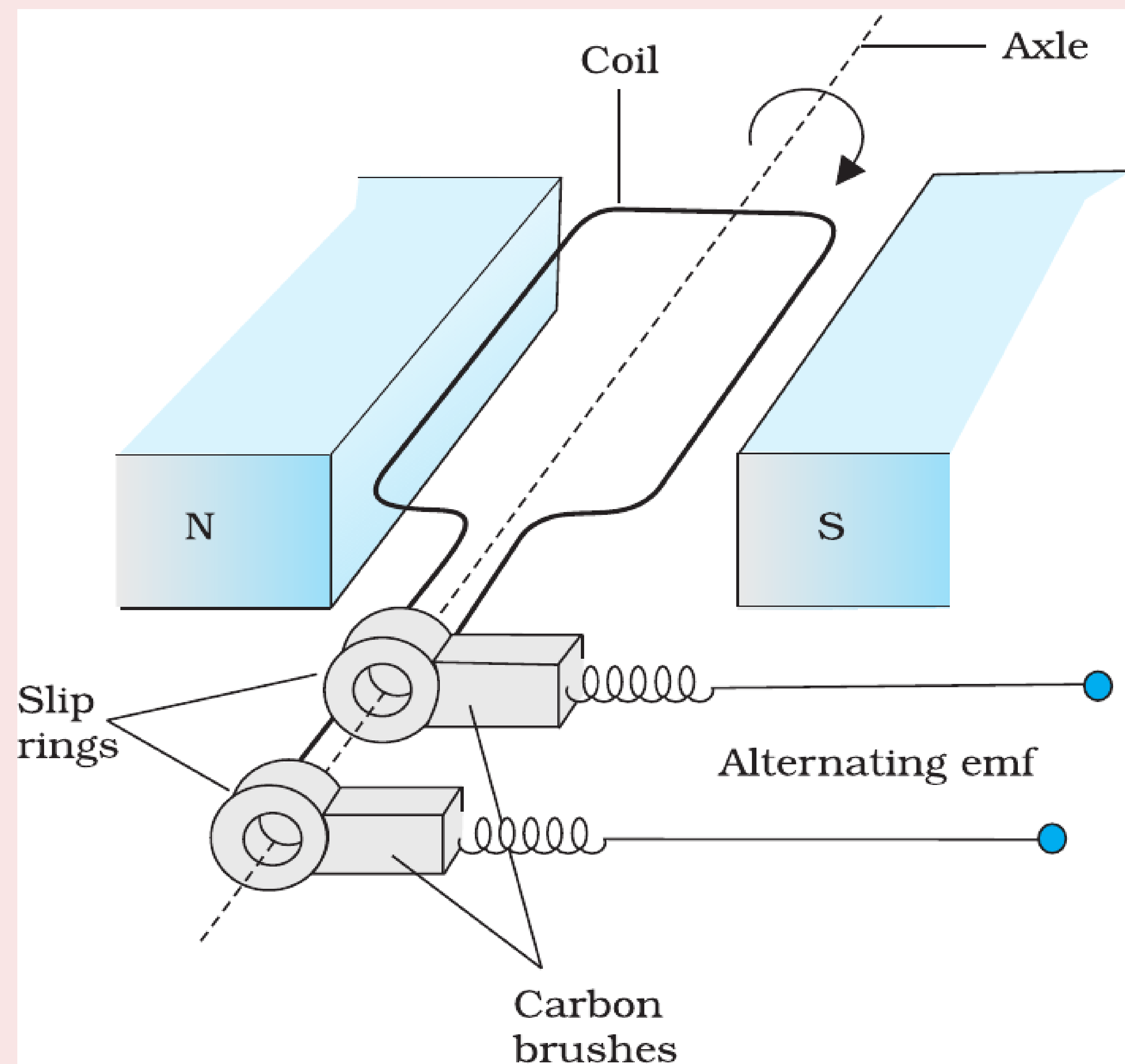
Theory and Working



- When the coil is rotated with a constant angular speed ω ,

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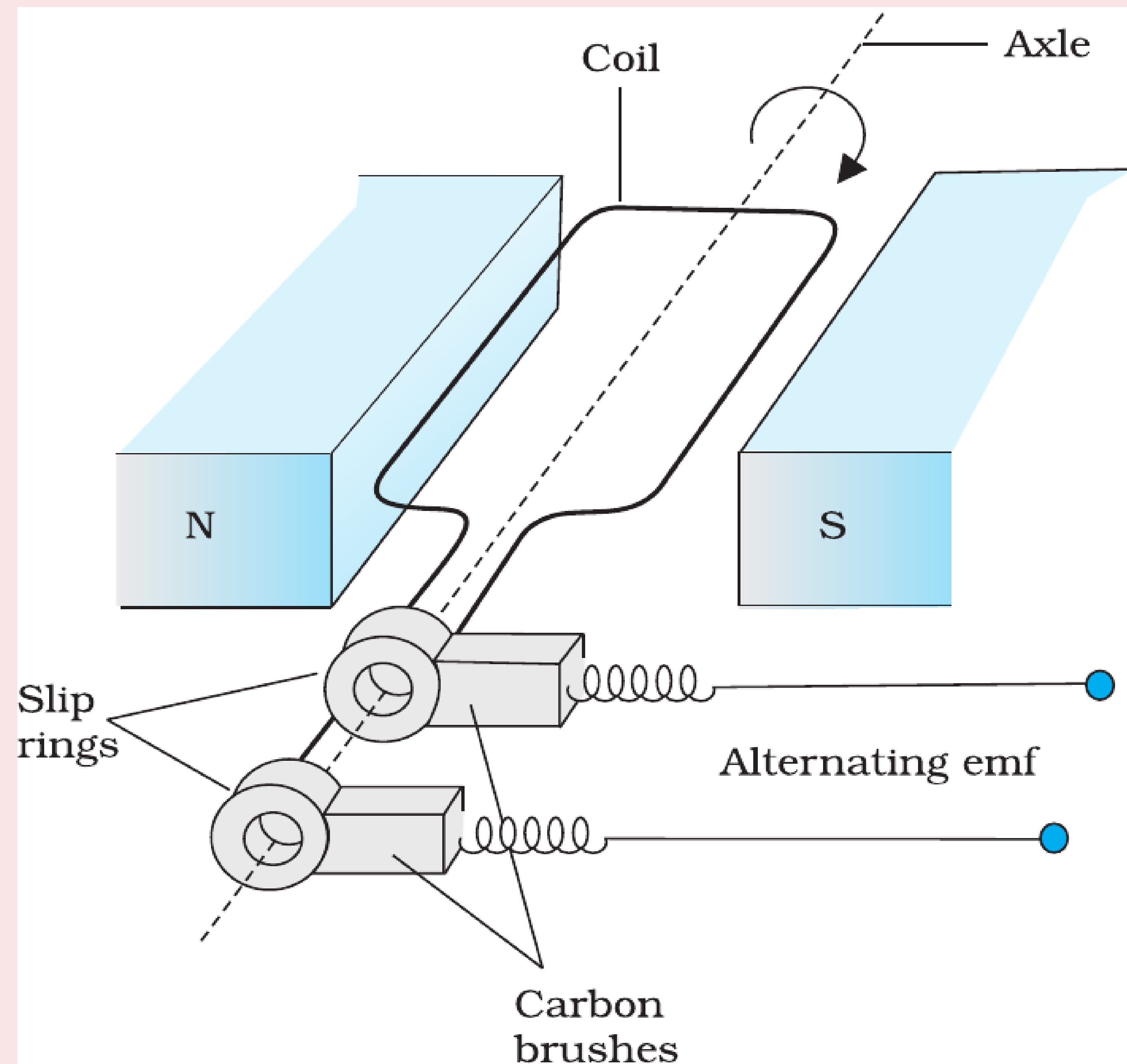
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AC GENERATOR

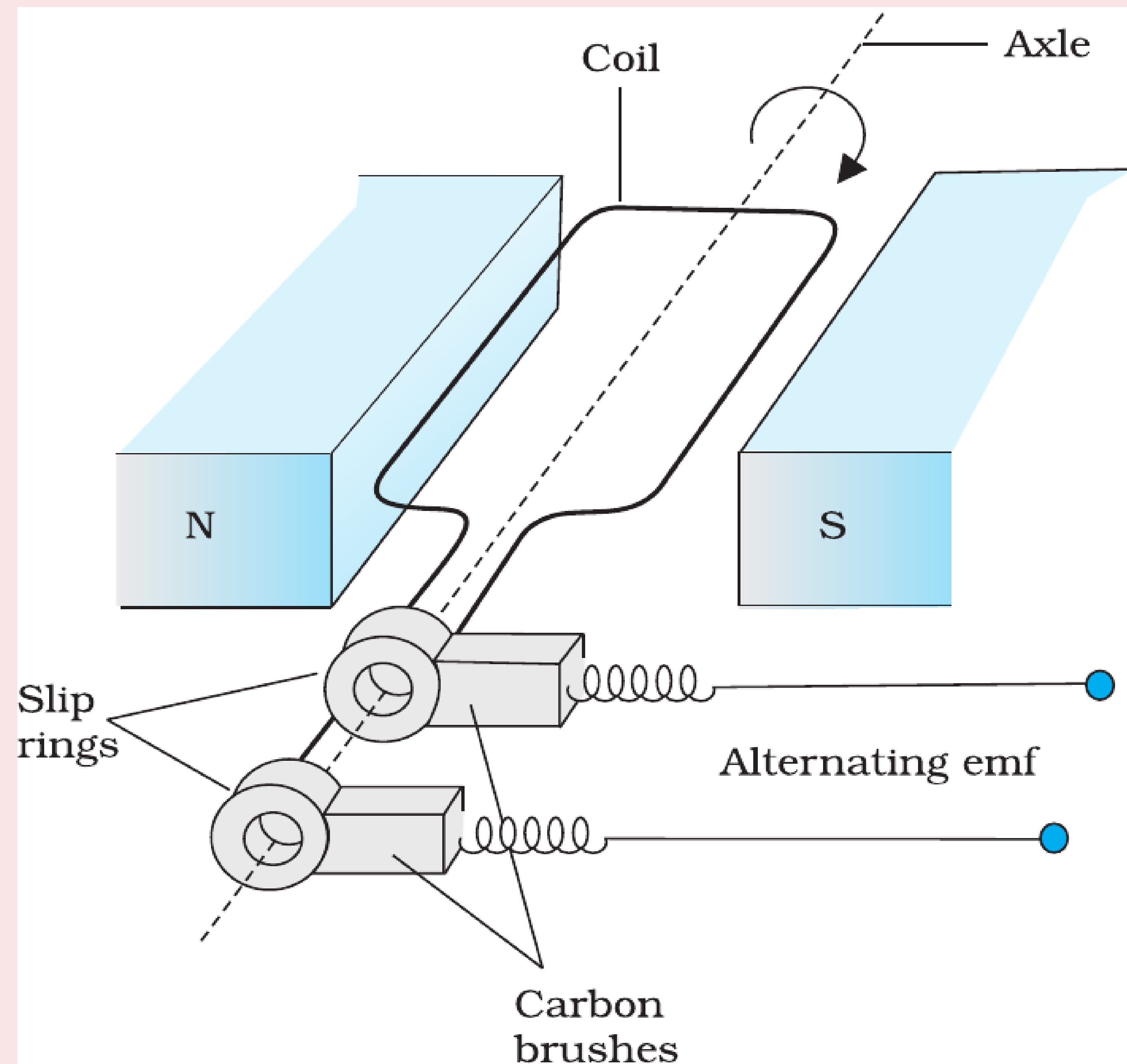
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AC GENERATOR

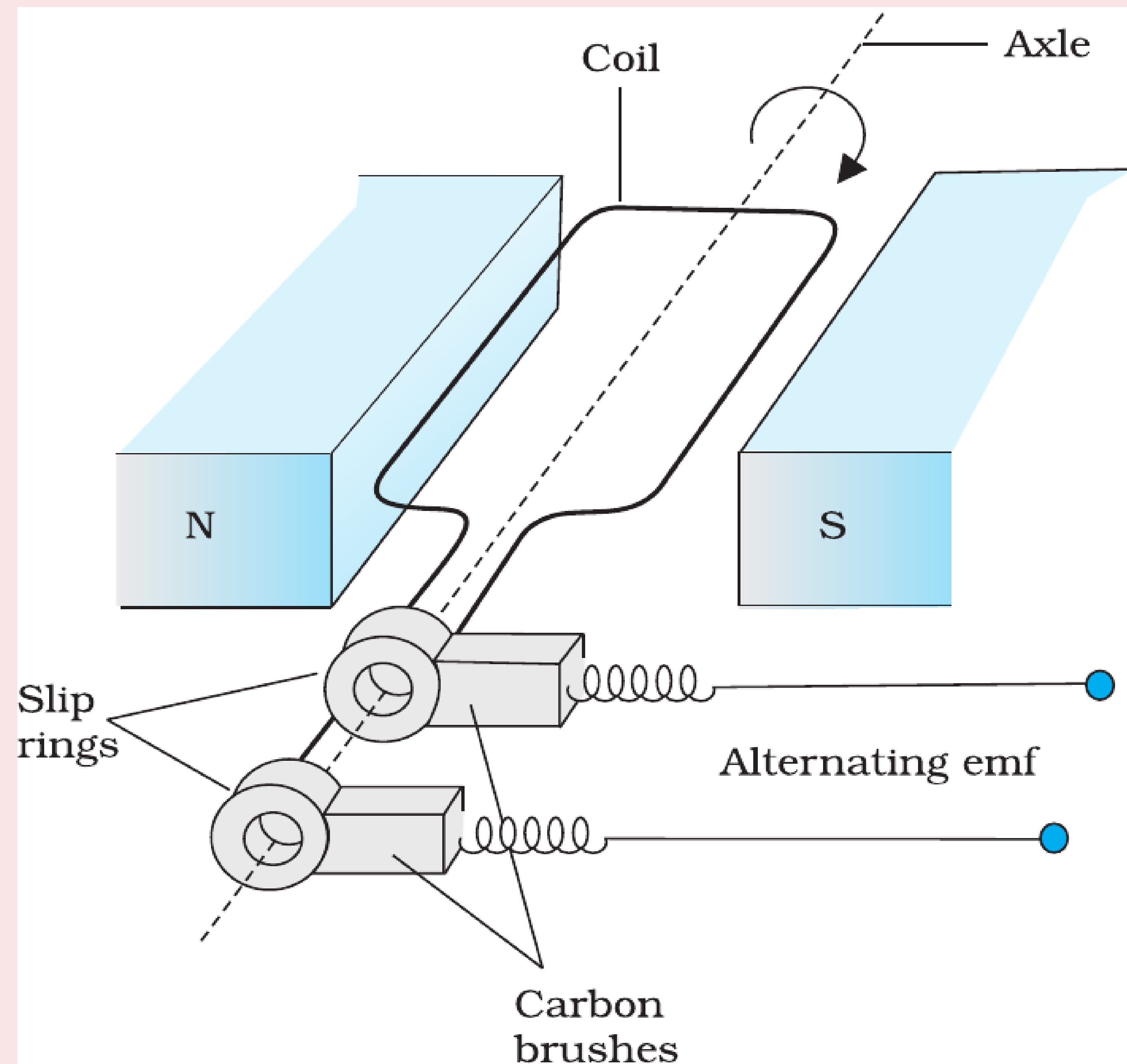
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- From Faraday's law, the induced emf for the rotating coil of N turns is
$$\varepsilon = - N \frac{d\phi}{dt}$$

AC GENERATOR

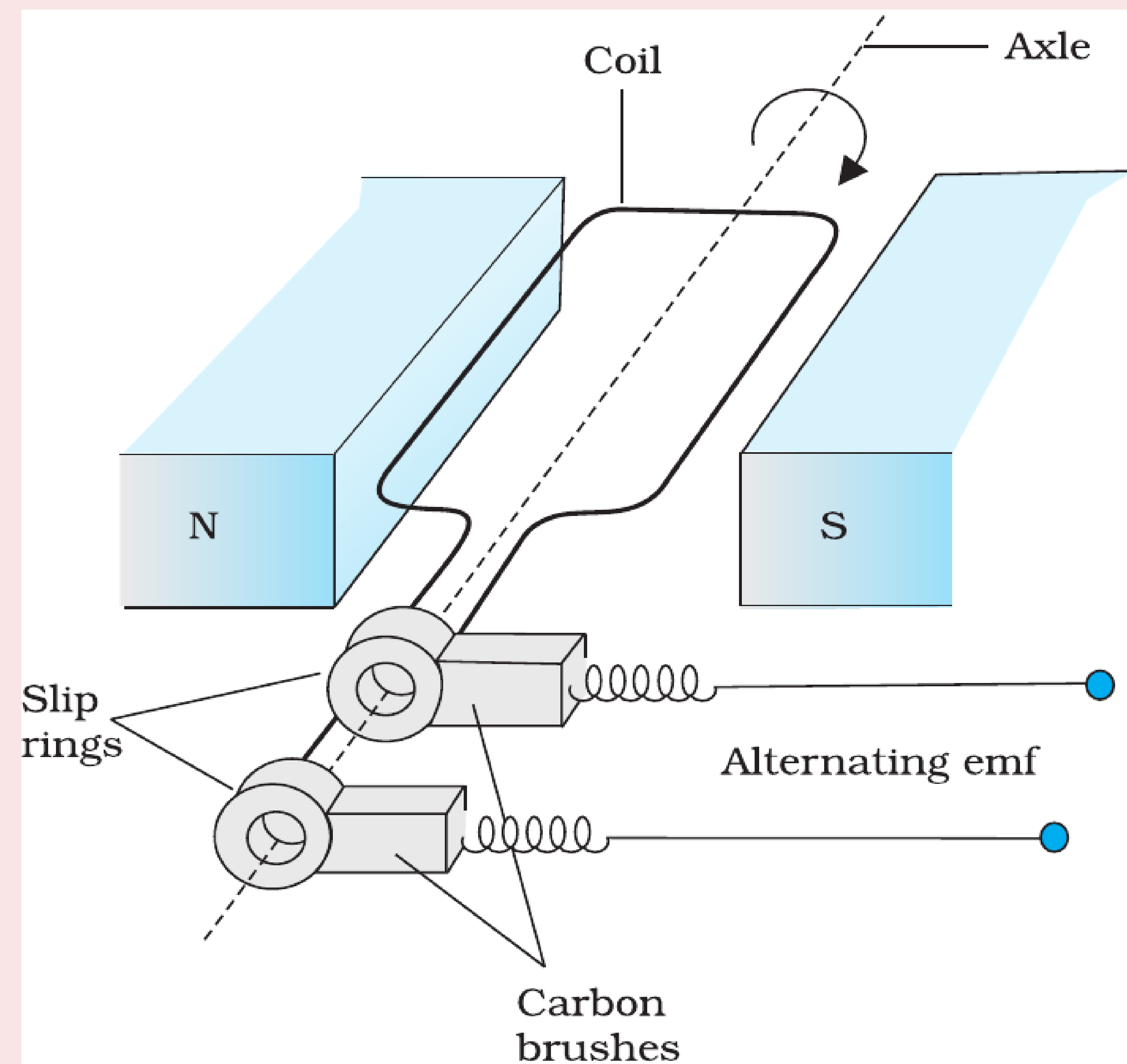
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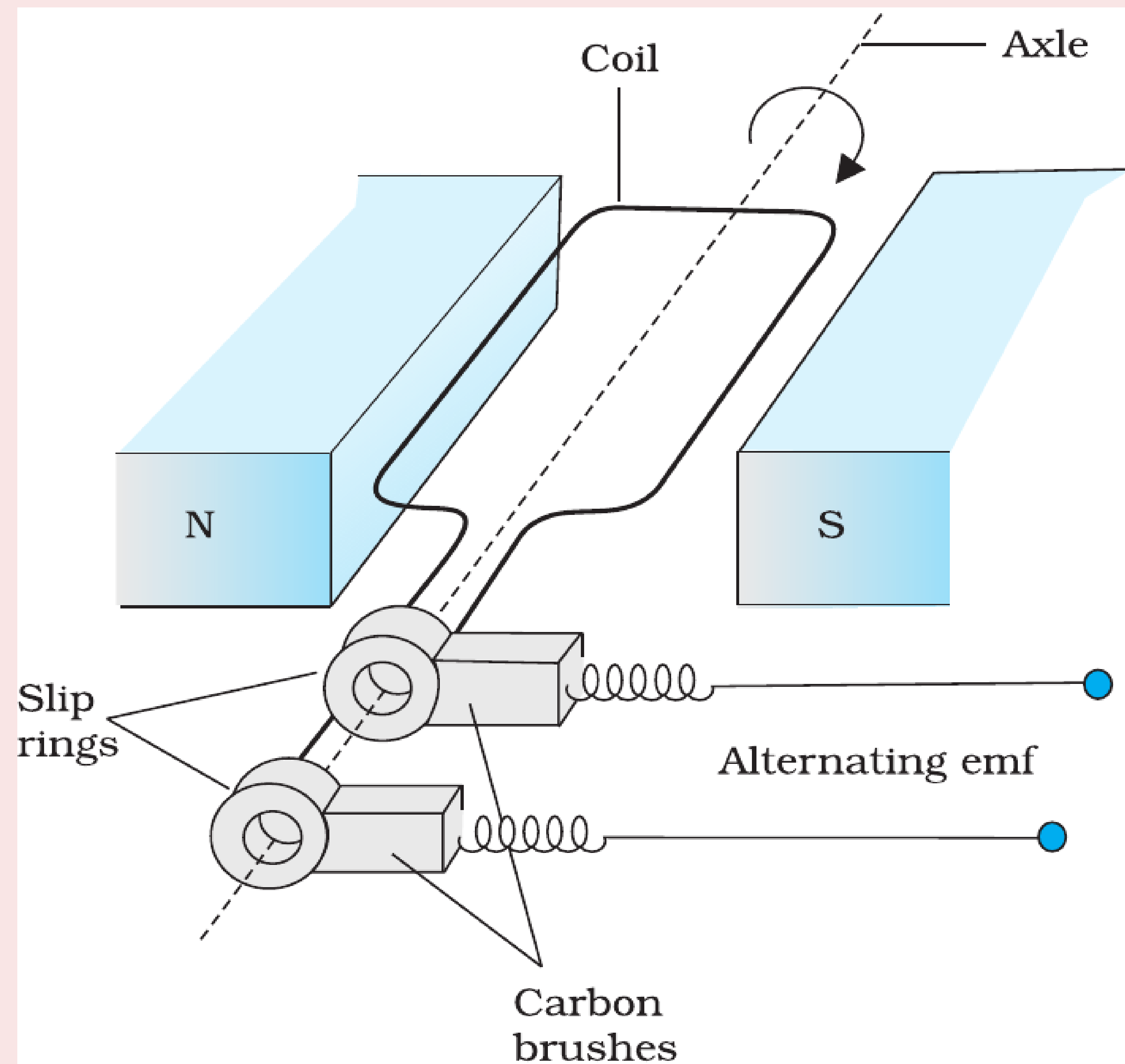
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AC GENERATOR



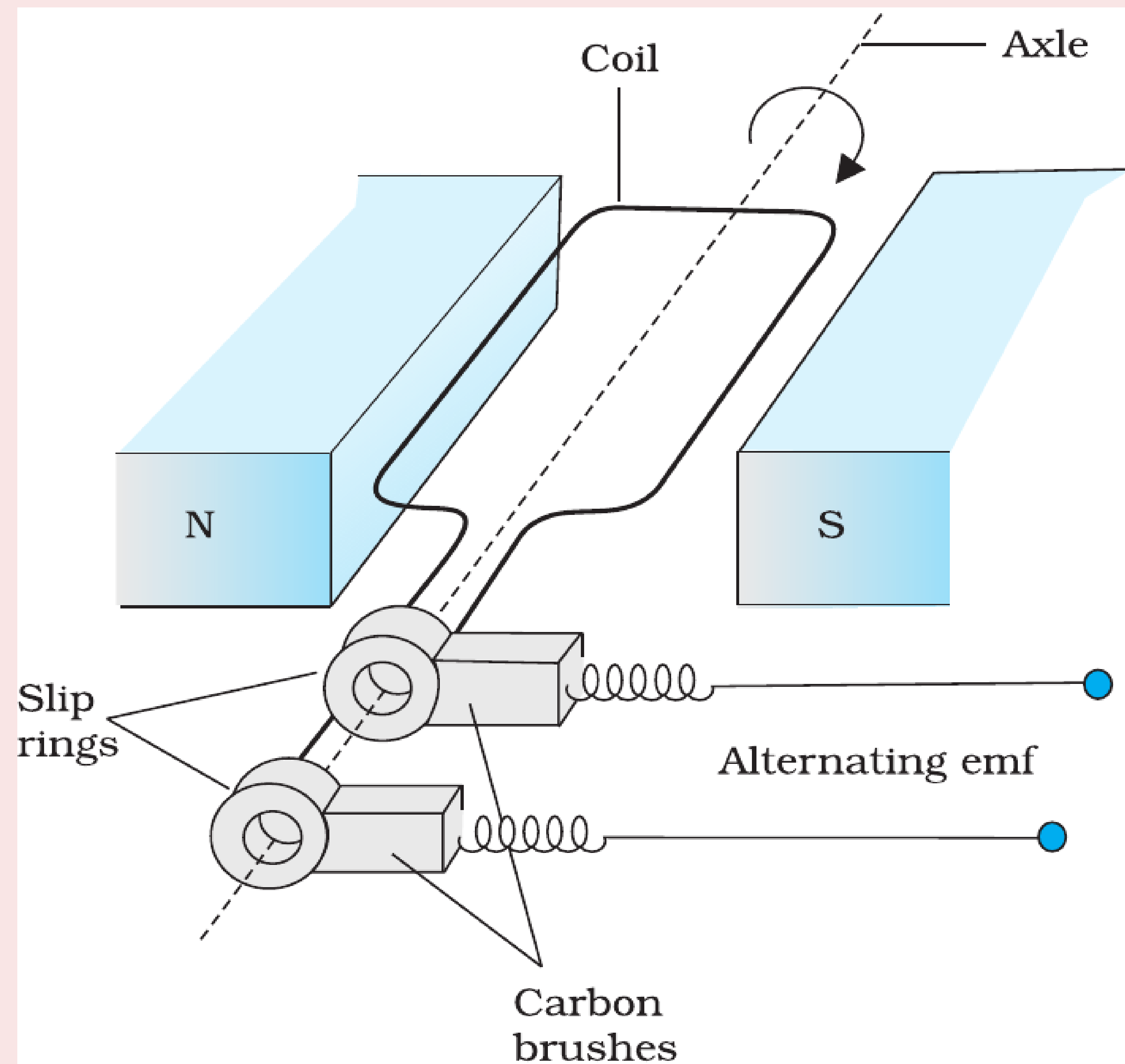
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AC GENERATOR



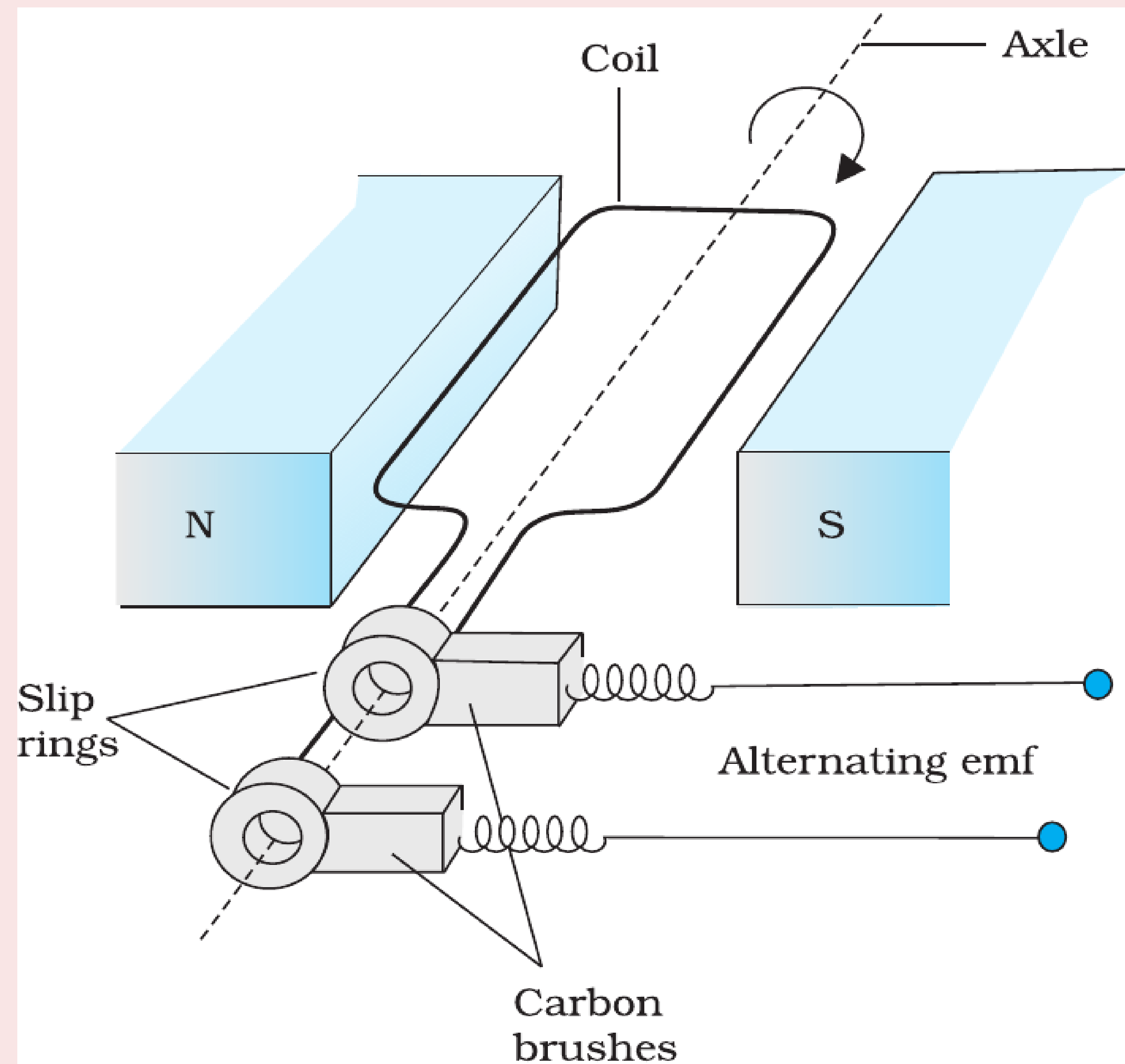
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AC GENERATOR



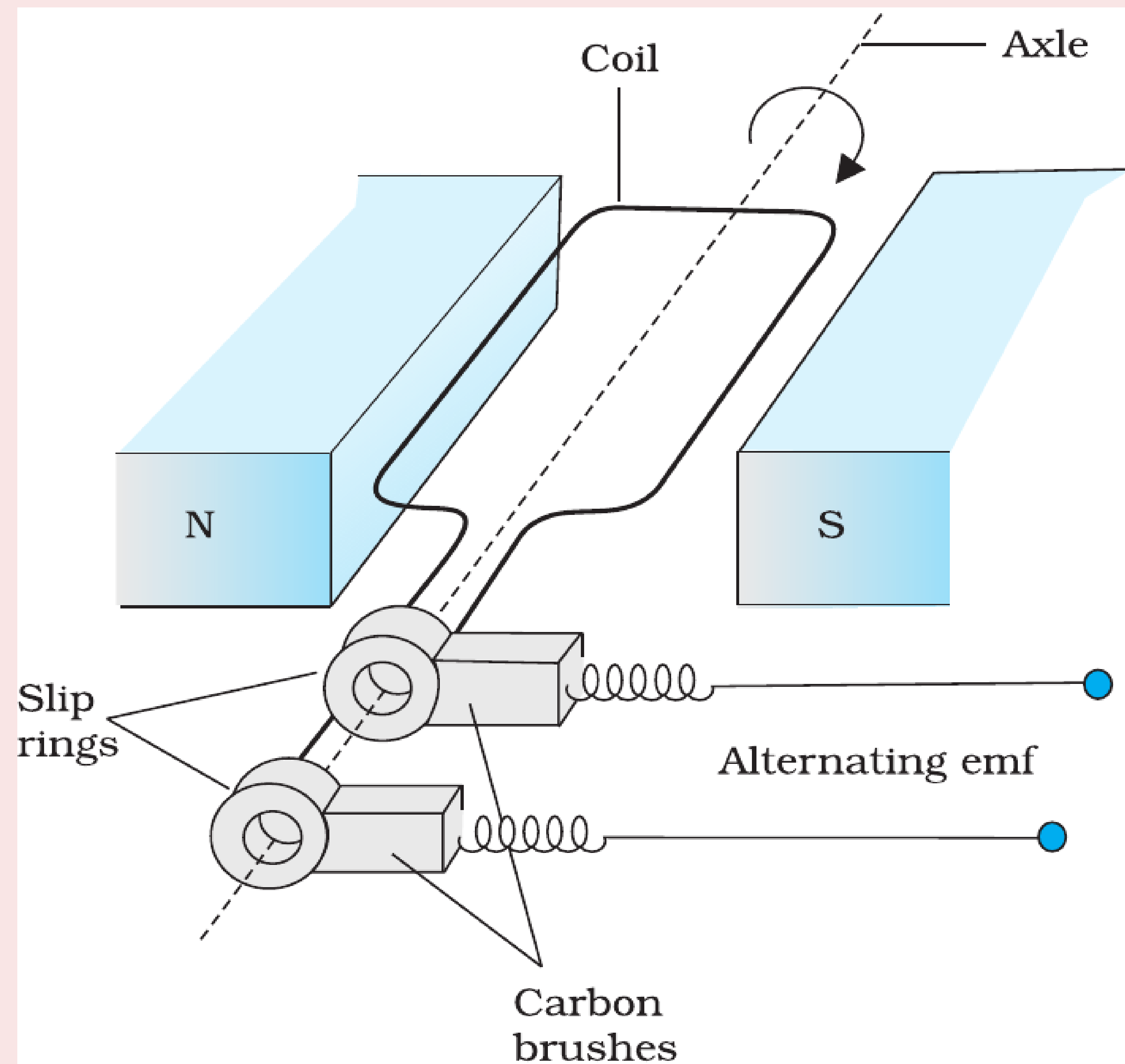
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AC GENERATOR



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- The **direction of the current changes periodically** and therefore the current is called **alternating current (ac)**.
- Since $\omega = 2\pi\nu$, Therefore, $\varepsilon = \varepsilon_0 \sin(2\pi\nu t)$ where ν is the **frequency of revolution of the generator's coil**.

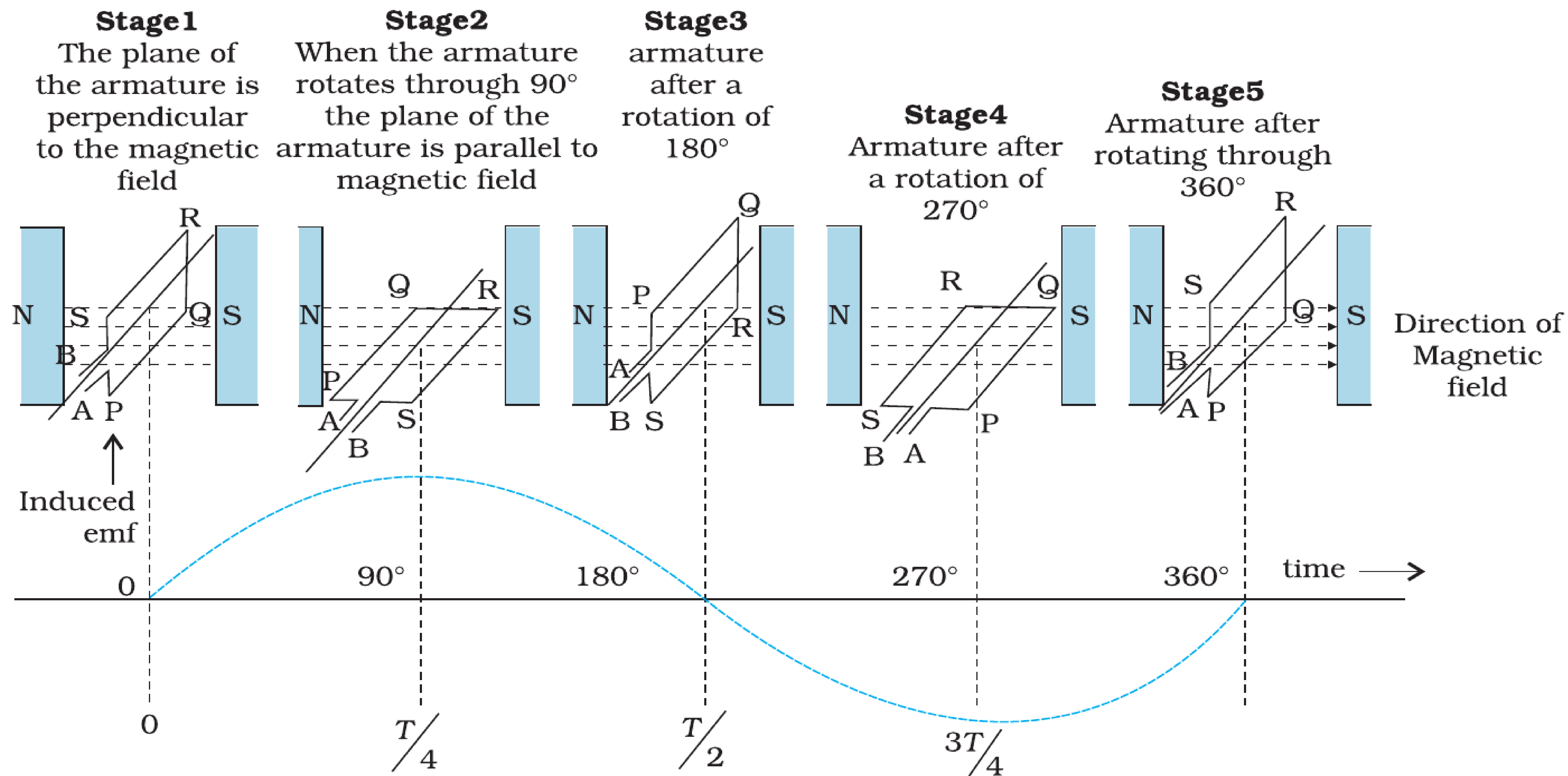


FIGURE 6.17 An alternating emf is generated by a loop of wire rotating in a magnetic field.

Example 6.11 Kamla peddles a stationary bicycle. The pedals of the bicycle are attached to a 100 turn coil of area 0.10 m^2 . The coil rotates at half a revolution per second and it is placed in a uniform magnetic field of 0.01 T perpendicular to the axis of rotation of the coil. What is the maximum voltage generated in the coil?

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Solution Here $\nu = 0.5 \text{ Hz}$; $N = 100$, $A = 0.1 \text{ m}^2$ and $B = 0.01 \text{ T}$. Employing Eq. (6.21)

$$\begin{aligned}\varepsilon_0 &= NBA (2 \pi \nu) \\ &= 100 \times 0.01 \times 0.1 \times 2 \times 3.14 \times 0.5 \\ &= 0.314 \text{ V}\end{aligned}$$

The maximum voltage is 0.314 V .

We urge you to explore such alternative possibilities for power generation.